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Introduction

Give a model an untreated cell and the name of a drug, and ask it what that drug will do to that cell. That is the task this thesis evaluates. It is worth answering because the answer is otherwise obtainable only by running the experiment.

The thesis takes one published cell-sentence checkpoint, fine-tunes it on a hundred-million-cell drug atlas, and asks whether the predictions it produces identify the drug that produced them. They do not. Most of what follows is the apparatus that makes that a finding and not an anecdote: a metric calibrated before it was trusted, controls a drug-blind predictor cannot pass, and a bar that needs no model at all.

1.1 The truth this task predicts is never observed

A drug that enters a cell changes what that cell transcribes. Which genes move, by how much, and in which direction is the readout the field treats as its most direct account of what a drug does. That is a working premise, not an established fact. A substantial share of the treated conditions in the atlas used here are statistically inert or not empirically discriminable at the tested pseudobulk and cell budget (chapter 3), and cannot support this evaluation.

Reading one response is routine; reading every combination that might matter is not, because a drug behaves differently across doses and cellular backgrounds, and even the largest screens sample a small corner of the space. A computed response would make the untested combinations inspectable, which is what prediction is for. The task is simple to state: given the transcriptome of an untreated cell and the identity of a drug, predict the transcriptome that cell would have shown had the drug been applied.

It is harder to grade than to state, because the ground truth is counterfactual. Sequencing destroys the cell it reads, so a prediction can only be checked against a *different* cell that received the drug. Paired training data must therefore be assembled by matching cells, evaluation is framed at population level, and distinguishing a drug effect from ordinary cell-to-cell variation requires explicit controls. The units, the frames and the controls that follow all descend from that one fact.

1.2 The atlas contributes breadth; the inferential unit decides what can be claimed

Tahoe-100M [1] makes that evaluation possible at unusual scale, with roughly one hundred million single-cell transcriptomes spanning about 1,100 small molecules across 50 cancer cell lines. Each condition, one drug at one dose in one cell line on one plate, is a population and not a single summary.

A population per condition admits two questions. The conditional one, what a drug does in a particular cellular context, is what the task asks. The prior one is whether the apparent response is there to be predicted at all, and how much of it reproduces across disjoint halves of the same treated well. Standard evaluations blur the two; much of this thesis keeps them apart.

Scale does not remove design limitations. Conditions are spread unevenly across drugs, doses, cell lines and treated wells, and plate membership can change which alternatives a rank-based score sees. Cells from the same well are not independent biological replicates, and neither are drug rows sharing a cell line. The analysis therefore builds pseudobulk truths from disjoint cell halves and keeps the well as a resampling unit.

SEE chapter 3: the atlas layout, and which unit each interval is clustered on

1.3 A model can be fluent in cell sentences without reading the drug

Cell2Sentence [2] writes a cell as its gene names in descending order of expression, turning it into text an ordinary language model can consume. It discards expression magnitudes and keeps ranks; in exchange it needs no specialised architecture. C2S-Scale [3] extends it to models pretrained on more than 50 million cells from 825 single-cell datasets, fine-tuned across classification, generation, captioning and perturbation tasks. Its published perturbation work covers cytokine stimulation and, for small molecules, bulk L1000 profiles and SciPlex3, evaluated primarily as distributions and not condition by condition.

CELL SENTENCE A cell written as its expressed genes, named in descending order of expression.

The checkpoint studied here is C2S-Scale-Pythia-1b-pt, the pretrained Pythia-1b model [4] from that family, with no drug-treated cells in its declared pre-training corpus. Fine-tuning uses several hundred thousand Tahoe cells, not the entire atlas. Every arm is cold-started from that checkpoint with identical hyperparameters, so within an experiment the training target or objective is the only variable. The prompt supplies the cell line, the drug, the dose, a mechanism annotation and the untreated control cell's own sentence, and what the model returns for that prompt is the prediction that gets scored.

The training target separates the two principal arms, and with it the frame each is scored in. The first arm predicts the treated cell's own sentence and is scored in the expression frame (■_{EXPR}), where a prediction is a cell profile ranked against other drugs on the same plate. The later arm predicts the drug-specific residual, meaning what survives once the drug-agnostic mean response, the *generic*, is subtracted from the treated-minus-control shift. It is scored in the residual frame (▲_{RESID}), where a prediction is a drug-specific signature ranked against other drugs in the same cell line. Both frames put

chance in the same place, and neither is interchangeable with the other. Which is in force is declared at the head of every section.

The appeal of the encoding is also the scientific risk. Housekeeping genes dominate every ranked profile, treated cells retain most of the state of their untreated lineage, and many perturbations share a generic response programme. A model can therefore produce a plausible treated cell while remaining almost indifferent to which drug was named.

SCOPE LIMIT | carried into every held-out result in chapter 4 and chapter 5

A held-out drug is not unseen by the whole pipeline. In natural-language pre-training the language backbone may have encountered drug names and their pharmacology, and every prompt supplies the drug name plus a Mechanism field, an annotation where one exists and the literal string `unclear` otherwise. “Held out” therefore means absent from Tahoe fine-tuning, not absent from the model’s prior world knowledge or from the inference prompt.

Holding data out comes at three levels of difficulty, not one. Trained conditions test whether the model can reproduce information available during fine-tuning. Unseen combinations hold out a whole treated well but often recombine a drug and a cell line each encountered separately. Unseen drugs remove all retained fine-tuning conditions for those drugs. These are not rungs on one clean learning curve. They differ in support and in empirical reliability, and only the last asks for a response without a Tahoe-trained per-drug memory. The thesis therefore avoids calling all held-out examples zero-shot and never pools the three strata into one transfer number.

1.4 Four questions, each asked because of the answer to the last

This thesis extends the cell-sentence line of work to chemical perturbation on Tahoe-100M and asks whether a model fine-tuned on this atlas learns what a particular drug does to a particular cell. A negative answer earns its place only if it can be made specific, so the investigation asks four questions in order, each one raised by the answer to the one before it.

- 1. Is the ruler trustworthy?** Can a predictor carrying no drug information score well on the field’s standard metrics, and which metrics survive that control?
- 2. Does the model read the drug?** Measured on whichever metric survives the first question, with the input cell fixed and only the prompt’s drug changed, does the prediction move at all?
- 3. Why does it behave that way?** Is the drug absent from the data, absent from the model’s internal representation, or present in the representation but not consulted when the model generates?
- 4. Does redefining the effect repair it?** If the target is re-encoded as the drug-specific residual instead of the treated cell’s own sentence, does the model become sensitive to the drug in its prompt?

The order is not presentational. A metric a zero-information predictor can saturate cannot grade anything asked after it, and four of the five metrics audited fail the corresponding control: the field's standard score is topped outright by a predictor carrying no drug information at all, and in both audited arms the other three rank a drug-agnostic baseline above a real positive control. The survivor is a retrieval test, defined next, and it is the repaired ruler every later answer is read from. The null the second question returns on it is worth little until the third locates it: drug identity is written into the activations and barely consulted when the model generates. That diagnosis is what motivates the fourth, which changes neither the model nor the data but the definition of what the model is asked to produce.

1.5 The question is identification, not resemblance

Distributional similarity and token likelihood reward a prediction for looking like a treated cell, even when the condition identity is wrong. The test left standing is harder, and it is a retrieval test: among the other drugs measured in the same context, does a prediction identify its own treatment?

DEFINITION Normalised inverse rank (NIR)

A prediction is compared with its own condition's truth and with the truths of the other conditions in a declared comparison set. NIR is the fraction of that set whose truth the prediction resembles *less* than it resembles its own. The comparison set is what fixes chance at 0.50, in both measurement frames and under an exchangeability condition that § 4.1 audits with empirical control arms. The score is tie-aware, and any component shared by every candidate cancels.

The metric is not infallible: its value depends on the gallery a prediction is ranked against, it can compress near one, and its uncertainty must respect the cell lines, drugs and wells that repeat in the design. What it will not do is pass a prediction for resembling a generic treated profile, because every treated profile does that.

1.6 The bar is a dictionary, not chance

Two questions are routinely conflated in perturbation prediction. One is whether a drug has a characteristic response that transfers across cellular contexts. The other is whether a model uses the untreated cell to predict how *this* context departs from that average. Beating a control-copy told nothing about the drug addresses neither, and approaching a split-half reference does not identify which was solved.

The first has a model-free answer. A per-drug lookup predicts a drug's residual as its mean residual over training conditions in other cell lines. It needs no checkpoint and no prompt, and it measures the shared component directly. What it predicts is no small remainder. Decomposing the treated-minus-control shift assigns 62.1% of its energy to the drug-specific residual and 39.0% to the generic programme, with a cross term of -0.011 closing the arithmetic.

FRAME ▲ **RESID**
scores here are on the
drug-specific residual,
§ 4.7

The second is what a lookup cannot answer, because a per-drug constant makes the same prediction in every context. How much of a drug's residual is shared across contexts is measured in § 4.8 by a disattenuated transfer coefficient, $\mathcal{T} = 0.553 [0.514, 0.593]$. The complementary 44.7% [40.7%, 48.6%] of that similarity scale is not shared across the compared contexts. It is not a drug-by-cell-line interaction and not a variance share; the estimator mixes cell line, dose, well and target-estimation differences at once. It bounds what a per-drug constant leaves unresolved without establishing that a model could recover any of it.

The win condition, fixed in advance, is therefore demanding. A conditional model must beat the training-only drug lookup on common support; only then would its access to the control cell have supplied information the dictionary lacked. Beating chance would not do it, and neither would beating a drug-agnostic control.

It was not met. On the 1,192 conditions where model and lookup are both defined, the lookup scores 0.913 against the model's 0.549. Its advantage is not merely that it averages many cell lines; a lookup restricted to one other cell line still scores 0.822. Re-encoding the target does make the model sensitive to the drug named in its prompt, which is the project's first measurable trained-drug effect. Responding to the drug is not the same as predicting what it does: on held-out treated wells the prompt sensitivity survives, and biological transfer is not established. Nothing built here reaches the dictionary.

SEE § 4.11: the full baseline ladder, scored through one identical pipeline

SCOPE LIMIT | carried into § 4.1, § 4.11 and chapter 5

The within-well reference is a measuring instrument, not an oracle ceiling. It asks how well one half of a treated population retrieves the other among competing drugs. A model or a lookup may exceed it because it averages more cells, borrows information across lines, or is compared with a noisy half-sample; NIR also compresses close to one. The reference establishes empirical discriminability at a stated cell budget and comparison set. It does not state the maximum predictable fraction of biology, and it is recomputed within each stratum and never transported across them.

Every conclusion that follows is re-tested under four changes to the measurement: comparison sets are plate-matched, uncertainty is re-clustered on drug, token budgets are equalised, and generic components are fitted without the holdout they later grade. Not every conclusion survives them. Mechanism annotation, for one, looks like a usable route to an unseen drug while uncertainty is clustered on cell line, and stops looking like one once it is re-clustered on the drug. A larger backbone was not tested, and a replication with one is informative only if it receives the same split, targets, sentinels and token budgets; is scored against the same within-plate galleries and training-only lookups; and carries the same validity and completion checks. Otherwise architecture, tokenisation, optimisation and evaluation change together. This thesis does not supply that replication, but it leaves an auditable ladder into which one can be inserted without weakening the controls that produced the negative result.

THE CHAPTERS. Chapter 2 places these questions against single-cell foundation models and the debate over whether deep perturbation models beat simple baselines. Chapter 3 sets out the atlas, the model and the rules of measurement; chapter 4 presents the investigation as one chronological argument; chapter 5 separates what is closed by measurement from what remains untested; and chapter 6 concludes. The symbols, terms and conventions are collected at the back in Reference apparatus, and nothing in the argument requires reading them first.

Background

This chapter is not a survey, but the shortest route to what this field measures, what those measurements assume, and where they give way.

2.1 The variation that matters is representational

A recipe built for text is now used to write cells. What made it portable, and what does adopting it commit you to? It asks almost nothing of language, and everything it does ask falls on how the data are written down.

The transformer recipe requires only sequences of discrete tokens, some predictable from others. Nothing there mentions words, which invited the transfer into biology. scGPT [5] and Geneformer [6] pretrain on tens of millions of cells written as gene-token sequences, toward some version of the *virtual cell* [7], whose simulated responses could substitute for an experiment.

That sets the bar for everything measured later, but it is not where the models differ. A cell is a vector over some twenty thousand genes; a transformer consumes discrete tokens. What must be discarded in between bounds what the model can learn.

2.2 A cell sentence keeps rank and throws magnitude away

If a cell has to be written as tokens, something must go. Which choice does the encoding make, and what argument says the loss is safe? Rank is kept, magnitude is thrown away, and the defence is a reconstruction argument.

Cell2Sentence [2] takes the most literal route available. Rank a cell's genes by expression, write the symbols out in descending order, and the cell is a string of ordinary text, a *cell sentence*. Rank is the whole of the representation, which symbols appear and in what order. Magnitude is not written down at all.

MALAT1 ACTB TMSB4X B2M EEF1A1 RPL13A FTL TPT1 RPS27 GAPDH . . .

WHAT THE DEFENCE COVERS. C2S defends the discarded magnitude by reconstruction. Expression can be recovered from rank through a fitted log-linear relationship $e_i = a_d \log(r_i) + b_d$, achieving $R^2 = 0.815$ against the original values on peripheral blood data. Rank, they argue, therefore encodes expression reversibly enough for the loss not to matter downstream.

The criterion is absolute, not discriminative. Reconstruction fidelity asks how close the recovered profile is to the true one, which is not the same as preserving what distinguishes one condition from another. That distinction runs through the whole thesis.

THREE PROPERTIES THAT MATTER LATER. C2S-Scale builds on decoder-only transformers of the GPT family, the checkpoint used here on Pythia [4]. Token embeddings form a *residual stream*, one vector per position that every layer reads from and writes back into additively, so a representation can be read at depth and modified. Training weights every token position equally, so a target largely shared between conditions gives any one condition only a small share of the gradient. And conditioning runs entirely through attention across a causal mask.

SEE § 4.5 performs the intervention this design permits

THE FAMILY AND THE CHECKPOINT. C2S-Scale [3] spans backbones from Pythia at 410 million and 1 billion parameters up to Gemma-2 at 27 billion, pretrained on over 50 million human and mouse cells from 825 datasets in CELLxGENE and the Human Cell Atlas. The perturbation task this thesis extends is evaluated on three benchmarks: cytokine stimulation of blood cells, bulk L1000 profiles, and SciPlex3.

Two features bear on the comparison later. Pairing is by group and not by cell, treated and untreated bulk profiles matched on cell line name alone, which is unavoidable once sequencing destroys the cell it reads. And single-cell results are scored as distributions, generated population against observed, which asks whether the model produces a plausible population and never whether it distinguishes one condition from another.

C2S-Scale-Pythia-1b-pt is the checkpoint fine-tuned below, trained on atlas data alone, containing no drug-treated cells.

2.3 The usual approach gives the drug a channel of its own

The obvious way to predict a treated cell is not this one. What do the standard methods do differently? In every one except the model this thesis fine-tunes, the drug arrives through a channel the architecture was built to receive it on.

The direct approach predicts a treated cell as a displacement in a learned latent space. scGen [8] estimates the vector between control and perturbed populations in a variational autoencoder's latent space and adds it to a context where the perturbation was never applied. CPA [9] makes that arithmetic

learnable, composing a basal cell state with additive embeddings for drug, dose and covariates; chemCPA swaps its one-hot drug embedding for a molecular-structure encoder, letting an unseen drug be embedded at all.¹ GEARS [10] reaches unseen genes through a gene–gene knowledge graph. CellOT instead fits a neural optimal-transport map from control to treated per perturbation.²

TWO PROPERTIES THEY SHARE. The prediction is a vector in a continuous space, emitted in one pass by a decoder trained on a loss written directly on that vector. And the drug enters on a channel built to receive it, an embedding in CPA, chemCPA and GEARS, a latent offset in scGen, a fitted map in CellOT.

A cell-sentence model does neither. It samples a token sequence left to right, carries the drug as a few words of prompt, and weights every token position equally in its loss. In the standard cell-sentence target, 83% of the top-200 gene positions are identical between any two drugs in the same context (§ 4.7); a gene spans several sub-tokens, so that count is not a count over loss positions. The string to be emitted is nearly the same string whichever drug was named, and a latent-transition model has no target with that property.

Whether that property is the binding constraint is not settled here. § 4.9 reports what re-encoding the target does, and stops short of establishing that tokenisation is what binds.

SCORED DIFFERENTLY. The latent-transition literature is evaluated condition by condition against the observed population: distributional distances, R^2 on pseudobulk, DE-gene overlap. Figure 2.1 draws where that diverges from a discriminative question.

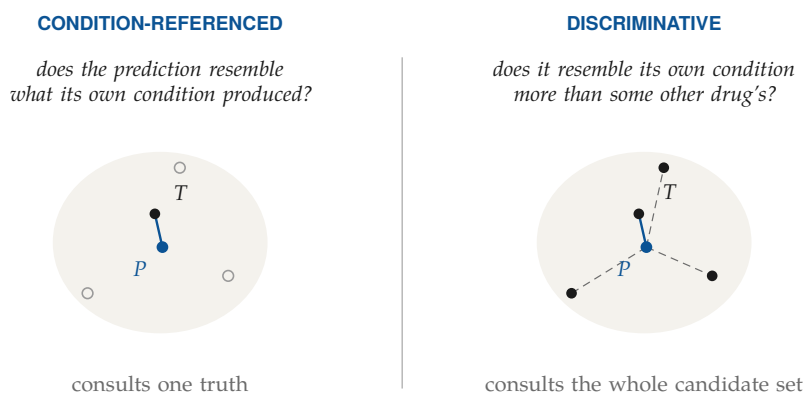


Figure 2.1. The same prediction, the same truths, two different questions. P is a prediction for one condition, T is that condition's own truth, and the other three points are rival candidates, all inside the pale region when the part of the shift shared across drugs dominates. A condition-referenced metric consults T alone, so the rivals are hollow on the left; a discriminative metric grades where T ranks among them. Point positions are identical in both panels. No axis carries a value.

¹Hetzel et al., *Predicting Cellular Responses to Novel Drug Perturbations at a Single-Cell Resolution*, NeurIPS 2022.

²Bunne et al., *Learning single-cell perturbation responses using neural optimal transport*, Nature Methods 2023; Bunne, Krause and Cuturi, *Supervised Training of Conditional Monge Maps*, NeurIPS 2022.

WHERE THE FIELD HAS CONSOLIDATED. scPerturb [11] harmonises the datasets and standardises distances between populations, and PerturBench [12] supplies the exception this thesis relies on, a rank-based score grading a prediction against a set of candidate conditions and not one truth. None of these is run as a baseline here; the results chapter concedes the missing head-to-head as a real gap.

SCOPE LIMIT | carried into § 4.11 and chapter 5

What is reported below is not evidence about CPA, chemCPA or CellOT, none of which was run here. The architectures named in this section supply the *motivation* for the baseline built in § 4.11 — they tell us what a drug-specific channel typically carries — and they are not competitors that have been scored and beaten.

WHAT IS OFFERED INSTEAD. Something narrower, measured and not argued. In scGen and in CPA the drug-specific content carried into a new context is a single per-drug vector, dose-scaled in CPA's case. § 4.11 measures what such a vector is worth by looking the drug up: the mean residual of the same drug in other cell lines, fitted from training conditions only.

On the conditions where both are defined it scores 0.913 against the fine-tuned model's 0.549 ($n = 1,192$). On the held-out wells, where the lookup cannot borrow the target's own well, it scores 0.890 against 0.533; that population runs to 893? **AUDIT: HELD-OUT SUPPORT** conditions. Chance is 0.50 under the exchangeability condition audited in § 4.1, so both comparisons sit far from chance, and what the section turns on is the ordering, not the size of either support.

That is not a score for CPA or chemCPA, whose decoders make the emitted shift context-dependent even when the drug channel is one vector. But it does say which question is informative. Not which architecture wins. Whether anything improves on a per-drug constant.

TWO GAPS. Tahoe-100M [1] is recent enough that we are aware of no published evaluation of these methods on it, so the reference points below are constructed here and not cited. And the atlas's own authors have since released Tahoe-x1 [13], pretrained on a corpus including this atlas. It is not a cell-sentence model, so it does not isolate the representation variable, but it is the obvious external check on whether this kind of readout failure survives perturbation-specific pretraining. It was not evaluated here, and nothing measured here speaks to it.

FRAME ▲ RESID
the scores below are
residual-frame; see § 4.7

SEE § 4.11 builds
the ladder these
scores sit on

2.4 The field disagrees about what a repaired metric would show

Does any of this work? The field has answered both ways, and what decides which answer to believe is not another model but an audit of the metrics. Both camps agree the standard metrics are defective; they disagree about what a repaired measurement would reveal.

ONE SIDE, TRIVIAL PREDICTORS ARE NOT BEATEN. Ahlmann-Eltze, Huber and Anders [14] find that deep-learning prediction of genetic perturbation effects does not outperform linear baselines. Csentes et al. [15] reach the same conclusion for foundation cell models, Bendidi et al. [16] report that a single PCA still rules them all, and Kernfeld et al. [17] concur across expression-forecasting methods. Nor is this confined to transcriptomic readouts. Hauptmann and Kramer [18] hold the protocol fixed and watch the differences between architectures dissolve; DrEval [19] finds deep models barely beating a mean drug and cell-line predictor.

THE OTHER SIDE, THE METRICS WERE AT FAULT. Miller et al. [20] argue that these conclusions “largely stem from limitations of benchmarking metrics, not from the models themselves”. They formalise positive and negative controls in perturbation benchmarking and introduce the *interpolated duplicate*, blending a technical duplicate with the mean baseline by per-gene differential-expression significance.

DEFINITION Dynamic range fraction (DRF)

A quantitative measure of whether a metric can see quality at all. Given reference points whose quality is known and tunable — the interpolated duplicate supplies them — DRF grades the metric on how much of that known difference in quality it registers. It is a property of the metric, not of any model scored with it.

INTERPOLATED DUPLICATE A positive control of known and tunable quality: a technical duplicate blended with the mean baseline.

Miller et al. apply it across 14 Perturb-seq datasets and 13 metrics. MSE and control-referenced Pearson correlation come out poorly calibrated; weighted MSE, weighted $R^2\Delta$ [21] and Normalised Inverse Rank [12] are consistently calibrated. On the metrics that survive, they report, deep models *do* beat mean, control and linear baselines.

WHAT IS ACTUALLY IN DISPUTE. One reading is that the models were always capturing real biology and the broken metrics hid it; the other is that they were never capturing much, and better measurement will show that failure more clearly.

Either way the question is empirical, and settled only in the setting where it is tested. Miller et al. and Ahlmann-Eltze et al. both study **genetic** perturbation, a designed near-binary knockout with a named target gene. DrEval studies **drug sensitivity** as a scalar potency such as IC_{50} . None takes as its object a full transcriptomic response to a drug, which acts at a dose and engages several

proteins at once, at single-cell resolution, from an autoregressive generative model. The models of § 2.3 do work there. What we are aware of no one having done is running the calibration audit first, letting it decide which metrics are admissible before any model is scored.

THE POSITION TAKEN HERE. § 4.1 ports the DRF framework onto drug perturbation, so the verdict cannot be dismissed by the objection Miller et al. themselves raise. The framework and its interpolated-duplicate positive control are inherited; not every choice inside them is. The surviving metric uses a similarity function we picked, Euclidean distance in expression space and not over rank vectors, and the two variants disagree. The family graded is our selection of five, including our own implementation of their weighted R^2 variant.

TWO SEPARATIONS FROM MILLER ET AL.. Only the first is a metric-level disagreement, and it begins with weighted $R^2\Delta$, which they endorse as calibrated but which is negative in both arms under our implementation. Its sign tracks cell count, so § 4.1 draws no directional conclusion from that negative, but this member of their calibrated set does not transfer intact. Their endorsed NIR does survive. `panel- $\tau$` and `spearman_expr`, neither in that set, both invert in this field-wide audit; that failure does not generalise to NIR or to rank-based metrics as a family.

The second separation concerns models. Miller et al. report that deep models beat uninformative baselines once the metric is fixed; on the metric their analysis endorses, the checkpoint evaluated here does not (§ 4.2). Both can stand. A repaired metric is a precondition for seeing skill, not a guarantee that there is skill to see. Which difference accounts for the split, genetic against chemical perturbation, regression against generation, or the per-drug baseline, is taken up in § 4.1 and § 4.11.

2.5 DrEval supplies the vocabulary this thesis audits itself with

A benchmark can report a number that is arithmetically correct and still mislead about everything a reader cares about. Which failure modes are catalogued? Four of DrEval's six shape the apparatus directly.

DrEval [19] identifies six obstacles to clinical translation: non-reproducible models, data leakage producing weak generalisation, pseudoreplication, biased evaluation metrics, missing ablation studies, and inconsistent response data.

PSEUDOREPLICATION. In Hurlbert's original sense [22], this is treating repeated measurements of one unit as if they were independent replicates. GDSC2's 395,025 measurements descend from only 969 cell lines and 287 drugs, so an interval computed over measurements is badly overconfident.

Our data has the same shape, hundreds of thousands of cells resolving to a few thousand conditions across 50 cell lines, and no interval here is computed

over cells. Three estimators stand in their place. An interval on a mean is clustered on the two units that repeat and are crossed, the cell line and the treatment well. A statistic over *pairs* of conditions uses a dyadic estimator, since neither grouping partitions a pair. And the two ratio statistics the two-way form is not written for use a one-way cluster bootstrap over cell lines, flagged where they appear.

BIASED EVALUATION. A version of Simpson’s paradox. Because drugs act at vastly different concentrations, most of the variance lies *between* drugs, so a model that learns nothing but each drug’s mean explains most of it. Their illustration is stark, a global Pearson of 0.90 alongside an average within-drug Pearson of 0.01. § 4.1 explains why a discrimination metric scored *within* a comparison set is immune by construction.

SPLITS. DrEval names four by application: Leave-Pairs-Out for imputation, Leave-Cell-Lines-Out for personalised medicine, Leave-Drugs-Out for drug design, Leave-Tissue-Out for repurposing.

Our holdout lands on one of those and deliberately misses another. `unseen_drug` is `Leave-Drugs-Out`. `unseen_combo` is *not* `Leave-Pairs-Out`, and that name is reserved here for the estimand this thesis declines: holding out a (drug, cell line) pair leaves the pair’s own treated well in training through its other ≈ 48 cell lines, measuring within-atlas matrix completion and not generalisation. Our split withholds the whole treatment well, strictly the stronger of the two, and answers a narrower question, an unseen well of a seen drug.

SEE § 4.10 names the estimand this thesis declines, and why

Citing Partin et al. [23], DrEval reports that good Leave-Pairs-Out and Leave-Cell-Lines-Out performance is reachable by predicting each drug’s average response across the training cell lines. A strong result on those splits therefore demands the per-drug average be measured explicitly, which is what § 4.11 discharges.

MISSING ABLATIONS. Their remedy is randomisation and robustness testing. The scramble ablation of § 4.2 is one, specialised to the drug. Hold every byte of the prompt fixed except the drug and see whether anything downstream notices.

2.6 The transfer premise is assumed everywhere and measured here

An entire drug-discovery methodology rests on the premise that a drug's characteristic transcriptional response travels between cellular contexts. The premise is normally assumed. Here it is measured, with the attenuation from sampling noise corrected.

The Connectivity Map programme [24, 25] built that methodology: a drug’s transcriptional response is taken to be stable enough across contexts that a

reference database can be queried by similarity to it. Its L1000 platform [25] ran the idea at scale, and its 978-gene landmark set defines the gene panel used here, intersected with Tahoe’s measured genes to give 946.

How much of that holds determines whether a conditional model has anything left to predict once a per-drug constant is accounted for.

§ 4.8 measures it on condition-level pseudobulk residuals. On a disattenuated cross-context similarity scale the shared part is $\mathcal{T} = 0.553$ [0.514, 0.593], a little over half, far above the structure-matched cross-line negative control (−0.008) and far below reproduction. The earlier same-plate, different-drug control (−0.017) is not a second matched control. It differs structurally from the estimand and was voided.

WHAT THE REMAINDER IS NOT. The remaining 44.7% [40.7%, 48.6%] is not thereby shown to be biology. The compared pairs differ in cell line, in dose and in treatment well at once, so the figure is not a variance share and not a drug×cell-line interaction, and the section says so where it reports it. What it delivers is a calibration of the transfer premise, not a verdict on it, and we are aware of no prior estimate of that quantity for chemical single-cell perturbation data.

TRANSFER COEFFICIENT How much of a drug’s residual in one context is recovered in another, on a disattenuated similarity scale.

CORRECTION the same-plate control, voided in § 4.8

2.7 Decodable does not mean used, and only intervention tells them apart

A probe can read a drug’s identity out of a model’s internals. Does that mean the model uses it? To learn whether a property is used, change it and watch what the model emits.

Linear probing establishes that information is *present* in a representation, not that it is *used*. Elazar et al. [26] make the case for keeping the two apart. Their amnesic-probing design is adopted here with a matched control added, since an intervention that fails to move the prediction proves nothing unless a comparable intervention does.

A SHARPER FORM OF THE SAME DISTINCTION. Gurnee et al. [27] report that language models maintain a small privileged set of representations, the *J-space*, found by a Jacobian Lens, and argue it functions as a global workspace. Roughly ten to twenty-five concepts are active at any moment, typically under 10% of activation variance. The defining property is causal. Swapping one active workspace vector for another changes the answer to match, while content outside may be richly represented and still inert.

WHAT THIS THESIS TAKES. The design is the swap. § 4.5 builds a domain analogue, substituting a drug’s learned representation for another drug’s and measuring whether the prediction moves toward the substituted drug. A property can be recoverable from the residual stream and still sit outside whatever the readout consults, in which case making it more explicit need not change what the model emits, an expectation § 4.6 tests and § 4.9 bounds.

SCOPE LIMIT | carried into § 4.5

What the thesis does not do is test the workspace claim itself. Nothing below identifies a J-space, measures its occupancy, or shows that a drug representation lies outside one; the substitution result is consistent with the workspace reading and equally consistent with a readout that reads the drug at low gain for some other reason, and § 4.5 declines to choose between them. The framework is cited for the instrument it justifies and the distinction it sharpens, not as a hypothesis this thesis adjudicates.

WHY ALLOCATE REPRESENTATIONS THIS WAY. Elhage et al. [28] show that networks reserve dedicated directions for features needed on every forward pass and push the rest into superposition, still linearly recoverable but without an axis. A property required to emit any plausible prediction should sit in the causally active set; one needed only for occasional composition need not. That asymmetry motivates building the instrument. It is not something the instrument returns.

2.8 Transport enters upstream of the model, as a target constructor

Sequencing destroys the cell it reads, so no cell is seen both before and after treatment and there is no paired training data. Where, then, does a training target come from? The instrument is optimal transport, used not as the predictor but as the thing that builds the target.

There is no ground-truth correspondence between the control and treated populations, only two clouds of points from the same well at different times, so the correspondence must be *inferred*.

COUPLINGS. A deterministic Monge map, one destination per source point, cannot split mass and is ill-posed for that reason. Kantorovich’s relaxation replaces it with a *coupling*, a joint distribution $\pi \in \Pi(\mu, \nu)$ with marginals μ and ν ,

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \, \mathrm{d}\pi(x, y), \quad (2.1)$$

for a ground cost c , typically squared Euclidean distance. A coupling may split a source point across destinations, which makes the problem solvable for noisy, overlapping empirical distributions. When c is the p -th power of a metric the optimal value defines $W_p(\mu, \nu)$, and $p = 2$ is used here.

REGULARISATION. Solved exactly, the discrete problem costs roughly $O(n^3 \log n)$, prohibitive at single-cell scale. Adding a negative entropy term,

$$\min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad (2.2)$$

renders it strictly convex, so the optimum is unique and takes a scaling form recovered by the Sinkhorn–Knopp algorithm through alternating matrix–vector products [29].

The parameter ε is not merely a computational convenience. It controls how sharply mass is assigned. As $\varepsilon \rightarrow 0$ the plan concentrates toward a sparse coupling; without further conditions, that limit need not send each control cell to one observed treated cell. As ε grows the plan blurs, until every control cell couples equally to every treated cell.

The *barycentric projection* maps each control cell x_i to the π -weighted average of its partners,

$$\hat{y}(x_i) = \frac{\sum_j \pi_{ij} y_j}{\sum_j \pi_{ij}}, \quad (2.3)$$

a convex combination inside the observed treated convex hull, not necessarily a point on the data manifold. Small ε therefore yields a target dominated by a sparse set of observed treated cells, large ε the group mean, and a sweep interpolates continuously between two otherwise unrelated target constructions.

A THIRD USE. Applied to biology this machinery mostly does trajectory inference, as in Waddington-OT [30], or prediction, taking the fitted map *as* the predictor. The use made here is neither. Transport supplies the correspondence from which a training target is built, and what then has to generalise is the language model trained on that target, not the coupling. Which is why what matters below is ε , the parameter deciding which target the coupling defines.

SEE § 4.6 sweeps this parameter and scores the targets it produces

TWO PRACTICAL POINTS. The ground cost needs a space where Euclidean distance is meaningful, and raw counts are not one, being sparse, overdispersed and dominated by library size. Couplings are therefore computed in the released Tahoe scVI latent [31]. And because a full coupling costs quadratically, transport is solved per condition, coupling its treated cells to the controls shared within its cell-line/plate group. A control cell from one cell line is not a plausible counterfactual partner for a treated cell from another.

The atlas, the model and the rules of measurement

□ □ □ □ □ □ □ □ □

■ EXPR / ▲ RESID

What has to be settled before a score on this atlas means anything? Four things. What is measured, on what population, against what reference, with what uncertainty. Tahoe answers none by convention, so each is settled below by a measurement, and each narrows a claim later.

3.1 The panel is a budget, not a biology

Tahoe-100M [1] holds drug-treated single-cell transcriptomes across cancer cell lines, with plate-matched DMSO controls. Metadata supplies cell line (Cellosaurus ID), drug, dose (drugname_drugconc), plate, and a fine-grained mechanism-of-action annotation (moa-fine), six fields that are the whole of what any prompt below can carry.

All analysis is restricted to the $G = 946$ genes in the intersection of the L1000 landmark set [25] with Tahoe’s measured genes, library-normalised to counts per 10,000 and transformed as $\log_{10}(1 + x)$. The restriction is a budget choice, not a biological one: the context is 8192 tokens and a prompt carries an instruction line and two cell sentences, so the transcriptome does not fit. Normalisation multiplies every gene in a cell by one constant and the logarithm is monotone, so within a cell the ordering, and therefore the sentence, is invariant to both.

WHAT IT DOES CHANGE. Everything downstream of a single cell. Scaling by library size improves comparability across cells of different depth, but the profiles remain compositional, not absolute, so pseudobulk means, the drug-specific residual, and every cosine and Euclidean quantity built on them depend on the normalisation even though the sentences do not. Depth also shapes which genes are detected, though never their order.

3.2 Only abundance checks gene identity

Tahoe’s per-cell `genes` field holds vocabulary token identifiers, not positional indices into the gene metadata table. Mapping them positionally scrambles gene identity quietly; the values exceed the metadata row count, so the error surfaces only as an out-of-bounds index at the tail. All expression construction here maps `gene_metadata['token_id']` to gene symbol to panel index.

The obvious check cannot fail. Permuting the panel’s columns is an orthogonal transform of the row space, so distances, principal components and accuracy are preserved exactly, and a PCA cell-line probe returns 0.864 against a chance rate of 0.02 on the correct panel and 0.864 on a scrambled one. What discriminates is abundance, a property of the gene and not of the panel. *GAPDH* is the most abundant of the 946 genes and *RPS6* the second, and across the canonical high-expressors the median abundance rank is 19.5, against 473.5 expected under a permuted index ($p = 0.00015$, 20,000 draws; `gene_identity_check.py`). The probe that survives the permutation is the probe that cannot detect it.

3.3 Every treatment sits in one well

Tahoe applies one drug at one dose to one *well* holding roughly 49 cell lines. This project’s cache spans more plates than the three its scored conditions fall on (`plate4`, `plate5`, `plate6`), and holds 289 treated wells, a mean of 48.6 cell lines each (median 49, range 43–50), with 286 of 287 distinct (drug, dose) treatments occurring in exactly one well.

Those figures describe this cache, not the atlas: Tahoe includes further plates, among them a replication plate, so “one well” below means one well *here*. The single exception, one treatment spread over three wells, is also the whole of the cross-plate structure at condition level, and it accounts for the 49 (drug, dose, cell line) groups the unit audit finds on more than one plate. Every other condition on more than one plate differs in dose, and each drug is assayed on its own plate or plates with plate-matched controls.

THREE CONSEQUENCES. Conditions from one well share a physical assignment, so they are not independent (§ 3.8); drug and plate are partially confounded, so discrimination is measured within plate (§ 4.2); and a (drug, cell line) pair cannot be held out cleanly (§ 4.10). Figure 3.1 draws the three in that order.

DEFINITION Condition, treated well, batch unit

A *condition* is a (drug, cell line, plate, dose) tuple, and it is the unit everything below is scored at. The physical assignment is the *treated well*, so a condition sits inside an assignment and is not one. Plates are numbered within cell line, not globally, so the batch unit is the pair (cell line, plate), never the plate label alone. A *pseudobulk* is a group’s mean expression profile; a *control pseudobulk* is the same over plate-matched DMSO cells.

Dose is a molar concentration resolved from `drugname_drugconc`; an earlier pipeline kept only the numeric part, which collides 1 μM with 1 nM. An audit of all 289 treated wells found 289 usable molar doses with zero unit colli-

(a) one drug at one dose enters one well



≈ 49 conditions, one assignment

(b) each drug is assayed on its own plate



a cross-plate drug contrast is also a plate contrast

(c) a held-out pair leaves its well in training



its other ≈ 48 conditions stay in

Figure 3.1. One assignment, many conditions. The roughly 49 conditions from one well share an assignment and a batch (a). Each drug is assayed on its own plate or plates, so a drug contrast across plates is also a plate contrast (b). Removing a (drug, cell line) pair leaves its well in training through the other ≈ 48 lines (c). Squares are conditions, twelve per well for legibility.

sions, so the defect was a demonstrated possibility that did not materialise. Combination treatments are dropped, not analysed as their first component.

SCOPE LIMIT | carried into § 4.10 and chapter 5

A (drug, cell line) pair cannot be held out cleanly on this cache. Removing the pair removes one condition, and its well remains in training through the other ≈ 48 lines. The residual arm therefore builds its own, larger holdout.

3.4 Barely half the atlas is identifiable

Table 3.1 states the quantities every n in the investigation should be located against. One symbol in it runs ahead of its derivation. NIR, the normalised inverse rank, is the fraction of a declared comparison set whose truth a prediction resembles less than its own, so chance is 0.50 under the exchangeability condition of § 4.1.

SEE § 4.1 derives NIR and audits its chance level with empirical control arms.

SCOPE LIMIT | carried into § 4.2 and § 4.9

Two populations are used and they are not interchangeable. The broad characterisation (§ 4.2) streams a wide sample of the atlas, over more plates and drugs than cached, so the drug and condition counts in its block of Table 3.1 count that sample, not the cache. The residual arm (§ 4.9 onward) works from a smaller cached subset. A count from one block does not bound a count in the other.

THE UPPER BLOCK IS A RESULT. It describes Tahoe-100M, not this project. Across the 5,014 control-reachable conditions the median has no differentially expressed gene at BH $q < 0.05$ and a signal-to-noise ratio of 0.75 against its own within-well split-half noise, while 72.7% are active on the whole profile, leaving 27.3% statistically inert. A per-gene test is far harsher, and 27.3% alone would make the atlas sound better resolved than it is. The typical condition carries a real response measured with too few cells, and not no response (§ 4.2).

Table 3.1. The populations behind every n quoted in the investigation. The upper block is a streamed sample of the atlas; the lower two are the cache built once and reused.

Task characterisation (streamed sample)	
conditions characterised	6,628
of which statistically active against DMSO	3,647 / 5,014 (72.7%)
empirically identifiable at the tested pseudobulk budget (own within-plate split-half NIR ≥ 0.8)	3,064 (46.2%)
with a plate-mate closer than their own split-half	78.7%
median cells per condition	44
median #DEGs per condition (BH $q < 0.05$)	0
median SNR (effect $\div$ within-well split-half noise)	0.75
drugs with a real dose series	218 / 351 (62.1%)
drugs seen in ≥ 3 cell lines	227
Residual-frame cache (rebuilt targets, § 4.7)	
cells (treated / control)	620,564 (600,000 / 20,564)
(cell line, plate) groups behind scored conditions	119
drugs / cell lines	106 / 50
conditions with ≥ 40 treated and ≥ 20 control cells	6,617
by split after inventory filters (train / held-out wells / held-out drugs)	4,999 / 900 / 711
retained by the data-resolved reliability line	2,523 of 4,999 train (50.5%)
training examples / well-held-out validation examples	147,710 / 3,600
Evaluation sets	
tier 1 (seen conditions) / tier 2 (unseen drugs) / tier 3 (unseen combinations)	4,325 / 2,188 / 99
conditions scored per tier, generation evaluations	400
tier-2 conditions in the expression-frame NIR benchmark (within plate)	606
conditions scored in the residual-frame evaluation	1,394
by split (train / unseen_combo / unseen_drug)	300 / 895 [?] / 199
treated wells behind them (no well spans two splits)	136 / 36 / 28
drugs per split (90 distinct in total)	76 / 34 / 10
cell lines per split (42 distinct in total)	39 / 42 / 40

The 72.7% figure is over the 5,014 conditions whose (cell line, plate) group carries plate-matched DMSO cells, not all 6,628.

The identifiability bar is per-condition, distinct from the aggregate ceilings of Table 3.2.

The unseen_combo count is flagged pending a cluster rerun; no argument turns on its exact value.

The identifiability bar is per-condition: a condition's own split-half NIR against its plate-mates has to clear 0.8. Of the 6,628 conditions, 46.2% clear it, and 78.7% have a plate-mate lying closer to them than their own second half does. Identifiability is a property of the pairing, not of the drug: over the 227 drugs assayed in at least three cell lines, the median drug's split-half reference spans 0.83 between its most and least identifiable context (drug_biology_atlas.json). A benchmark omitting those figures quotes accuracies over a population barely half of which is empirically identifiable at the tested pseudobulk budget, and a per-drug leaderboard on it ranks pairings.

A FILTER THIS TABLE DOES NOT TOTAL. The middle block records 2,523 of 4,999 training conditions as *retained by the data-resolved reliability line*, which § 4.7 resolves from the data and applies to the training split alone. It does not state the pool once the held-out splits join that filtered set. The channel gate of § 4.12 later scores 4,131[?] **AUDIT: THE RETAINED POOL** retained conditions; a cluster rerun settles whether that is this table's three-way tally under another name, and no figure here depends on it.

RELIABILITY LINE

The half-split cosine threshold above which a condition's residual counts as reproducible.

The three tiers of the last block are defined by what training contained. *Tier 1* holds conditions whose (drug, cell line) pairing was seen in training, so it measures fit and not generalisation. *Tier 2* holds conditions whose drug never appeared, Leave-Drugs-Out in the vocabulary of § 2.5. *Tier 3* holds pairings whose drug and cell line were each seen but never together, Leave-Pairs-Out; it is small (99 conditions) and is reported only where adequately powered. The residual arm builds its own, larger holdout instead: § 3.6 names its three splits, and § 4.10 settles the unit they are keyed at.

3.5 The backbone has seen no perturbation

A cell is written as a *cell sentence*, its expressed panel genes in descending order, terminated by [END_CELL]. The prompt supplies cell line, drug, dose, mechanism and the control cell's sentence; what the model must produce is the treated cell's. [END_CELL] and (for the residual arm) [DOWN] are registered as atomic special tokens; unregistered, [DOWN] splits into sub-words and the up/down boundary stops being a clean symbol.

The backbone is C2S-Scale-Pythia-1b-pt [3], a C2S-Scale checkpoint on the Pythia-1b decoder [4], cold-started for every arm from the same base checkpoint with identical hyperparameters (1 epoch, learning rate 10^{-5} , gradient accumulation 16, maximum length 8192, bf16). Cold-starting makes the arms comparable; the training target or objective is the only variable.

SEE chapter A tabulates the training arms.

WHAT THE CHECKPOINT HAS SEEN. It is pretrained on over 50 million human and mouse cells from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas, an atlas corpus with no chemical or genetic perturbation data. Its model card lists cell and tissue prediction, conditioned generation and gene-set conversion, but not perturbation prediction. The base model arrives at fine-tuning here having never seen a drug-treated cell.

TWO CHANNELS STAY OPEN. The decoder underneath is Pythia-1b, pretrained on text in which drug names, targets and mechanisms occur, and every prompt supplies the drug's *moa-fine* annotation. A drug the fine-tuning never saw is not a drug the system knows nothing about.

SCOPE LIMIT | carried into § 4.12 and chapter 5

"Held out" therefore means held out from Tahoe fine-tuning, with mechanism supplied, and every unseen-drug result below is read under that narrower scope.

3.6 The generic replaces the target

Two *measurement frames* run through the investigation and are never interchangeable (Figure 3.2): the *expression frame* scores a predicted cell profile against other drugs on the same plate, the *residual frame* (§ 4.7) scores the drug-specific residual against other drugs in the same cell line. Both put chance at 0.50. The *comparison set J*, the conditions a prediction is ranked against, is what fixes it there, and it is declared at every score below. A score without its comparison set is not a score.

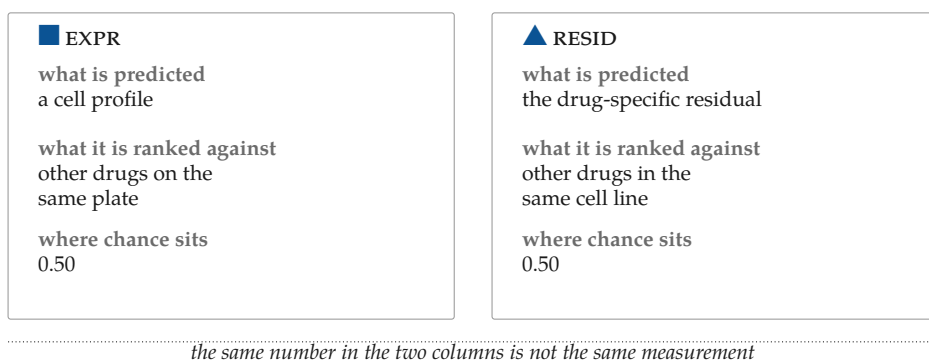


Figure 3.2. Two frames, one chance level, and no exchange rate between them. The prediction, the comparison set and the meaning of a score change between frames; the chance level does not.

DEFINITION The generic and the residual target

The *generic* is the drug-agnostic mean response subtracted to form the residual, operationally the mean shift over the other drugs at the stated scope, conventionally read as the stress and cell-cycle programme common to all drugs — a reading used for intuition and nowhere tested. The *residual target* is what remains. It is built *split-before-fit* (holdout fixed from metadata before any statistic is estimated) and *leave-one-drug-out* (no condition's generic contains its own drug). The *scope*, *cell_line* or *plate*, is the set of drugs the mean is taken over.

The generic carries 39.0% of the perturbation shift's energy (§ 4.8), so subtracting it does not clean the target up but replaces it: every residual-frame number is a score on what is left.

BASELINES AND SPLITS. *Control-copy* predicts the untreated cell unchanged; carrying zero drug information, it is the floor a drug-aware model has to clear. The residual arm's split has three parts, *train*, *unseen_combo* (an unseen *well* of a seen drug) and *unseen_drug*, sampled under an explicit per-split *quota*, because a uniform draw reproduces the pool's proportions and starves the held-out splits (§ 4.10). Five training *arms* differ only in target or objective; where a claim is multiplicity corrected, the family is those five by up to seven depths.

3.7 One estimator, and not a replicate

Almost every negative result below is stated as a distance from a *ceiling*. The word names one estimator, not seven.

DEFINITION Within-well split-half precision reference

The score obtained when a disjoint cell subset of the same treated well is substituted for the prediction, recomputed for every stratum scored, so its value moves with the stratum. It is not a replicate ceiling. What varies across the rows of Table 3.2 is the frame, the comparison set and the cell budget, never the estimator.

Two properties limit how the ceilings read. First, they are *split-half* estimates and not biological replicates, because this cache places 286 of 287 treatments in exactly one well. Second, a split-half ceiling estimates within-well sampling precision, which is optimistic relative to true biological reproducibility, and every coverage figure in § 4.11 inherits that optimism.

THE ONE EXCEPTION. One ceiling is not that estimator and not a row in Table 3.2. The positive control inside the expression-frame DRF audit of § 4.1 is a denoised interpolated duplicate, not a raw split-half, and that section turns on the difference.

Table 3.2. One estimator, seven settings. Every “ceiling” here is a within-well split-half precision reference, the score of a disjoint subset of the same treated well’s cells. They differ in frame, comparison set, cell budget and split contents.

Ceiling	Frame and metric	Comparison set and budget	Where used
0.76	expression, DE- Δr	two real cells	metric audit (§ 4.1)
0.625 / 0.575	expression, NIR	leak audit; cross-plate / within plate	plate leak (§ 4.2)
0.67–0.83	expression, forced choice	within plate, by cell budget	§ 4.2
0.576	expression, NIR	within plate, tier 2	the blindness verdict
0.61 / 0.76 / 0.85	expression, NIR	within plate; 10/40/120 cells	identifiability curve
0.956/0.824/0.836	residual, NIR	within cell line; train/combo/drug	§ 4.11
0.742 / 0.7432	token retrieval	within condition, split-half	§ 4.7

The DRF positive control of § 4.1 is the investigation’s one exception and is not a row here.

None of these is a biological replicate (286 of 287 treatments sit in a single well).

In the residual frame the raw reference is not constant across splits (a spread of 0.132), but that spread is an artefact of how the splits were filtered and not a difference in recoverable signal; matched at the training reliability line the three agree to 0.006 (chapter 5).

SCOPE LIMIT | carried into § 4.11 and chapter 5

A distance to the reference is comparable within a split and not between them, because the splits are scored on populations selected by different rules. And because the reference is a sampling-precision estimate and not a biological replicate, every coverage figure computed against it is a distance to an optimistic bar by an unknown amount. Where one is quoted, the ordering carries the claim.

3.8 Two crossed dependencies, not one

Two rules apply to every result below, plus one statement of provisionality. Conditions are not independent, and not independent in two crossed ways at once (Figure 3.3): two from one treated well share a physical assignment and a batch, two from one cell line share the biology. Treating them as independent gives intervals about 1.5 times too narrow on the scramble strata and 1.7 to 2.5 times too narrow on the contrasts against the lookup baselines. Clustering on the assignment unit alone uses the larger cluster count (289 wells against ≈ 50 lines) and would tighten every interval while sounding more rigorous. That is the failure the estimator below avoids.

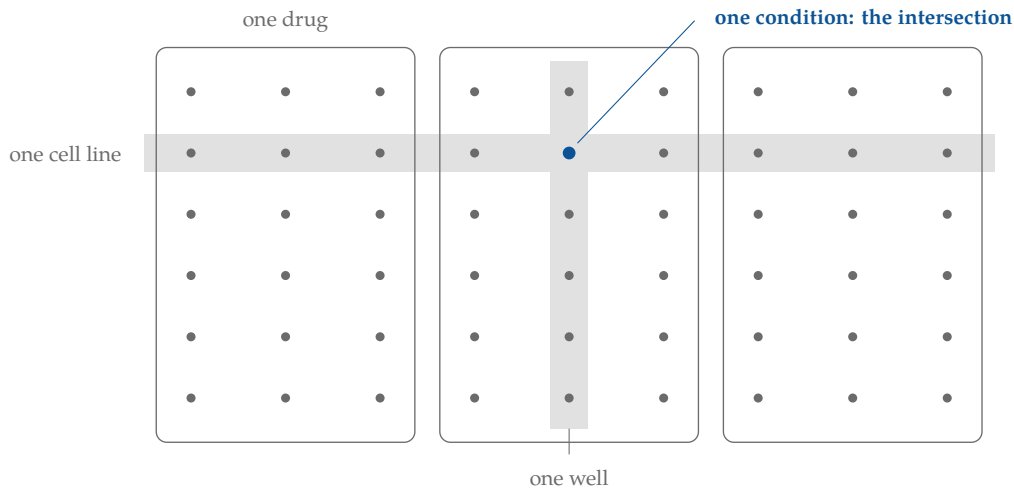


Figure 3.3. A condition belongs to a well and to a cell line at once, and the two groupings cross. Each dot is a condition; the horizontal band is one cell line, the vertical band one well. The dot where they meet is one condition, which is why the intersection term is the ordinary independent-sampling variance. The rounded envelopes are drugs: a well carries one drug at one dose, so wells nest inside drugs, while the cell-line band crosses every drug and neither of the two bands is nested in the other. The lattice is schematic — nine columns of six, for legibility, and not a count of anything in the cache.

Every interval on a *mean* is therefore two-way cluster-robust (Cameron–Gelbach–Miller),

$$V = V_{\text{line}} + V_{\text{well}} - V_{\text{intersection}}, \quad (3.1)$$

where the intersection of the two groupings is the single condition, so $V_{\text{intersection}}$ is the ordinary independent-sampling variance. When it goes non-positive in a finite sample, the wider one-way variance is used and the branch recorded. The formula above is valid only where the two groupings genuinely cross, which cell line and well do: a treated well spans ≈ 49 cell lines and a cell line recurs across wells. Drug and well do *not* cross, because one drug is assigned per well and wells therefore nest strictly inside drugs. Where an interval is reported on the drug axis the estimator tests the two groupings for nesting first and, finding it, clusters one-way on the drug rather than applying the expression above to a nested pair; the branch taken is

recorded beside the interval. Reading “two-way” as “the equation above was applied to drug and well” would therefore be wrong. It is asymptotic in clusters, so the reference distribution is Student’s t with one fewer degree of freedom than the smaller cluster count. The `unseen_drug` arm shows what that costs: Table 3.1 puts 199 scored conditions behind it, drawn from 28 treated wells and 40 cell lines, so the smaller cluster count is the 28 wells, the reference is t_{27} and not the normal, and the interval widens by 4.7%.

For statistics over pairs of conditions, such as the transfer coefficient, neither grouping partitions the pairs, so those intervals use the Fafchamps–Gubert dyadic estimator, under which two pairs are dependent when they share either member.

Throughout, an estimate is quoted as value [lower, upper], a two-way cluster-robust 95% interval unless stated otherwise. DrEval [19] identifies the same structure more broadly, citing Hurlbert [22]: hundreds of thousands of cells resolving to a few hundred conditions over 50 cell lines.

3.9 The generation budget is part of the claim

The second rule concerns generation validity. The budget is a claim-level variable, not a runtime setting.

A 600-token cap truncated 26% of predictions in an early run, silently, because nothing measured them. A prediction cut at the cap never reaches [DOWN] and is scored with its whole down-block missing, and no completion rate was quoted beside the score.

WHAT IT COST. The optimal-transport arm of § 4.6, scored at that cap, gave a gap of +0.0144 [−0.0051, +0.0338] against its geometry-defined orth partner, whose observed score is near chance. That interval spanned zero and the arm was printed as inert. Re-run at the 1400 tokens the residual arm had, it gives +0.0292 [+0.0144, +0.0441], and the negative had to be withdrawn (§ 4.9). The same cap had already halved the residual arm’s own effect. Wherever a paired comparison is scored on variable-length predictions, then, the token budget is part of the claim, and a null quoted without its budget cannot be read.

CORRECTION The printed negative is retracted in § 4.9, in full and where it was made.

THE RULE THAT FOLLOWS. Every generation-based evaluation computes the [END_CELL]/[DOWN] completion rate, the fraction of emitted symbols that are valid panel genes, and the duplicate rate before any score is read from it, and no score below comes from a run whose validity block was not inspected.

The intervals do not cover the generation configuration. Sampling is at temperature 0.8 with top- p 0.9 throughout, over four budgets: 1200 tokens with eight predictions per condition in the expression-frame evaluations feeding § 4.2 and Table 4.6; 1400 with four in the residual-frame evaluations of § 4.9; 600 with four in the control arms until re-run at the residual arm’s budget; and 3800 for output invariance, which compares whole predictions and is not truncated to a scoring panel. Where a null turns on its budget the budget is named there, and every contrast uses a fixed per-(condition, arm) seed.

SCOPE LIMIT | carried into every residual-frame number below

Every residual-frame number comes from one generation run and its companion audits (re_v3 with the scramble-stratum audit, the matched variance decomposition, the two calibration runs and the channel gate). Numbers from the two earlier runs are not reconciled with these and are not quoted, except where a superseded estimate is named as such and the conclusion drawn from it withdrawn, because between runs the sign of several estimates changed and not only their magnitude.

Two components carried defects found after that run and have since been repaired and re-run, the DRF calibration bootstrap (§ 4.1, both arms) and the channel-gate partner pool (§ 4.12); both are reported on the corrected runs.

That is the apparatus. It settles what is measured, on what population, against what reference and with what uncertainty, and reaches no verdict about the model.

The investigation, in the order it happened

□ □ □ □ □ □ □ □ □

■ EXPR / ▲ RESID

Does a model that has learned something about a drug answer differently when it is told a different drug? The question is cheap to ask and expensive to ask honestly, because the instrument had to be measured before the model could be. Each beat below builds the tool that the measurement before it made necessary.

The order is chronological, not the order that would argue best. That is load-bearing: a design choice appears as the failure it avoids, and a dead end appears where it was walked into. The apparatus is chapter 3; the terms, symbols and conventions are fixed in Reference apparatus.

THE FINDINGS, BEFORE THE BEATS THAT PRODUCED THEM. Six, in the order they arrived, with no numbers attached. Each one is established, qualified and scoped in the beat that owns it, and is quotable only from there. What follows is a skeleton, not a record.

- The ruler was wrong. The field's standard perturbation metric is saturated by a predictor that carries no drug information at all, so the instrument was repaired before the model was measured.
- On the repaired instrument the model is drug-blind in the expression frame. It returns the same answer whether told the true drug or a different one, and a baseline that copies the untreated cell matches it.
- The blindness is not a failure to encode. Drug identity is written into the activations at every layer measured, and generation barely consults it. This is the chapter's least well-evidenced measurement.
- Re-encoding the training target as the drug-specific residual makes the model demonstrably sensitive to the drug in its prompt. That is the project's first non-null drug effect: first in sequence, and not the only target construction that produces one.
- Responding to the drug is not predicting it. Prompt sensitivity survives a holdout of whole treated wells; biological transfer to those wells is not established on the axis a drug-specific claim has to generalise across, and a lookup table of training averages outscores the model. That dissociation is this chapter's argument.
- For a drug the model has never seen, same-target retrieval is the strongest observed lead on its own restricted support; mechanism and chemical structure are inconclusive, not dead.

Figure 4.1 is the map. Nine beats produce those six findings; the ninth produces none, and that is a decision, not work left undone.

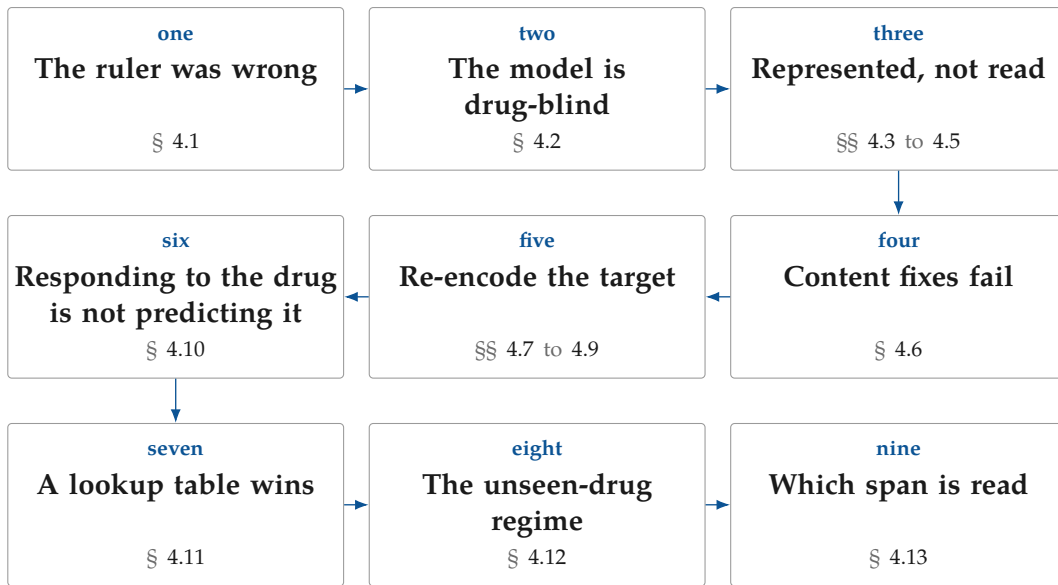


Figure 4.1. The investigation in nine beats, each told with the instrument it required. The arrows are the order of discovery. The ordinal on each beat names the cell the strip fills in that beat’s opening box. Neither the apparatus (chapter 3) nor the summary table (§ 4.14) is a beat. The ninth beat is the one the chapter does not close.

4.1 The metric is the first result

■ ■ ■ ■ ■ ■ ■ ■ ■ ■

■ EXPR

Every claim in this chapter is a number handed down by a scoring rule, so the scoring rule is what has to be audited first. The instrument grades metrics, not models. It asks each candidate to separate a positive control from a predictor that knows no drug, and four of the five candidates fail.

VERDICT The field’s standard score is saturated by a predictor that carries no information, and of the five metrics audited only NIR grades in the direction its name implies.

A reported score is two claims at once, one about the model and one about the rule that graded it. Nobody checks the second. Two questions decide it. Does the score reward the thing it is named after, or can it be earned another way? And does it register a true positive when there is one?

THE SCORE THE FIELD REPORTS. That score is a correlation between predicted and true change from control, restricted to the genes that actually move. The benchmarks of § 2.4 report it, Miller et al. [20] find it poorly calibrated, DrEval [19] counts it among six standing obstacles. Write x for the control profile, y for the treated truth and $\hat{y}$ for the prediction, and let

$$S_K = \arg \text{top-}K_g |y_g - x_g| \quad (4.1)$$

SEE chapter 3 fixes G.

be the K genes with the largest true change. Then

$$\text{DE-}\Delta r = \text{corr}((\hat{y} - x) \mid_{S_K}, (y - x) \mid_{S_K}). \quad (4.2)$$

All three are the panel profiles converted to ranks, absent genes tied at the worst rank G (§ A.2), so S_K is selected on rank change too; the *expression frame* label used elsewhere names the object scored, a cell profile rather than a residual, and not the units. We report $K = 50$, Pearson; $K \in \{20, 100, 200\}$ and Spearman move the numbers by under 0.02, except at $K = 200$, where the top- K set starts to include genes that do not move.

WHAT THE METRIC ACTUALLY REWARDS. Score a predictor that contains nothing. Place every gene at the middle rank. It scores $\text{DE-}\Delta r(K=50) = 0.999913$ $[0.999908, 0.999919]$, above the two-real-cell positive control of 0.76 and above the model’s 0.73. Less information yields a higher score, the hallmark of an artefact; Figure 4.3, panel a, sets the five arms against that reference on one axis.

Its partial- $\text{DE-}\Delta r$ and panel- τ are undefined, not merely poor, for two separate reasons. The prediction is the constant midpoint rank $\hat{y} = G/2$, so the shift from control is $G/2 - x$, an exact affine function of x from which nothing survives partialling. And every predicted gene being tied, no rank ordering exists whose Kendall τ can be computed. The zero-shift construction, easily confused with this one, is control_copy , $\hat{y} = x$; `revert_center` is not it.

REVERT_CENTER
Predicts every panel gene at the panel’s middle rank, defined from the metric alone and needing no model, training set or atlas.

Table 4.1. $\text{DE-}\Delta r$ is saturated by a control artefact. The two arms needing no drug information sit above the positive control and above the model; regressing the control out leaves ≈ 0.26 of drug-agnostic skill in the two rows where the partial is defined.

Predictor	Information	DE- Δr (K50)	partial-DE- Δr
<code>revert_center</code> (genes $\rightarrow G/2$)	none	0.9999	undefined
<code>revert_mean</code> (predict mean control)	none, no fit	0.961	0.25
ridge linear (control $\rightarrow$ shift)	control	0.947	0.28
positive control (two real cells)	–	0.76	–
model (1B LLM)	control + drug	0.73	not measured

The $\text{DE-}\Delta r$ column is tier 2; the partial column, tier 1.

TWO INNOCENT PROPERTIES COMPOSE INTO THE FAILURE. S_K is selected using the truth, so the exam questions come from the answer key. And treated cells in this atlas regress toward a shared response, so a prediction reverting toward the population centre correlates with $y - x$ by construction. Partial- $\text{DE-}\Delta r$ regresses the control out of both arguments first, closing that route and collapsing the raw 0.95 to ≈ 0.26 in the two baselines with a defined partial value, one of which performs no fitting at all.

The model’s own partial- $\text{DE-}\Delta r$ was never computed,¹ so the comparison anchors to panel- τ , Kendall’s τ over the full 946-gene panel, measured for

¹It requires a GPU pass that was not run, so the cell in Table 4.1 is marked, not filled with an expected value.

every arm. The calibration audit finds panel- τ inverted, so neither an absolute claim nor an ordering read off it counts as evidence of skill. What it can still show is an *absence* of separation. `revert_mean` scores 0.268, ridge 0.265 and the model 0.257 (tier 2, worst convention), three arms inside a spread of 0.011.

The shift therefore decomposes into three parts: control reversion, an artefact; a generic drug programme worth ≈ 0.26 , trivially matched; and a drug-specific component no arm is shown capturing. Later residual-frame instruments show the checkpoint reads trained-drug information while still failing condition-specific prediction.

ONE CONVENTION, DISCLOSED AND THEN DROPPED. A cell sentence lists only expressed genes, so a rank-based metric needs a convention for the unranked panel genes: worst gives them the last rank G , francesca a fixed middle rank $G/2$. Spike-in DE- Δr reads 0.952 against 0.954 and nothing here moves by more than 0.01, so all figures are quoted under worst. That diagnostic went to no artifact, so it is a check that was run, not a reproducible result.

WHAT TRAVELS BEYOND THIS ATLAS IS THE AUDIT, NOT THE VERDICT. The saturating predictor is defined from the metric alone, so any result on this family can be checked the same way. Whether the reversion property is strong enough to invert the score elsewhere, or in the log-expression units the benchmarks compute the same functional form in, this cache cannot answer.

THE METRIC THIS INVESTIGATION RELIES ON INSTEAD. NIR is Wu et al.’s “transposed-rank” metric from PerturbBench [12], which Miller et al. [20] identify as one of the few consistently calibrated metrics across 14 Perturb-seq datasets. It asks not how closely a prediction matches its own truth, but whether it matches its own truth *more closely than it matches anyone else’s* — much harder to answer by accident.

DEFINITION Normalised Inverse Rank (NIR)

For a prediction $\hat{y}$ of condition i , with truths $\{y_j\}$ over a declared comparison set J ,

$$\text{NIR}(\hat{y}, i) = \frac{1}{|J|} \sum_{j \in J} \left[\mathbf{1}\{\sigma(\hat{y}, y_i) > \sigma(\hat{y}, y_j)\} + \frac{1}{2} \mathbf{1}\{\sigma(\hat{y}, y_i) = \sigma(\hat{y}, y_j)\} \right], \quad (4.3)$$

the fraction of the other conditions in J whose truth the prediction resembles less than its own, ties counted at one half. σ is a *similarity*, so higher means closer; built from a distance, as with $\sigma(\hat{y}, y) = -\|\hat{y} - y\|_2$ for `nir_expr`, it is that distance negated. Only σ and J change across frames, both declared where a figure is quoted.

Figure 4.2 draws one comparison set under this definition, with the three reading rules that follow from it, the half-tie rule among them.

WHY CHANCE IS 0.50, AND WHAT THAT CLAIM RESTS ON. Chance sits at 0.50 under a symmetry assumption, not as a consequence of using the same truth set, which is necessary but insufficient. Three things must hold for an uninformative prediction: y_i exchangeable with the comparison truths $\{y_j\}$; its

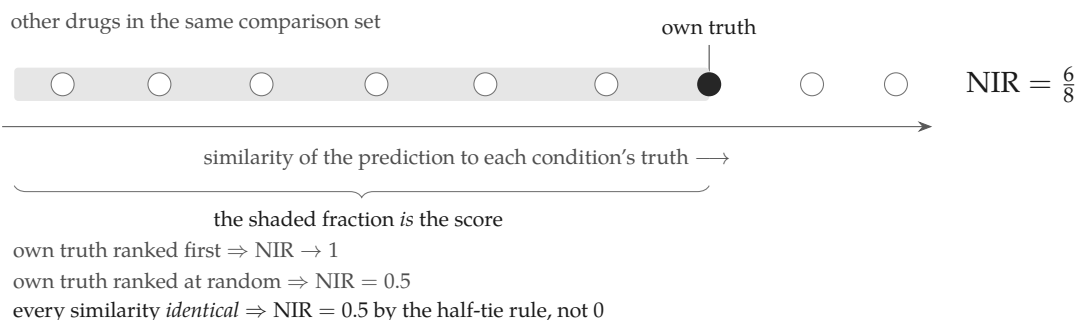


Figure 4.2. NIR asks whether a prediction resembles its own condition's truth more than it resembles the truths of the other conditions in the comparison set J . Own truth ranked first gives $NIR \rightarrow 1$; ranked at random gives 0.5; and a predictor whose similarities are all identical — the zero vector, which is what “predict the average drug response” becomes in residual space — scores 0.5 by the half-tie rule, not 0.

label conferring no privileged relation to the prediction; and symmetric tie handling. The reference is assumed, not proved, so it is checked empirically in each frame. In the expression frame the drug-agnostic linear map and control-copy score 0.500 and 0.504; in the residual frame random, control-copy and orth score 0.501, 0.506 and 0.501. One asymmetry survives *between* arms. The split-half reference is scored against half-sample truths, every baseline against full-sample truths, which makes the comparison conservative.

THE TIE TERM IS NOT COSMETIC. Discard it as bookkeeping and the metric breaks in residual space (§ 4.7), where a predictor of the generic response is exactly the zero vector, so every comparison is a tie. A strict inequality scores it 0.000 instead of the 0.500 it deserves, below every random predictor.

THE COMPARISON SET MUST BE DECLARED TOO. Conditions are grouped by (cell line, plate) in the expression frame and by cell line in the residual frame, and the two are not interchangeable. Widening J across plates moves a drug-agnostic baseline from 0.161 to 0.399 (§ 4.2), more than double the score for the same predictor. And because the generic response sits in every truth in equal measure, it cancels, leaving only the drug-specific difference to decide the score — the component the reversion artefact exploits.

THREE SIMILARITIES, ONE OF THEM GRADED. Three similarity functions are computed and stored: `nir_expr`, Euclidean distance in expression space; `nir_rank`, the same over rank vectors; and, in the residual frame of § 4.7, cosine similarity between signed rank-discounted residual vectors. Every *expression-frame* NIR figure here is `nir_expr`.

`nir_rank` never entered the graded family; a cheaper admissibility screen excludes it. On the within-plate tier-2 run that later delivers the drug-blindness verdict ($n = 606$, `nir_sameplate.json`), every arm sits below the 0.50 chance line, the positive control included at 0.380, against control-copy 0.337, model 0.313, drug-agnostic linear map 0.285, mean baseline 0.070. That is no argument against rank-based metrics; NIR's rank orders *conditions*, not genes.

SEE chapter 3 lists every reference in the chapter.

ADMISSIBILITY SCREEN A pass/fail gate applied before grading. Can a disjoint sample of a condition's own cells be retrieved above chance at all?

GRADING THE METRICS, NOT THE MODELS. The framework is Miller et al.'s [20]. Ask not which model scores highest but which *metric* can see quality at all.

DEFINITION Dynamic Range Fraction (DRF)

$$\text{DRF}(m) = \frac{m(\text{pos}) - m(\text{neg})}{m(\text{perfect}) - m(\text{neg})}, \quad (4.4)$$

on a scale where 0 is the drug-agnostic baseline and 1 a perfect predictor. Here pos is an interpolated-duplicate positive control, a technical duplicate interpolated toward the mean baseline with per-gene weights from differential-expression significance, and neg is a drug-agnostic leave-one-out mean. A negative DRF means the metric scored the uninformative baseline above the positive control in that run.

Reading a negative value as *inversion* takes one further step, that the sign belongs to the metric and not to the run. Three of the four negatives survive it; the fourth is carved out where it appears.

A drug-agnostic mean of that kind also appears in the leak audit of § 4.2, scoring 0.161 within plate and 0.399 across; the DRF audit's own negative scores 0.457 across plates and 0.272 within. Whether the two constructions coincide is **under audit**, and nothing depends on it. DRF is formed inside one run from that run's own positive and negative, so all it needs is that the negative carry no drug information, which both do by design.

FOUR CHOICES DECIDE WHAT THE NUMBER MEANS. Three govern the point estimate. Ties take average ranks, since otherwise a tied gene's rank would follow its position in the panel. The ratio is recomputed whole inside every bootstrap resample, since holding a denominator estimated from the same data understates the interval. And Holm control runs across the five metrics at once.

The fourth governs whether the interval means anything, and in the stored runs it was not made. The resampler treated each of the all-plates run's 1,820 rows (one drug within one cell line) as its own cell line, against the 25 that exist, and the same-plate run carried the identical defect. The one-sided p was read off the observed resampling distribution with no null imposed, so it degenerates to the floor $1/2001$ where $\text{DRF} > 0$ and to 1 where $\text{DRF} < 0$; no metric was in fact tested. Both are repaired and both arms rerun. The rerun also moved the cluster to the cell line, since clustering on the (cell line, plate) pair repeats the original error one level up; grouping unit and cluster unit are reported separately.

ONLY ONE CANDIDATE POINTS FORWARD. Table 4.2 carries the values; three things it does not. Against a null imposed by centring the resampling distribution at $\text{DRF} = 0$, no draw of 2,000 reaches NIR's observed value, so its p sits at the resampling floor and the estimate is some forty cluster-robust standard errors from zero. Restricting swap partners to the same plate changes no sign and matches the arms at 27 cell lines within plate against 25 across; an earlier configuration capped the run at 25 *groups* and admitted only 18 distinct cell lines, and still gives NIR $+0.519$ [$+0.479$, $+0.546$] over 335 rows. And the

INVERSION A metric placing the drug-agnostic baseline above the positive control, as a property of the metric and not of one run.

CORRECTION The DRF bootstrap resampled the wrong unit and imposed no null; both repaired and both arms rerun before any figure here was read.

Table 4.2. Under the hardened protocol, NIR is the only one of the five whose DRF is positive, and the sign pattern is identical in both arms. All four rivals score the drug-agnostic baseline above the positive control and exclude zero in both arms. The second column is what separates NIR.

Metric	Scored against	DRF all plates	DRF same plate	Reading
NIR (nir_expr)	other conditions' truths	+0.635	+0.545	the only positive
weighted_r2	its own truth	−0.163	−0.104	no directional claim
panel- τ	its own truth	−0.319	−0.250	inverted
spearman_expr	its own truth	−0.650	−0.500	inverted
DE- Δr	its own truth	−0.686	−0.515	inverted; the default

The all-plates run has 1,820 rows (one drug within one cell line) clustered on the 25 cell lines they nest in; the same-plate run has 586 rows over 44 cell-line $\times$ plate groups spanning 27 cell lines, clustered on cell line. NIR reads +0.635 [+0.604, +0.663] and +0.545 [+0.516, +0.569], Holm $p = 0.002$ at the resampling floor in both arms. The four negatives are point estimates, each excluding zero on the same clustering in both arms; weighted_r2 sits closest to zero and its sign tracks cell count, hence no directional claim.

inversion belongs to the data, not to a leak. A leave-one-out correction barely moves the numbers, real cell lines carrying 20–204 drugs, and it survives holding the plate constant. What the argument uses is the sign pattern, drawn for the same-plate arm in Figure 4.3, panel b; no ranking among the negative metrics is claimed.

WHAT THIS REPLICATES, AND WHERE IT DOES NOT. The methodological core of Miller et al. replicates, and this audit sharpens it. Metric choice dominates, and the control-referenced correlation they flag is the worst performer in both arms (DE- Δr , −0.686 across plates and −0.515 within). Their calibrated set does not transfer intact. weighted_r2, this project's counterpart of the weighted $R^2\Delta$ [21] they endorse, is negative in both audited arms, though closest to zero and sign-tracking cell count, so it carries no directional claim; the inversions of panel- τ and spearman_expr belong to neither calibrated set. Their endorsed NIR does survive. Being rank-based is not what protects it; refusing to score anything the generic response can supply is.

ADOPTING THE INSTRUMENT IS ALSO A DEFENCE. Miller et al. conclude that under calibrated metrics deep models outperform uninformative baselines; § 4.2 finds ours at chance on that very metric. Genetic against drug perturbation and a discriminative against a generative model are candidate explanations; § 4.11 supplies a third. Adopting their metric also puts the verdict beyond the objections their own paper raises.

SEE § 2.4 sets out the dispute this audit enters.

THE RESIDUAL FRAME IS AUDITED SEPARATELY. A metric is not calibrated by having a relative calibrated, so the residual frame carries its own DRF. With a split-half replicate as positive control and the evaluation's drug-agnostic arms as negatives, it reads +0.704 [+0.667, +0.736] against control_copy, clustered on the 42 cell lines. Which negative is chosen barely matters. A random vector and orth both give +0.707, and all three sit at chance as a matter of measurement (0.506, 0.501, 0.501). The positive control is the more conservative of the two, a raw split-half replicate instead of the denoised interpolated duplicate, so the residual frame is the better separated (+0.704 against +0.635) despite the

FRAME ▲ RESID the second audit is scored in the residual frame, § 4.7

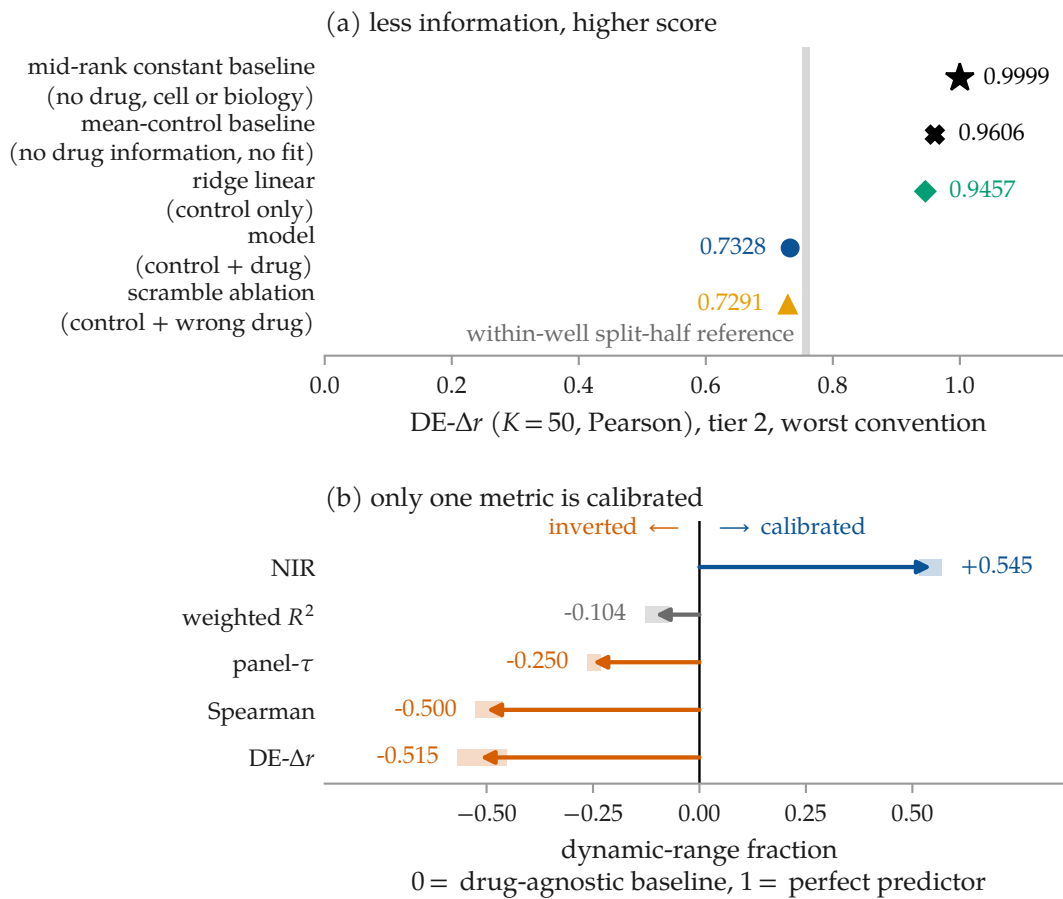


Figure 4.3. A predictor carrying nothing about drug, cell or biology scores the field's standard metric above both the within-well reference and the model, and of five metrics only NIR grades in the direction its name implies. (a) DE- Δr , $K = 50$, Pearson, tier 2 unseen drugs, worst convention, $n = 400$ conditions; the axis runs from 0 and is not truncated. The grey band marks the within-well split-half precision reference at 0.76, the two-real-cell positive control of Table 4.1: two halves of the same well, a precision figure and not a biological replicate. Its width is a drawing device, not an interval, and the arm marks are bare — the source carries a normal-approximation interval over conditions, and this document draws only the clustered bootstrap, so nothing is drawn rather than an interval of another kind. (b) DRF for the five metrics of Table 4.2, the same-plate arm only: 586 drugs over 44 cell-line $\times$ plate groups spanning 27 cell lines, the pale block at each arrowhead the 95% percentile bootstrap interval clustered on cell line quoted there.

harder reference. generic scores exactly 0.500 with zero variance, the residual being defined by subtracting it, and carries no weight.

ONE DECLARED EXCEPTION ON THE INTERVALS. DRF is a ratio of means; the two-way estimator used everywhere else here is written for a mean. Both DRF intervals are therefore *one-way* percentile cluster bootstraps that resample whole cell lines and recompute the ratio inside each draw, respecting the cell-line dependence and ignoring the crossed well dimension, so they are narrower than fully crossed intervals. The separation they grade is far too large for that to change the sign, but the caveat belongs beside the number.

RECALL two-way cluster-robust intervals on every mean chapter 3

TWO QUESTIONS ABOUT THE INSTRUMENT REMAIN, AND THEY DIFFER. The first is whether the conditions are empirically discriminable at the tested budget. A spike-in forced choice answers it. Within a plate, a reference pseudobulk from drug A's cells must be scored closer to a disjoint sample of A's own cells than to drug B's, chance 0.50, with contamination of B by A titrating the difficulty. At zero contamination three of the six metrics run on it separate the two populations essentially perfectly (panel- $\tau = 1.000$, DE- $\Delta r = 0.995$, top- $n \tau = 0.987$, plate held constant); at full contamination those same three return 0.501–0.502. The whole-profile similarities are much weaker even at zero contamination, cosine 0.740, Spearman 0.680, cosine on the shift 0.620, which matters because a whole-profile similarity is what NIR uses here. Split-half retrieval establishes discriminability, not predictability from untreated input, so discriminability is not the sole bottleneck at this budget.

The spike-in cannot answer the second question, whether NIR registers a true positive, since NIR was not among the metrics run on it. The evidence is the DRF audit's own positive control. Under NIR the interpolated duplicate scores 0.802 against the drug-agnostic mean's 0.457 across plates, and 0.668 against 0.272 within plate; in the residual frame a split-half replicate reaches 0.854 against negatives at 0.501–0.506. None of those is the reference the verdict is measured against. That reference is the within-well split-half precision estimate recomputed on the tier-2 within-plate benchmark, where it sits above chance but not far above it; that, and not 1.0, is the bar § 4.2 states the model against.

SCOPE LIMIT | carried into § 4.2 and the rest of the chapter

The audit licenses NIR and nothing else. panel- τ is inverted in both audited arms, so no absolute claim rests on it and no ordering read off it counts as evidence of skill; an inverted metric preserves neither. Where DE- Δr appears later it is only as a paired comparison of two arms on one scale, never as an absolute score. And a metric is not calibrated by having a relative calibrated. What is graded here is `nir_expr` in the expression frame and the residual frame's cosine similarity in its own frame, each against its own controls. A third σ would need its own audit.

THE CLAIM. The field's standard score is saturated by a predictor that carries no information, and of the five metrics audited only NIR grades in the direction its name implies. This audit grades five candidates, and NIR is the only one with a positive DRF (+0.635 [+0.604, +0.663] on a one-way cluster bootstrap over the 25 cell lines, the declared exception to this chapter's two-way rule, Holm $p = 0.002$; +0.545 [+0.516, +0.569] over 27 cell lines with same-plate partners, Holm $p = 0.002$), and the other four are negative with intervals excluding zero in both audited arms. Three of them (panel- τ , `spearman_expr` and DE- Δr) are inverted rather than merely uninformative, each ranking the drug-agnostic baseline above the positive control; `weighted_r2` sits closest to zero and its sign tracks cell count, so no directional claim is made for it. The saturating predictor is not a curiosity. It carries no information about drug, cell or biology and scores DE- $\Delta r =$

0.9999 in rank space, above the two-real-cell positive control of 0.76 and above the model's 0.73. Under NIR the positive control is retrieved well clear of the negative (0.802 against 0.457 across plates, 0.668 against 0.272 within), so the instrument registers a true positive. Every drug-use claim in the rest of this chapter that rests on a continuous prediction score is therefore made on the calibrated metric; DE- Δr appears below only where two arms are compared with each other on the same scale, never as an absolute score of the model's drug use. The remaining instruments (forced-choice grading, output invariance, and the logit-space activation tests of § 4.5) are deliberately on scales of their own, each with a fixed chance value or a paired control.

4.2 The model is drug-blind



■ **EXPR**

With a ruler that cannot be gamed in hand, the investigation can ask the question it exists to answer. Does the model use the drug at all? Four instruments answer it, and each exists because one specific objection would otherwise be fatal.

VERDICT On the calibrated metric the model is drug-blind in the expression frame, including on the identifiable stratum.

The answer is no, and the instruments matter only as the objections they close. On the calibrated metric, within plate on tier 2 ($n = 606$), the model scores 0.498 against a within-well split-half precision reference of 0.576. It is matched by predictors that know nothing about the drug: a drug-agnostic linear map scores 0.500, a control-copy 0.504, the global mean 0.180. And on the identifiable stratum, whose own within-well reference clears the pre-set 0.80 bar, a zero-drug-information control-copy reaches 0.766 against the model’s 0.768.

FOUR INSTRUMENTS, AND EACH CLOSES A DIFFERENT OBJECTION. The benchmark itself, within plate; the scramble ablation, the only manipulation that isolates drug use; forced-choice grading, whose chance value is fixed; and output invariance, which no scoring choice can confound. They are exposed to different artefacts, so no single one explains all four.

ONE MEASUREMENT RULE COMES FIRST. Drug and plate are partially confounded by the atlas's design (chapter 3), so grouping by cell line alone lets the batch effect stand in for drug identity. A drug-agnostic mean baseline, containing no drug information whatsoever, scores 0.399 when the comparison set may span plates against 0.161 when it may not, and the precision reference falls from 0.625 to 0.575 alongside it, the batch effect having inflated that too. That is plate leakage, measured and not assumed. Being wider, its within-plate figures sit slightly off the ones above; what it licenses is the rule, not a substitute set of scores.

RECALL one estimator, seven settings Table 3.2

FRAME ■ **EXPR** all
four instruments are
scored in the expression
frame

SEE Table 5.1 is the leak audit, a wider run than the benchmark

SCOPE LIMIT | carried into every discrimination result in this chapter

Every discrimination instrument gained a `-same_plate_only` mode and every discrimination result in this chapter was re-run within plate, except where a departure from that rule is named at the point of use, as with the swap-distance sweep that answers the fourth objection below, whose swap partners were not plate-restricted (§ A.3), and the residual-frame scores of § 4.7, whose comparison set is grouped by cell line and whose justification is given there. Conclusions are unchanged; some magnitudes shrink.

THE CENTRAL MANIPULATION IS THE SCRAMBLE ABLATION. The prompt is left byte-identical except that the drug name and mechanism are replaced by another drug's, the control cell and the scoring truth unchanged (Figure 4.4). Control- and batch-derived leakage therefore cancels in the scramble gap $\Delta_{\text{scr}} = \text{NIR}_{\text{model}} - \text{NIR}_{\text{scramble}}$, the only difference in this chapter whose leakage cancels by construction. It does not displace the drug-agnostic comparisons above and below, which rest on the within-plate rule instead.

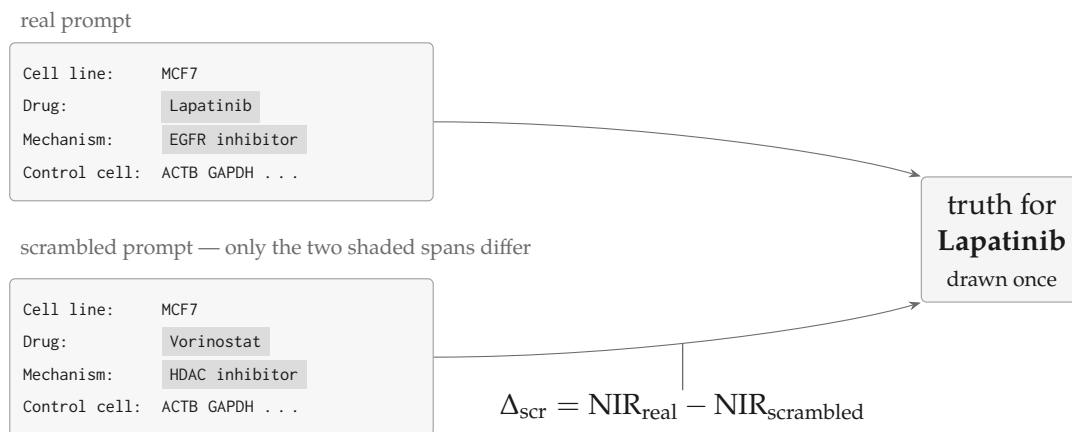
TWO CONSTRAINTS ON THE SWAP PARTNER. The partner must be a different drug. Allowing the same drug at another dose or plate substitutes the drug for itself, an unchanged prediction is then correct and not blind, and the contamination concentrates in the most similar stratum. It must also be on the same plate, since residuals retain plate structure and a cross-plate partner would differ in batch as well as in drug.

A scramble that swaps in a drug with a similar true response is a weak test, and that risk is structural here. In full-profile space the mechanism annotation is close to uninformative about transcriptional response. Same-mechanism drugs in Tahoe are just over two per cent closer to each other than arbitrary pairs are. Hence conditioning on mechanism fails to move $\text{DE-}\Delta r$ in the grouping ladder of § 4.11, and a partner of different mechanism cannot be relied on for a different response. Mechanism does carry signal in the residual frame (§ 4.12); in full-profile space the generic response swamps it, a frame difference and not a contradiction.

SEE § 4.9 quotes the distances behind that statement

Two remedies are used. A swap-distance sweep bins (drug, swap) pairs by the response distance $\|y_A - y_B\|$ between real and swapped-in drug, turning the single test into a curve (§ A.3). And a stratified scramble defines three partners per condition by residual cosine within the cell line: *near* (most similar), *orth* ($\cos \approx 0$) and *opposite* (most anti-correlated). The strata expose what a single comparator hides (Figure 4.5): the gap it defines has two parts, how much naming the truth helps and how much naming a lie hurts, and only the first is wanted, so a usable null is a stratum scoring at chance. § 4.10 shows exactly one qualifies, and not the one an evaluation would default to.

THE THIRD INSTRUMENT IS FORCED-CHOICE GRADING. A prediction is scored against its own condition's truth and against another drug's truth from the same comparison set, recording how often its own wins. Chance is 0.50 by construction, and the precision reference lands between 0.67 and 0.83 depending on cells per condition.



the shared control cell enters *both* arms, so control- and batch-derived leakage cancels in the difference

Figure 4.4. The scramble replaces the drug name and its mechanism and nothing else. Both arms are conditioned on the same control cell and scored against the same truth, so any leakage through the control cell or the plate enters both arms in equal measure and cancels in the difference Δ_{scr} . This is why Δ_{scr} is the only difference in the chapter whose leakage cancels by construction, while a comparison against a drug-agnostic predictor is not leak-immune — such a baseline inherits whatever plate structure the comparison set carries (Table 5.1). Comparisons of that kind are used, and used centrally, with the comparison set declared at every score: within plate in the expression frame, where the leakage is controlled rather than cancelled, and by cell line in the residual frame, where the plate-scoped generic has already subtracted each plate’s mean response from truths and predictions alike — in expectation, with however much survives in an individual residual left unmeasured (§4.7, and §5 for what that leaves open). A refinement matters here: if the swapped-in drug happens to have a *similar* true response then an unchanged output is correct rather than blind, so the swap partner is chosen from three defined similarity strata by residual cosine. Which stratum can carry the canonical magnitude associated with this figure is constrained by the comparator’s own score. §4.10 finds the geometry-defined *orth* partner empirically near chance at 0.501 in this sample, making it the least contaminated available comparator rather than an independently established null. That restriction belongs to this canonical scramble comparison: the three-stratum repair comparison differences each model score against its own near/*orth*/opposite partner, while field attribution uses its stated span-specific partner and retains the resulting comparator caveat.

THE FOURTH IS OUTPUT INVARIANCE. It asks whether the drug token changes the generated text at all, without reference to any truth. It is run at 3800 tokens, the longest budget in the chapter, because truncation would cut the comparison itself. For a fixed control cell it compares

$$\text{invariance gap} = \underbrace{\overline{\text{sim}}(\text{same prompt, resampled})}_{\text{sampling noise alone}} - \underbrace{\overline{\text{sim}}(\text{prompts differing in the drug})}_{\text{noise plus any drug effect}}, \quad (4.5)$$

with similarity as top-100 Kendall τ and as Jaccard overlap, over 8 contexts $\times$ 12 drugs at temperature 0.8. It is the least informative of the four and the least confoundable.

SEE chapter 3
records the four
generation budgets

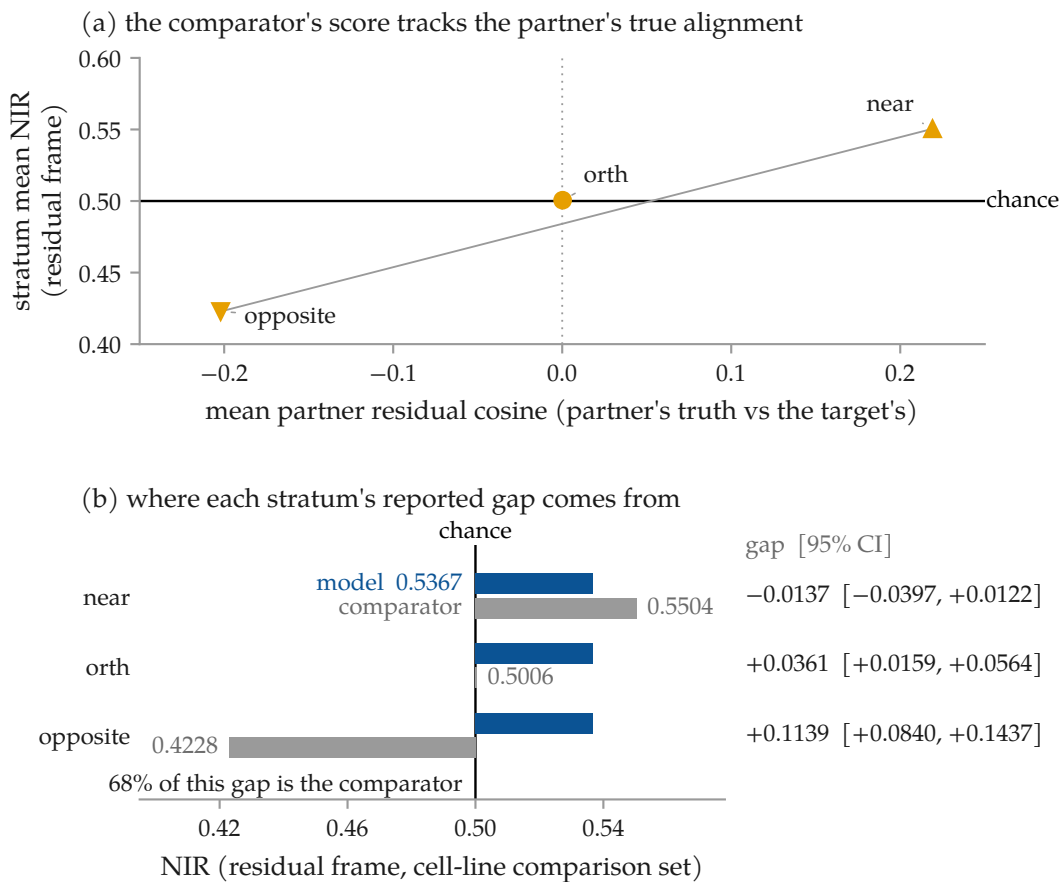


Figure 4.5. The comparator's own score is a monotone function of how correct the swapped-in partner is, so most of the opposite stratum's headline gap is the comparator failing rather than the model knowing. All quantities are residual-frame NIR scored against the cell-line comparison set, pooled over the 1,394 scored conditions. (a) Each mark is one swap stratum, at its mean partner residual cosine and its mean NIR. The score falls from 0.5504 through 0.5006 to 0.4228 as the partner's true residual moves from aligned to anti-aligned, so *near* hands the model a partly correct answer and *opposite* an actively wrong one. The straight line is the chord joining the two extreme marks, drawn as a guide to the eye; it is not a fit, and *orth* lies 0.016 above it. Because the strata are defined using the *true* residual geometry, *orth*'s sitting on chance is an empirical property of this sample and not an independently established null: it is the least contaminated comparator available, which is a weaker statement (§ 4.10). (b) The same gap Δ_{scr} split at chance into its two terms, $\Delta_{scr} = (NIR_{model} - 0.5) + (0.5 - NIR_{scramble})$. The accent bar is the model's own advantage over chance, +0.0367, and is identical on all three rows because it is the same model arm; the strata differ only in the grey bar, the comparator's own displacement from chance, drawn on the side it actually falls — right of the rule it subtracts from the gap, left of the rule it adds. On *opposite* the grey term is +0.0772 of the +0.1139 reported, 68% of it, which is the annotation on that row. Intervals are 95% two-way cluster-robust on cell line and treatment well (42 cell lines, 200 wells); the one-way variants stored alongside them are not drawn.

Table 4.3. Four independent instruments agree, and the two that return intervals bound the drug's effect instead of merely covering zero. Near-ceiling $DE-\Delta r$ is achieved identically with a wrong-mechanism drug, so that score is drug-independent.

Instrument	Quantity	Value	Reading
Forced-choice grading	own-truth win rate	0.48 ($\approx$ scramble; reference 0.67–0.83)	chance
Output invariance	invariance gap	0.000 [−0.019, +0.016]	bounded, ≤ 0.016
Scramble, tier 1	$DE-\Delta r$	0.740 real vs 0.739 scrambled	indistinguishable
Scramble, tier 2	$DE-\Delta r$	0.733 real vs 0.729 scrambled	indistinguishable
Scramble (identifiable subset)	Δ_{scr} (NIR)	+0.014 [−0.016, +0.042]	bounded, $\leq +0.042$

The two $DE-\Delta r$ rows are paired comparisons on one scale, the only use § 4.1 licenses for that metric.

AN INTERVAL THAT COVERS ZERO EARNS ITS KEEP BY WHAT IT EXCLUDES. Not by the word *null*. Two predictions from a byte-identical prompt agree at $\tau = 0.198$, so the invariance interval caps any drug-driven divergence at roughly a twelfth of that agreement, and the Jaccard version is tighter still, +0.000 [−0.008, +0.008]. On the identifiable stratum the scramble gap of +0.014 [−0.016, +0.042] runs against the 0.170 separating the model's 0.768 from that stratum's 0.938 reference; at the very top of the interval, naming the right drug would buy a quarter of what is there to be bought, and the point estimate puts it at 8% of that. Neither figure shows the effect is zero, and no experiment here could. Both put it far below the headroom these conditions leave.

FOUR OBJECTIONS CLOSE ON MEASUREMENT, NOT ARGUMENT. The first is whether there is anything to know at all. A ladder of mean-shift baselines answers it, and the informative conditioning variable is the cellular context, not the drug class.

The second decides whether this section reports a result or an instrument failure. Barely half the atlas clears the pre-set identifiability bar (Table 3.1), so the verdict is re-scored where the wells are demonstrably adequate. That does not rescue the model. It appears to beat the linear baseline there (0.768 against 0.531), yet a zero-drug-information control-copy reaches 0.766 on the same conditions, the paired contrast is +0.002 [−0.026, +0.028], and the leak-immune Δ_{scr} stays bounded as above.

SEE § 4.11 gives the ladder

The control arm carries a second result that is easy to walk past. Across the three strata its score tracks the reference almost step for step (Figure 4.6, panel b): 0.310 against a reference of 0.287 on the low-reference stratum, 0.540 against 0.689 on the marginal one, 0.766 against 0.938 on the identifiable one. A predictor knowing nothing whatever about any drug does best exactly where the ranking problem is easiest, so what makes a condition identifiable is largely what makes its untreated cells distinctive among their plate-mates, and the distance to a high reference is not all room that better drug modelling would

fill. The reference rises with aggregation (Figure 4.6, panel a: 0.61 at 10 cells, 0.76 at 40, 0.85 at 120); what the strata add is that most of the ranking can be done without the drug.

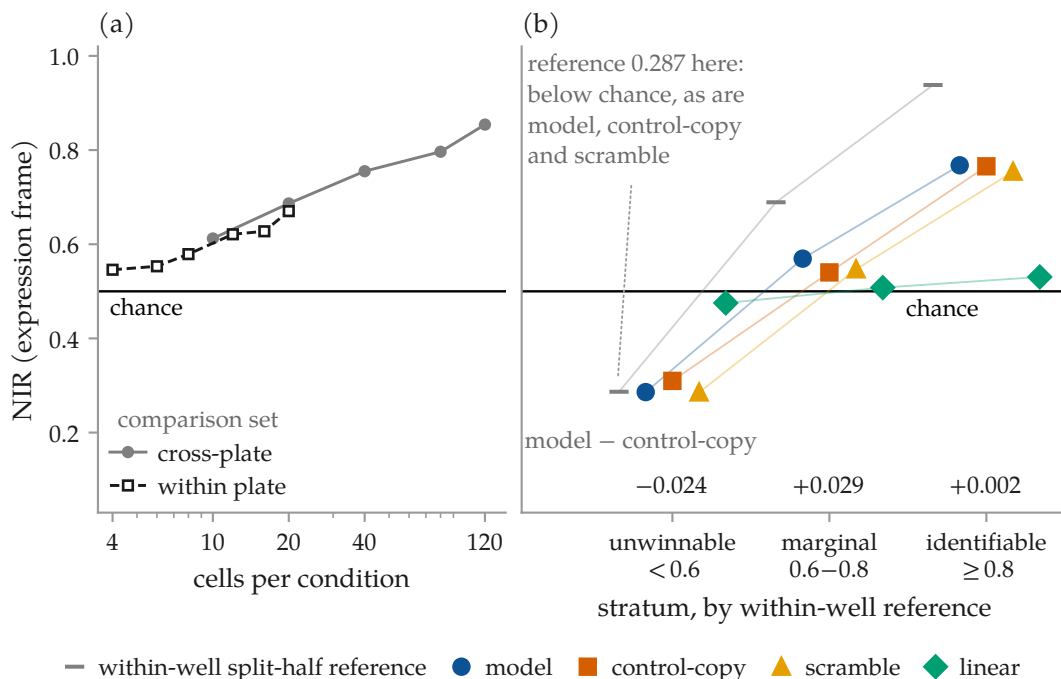


Figure 4.6. Conditions are more empirically discriminable at the tested pseudobulk budget where the precision reference is high, and precisely there, a predictor containing no drug information matches the model. Both panels are NIR in the expression frame, on one shared axis. (a) The within-well split-half precision reference against cells per condition, for two comparison sets: cross-plate, and within plate. (b) The same reference used to bin conditions, five arms per stratum, $n = 296$ unwinnable, 106 marginal, 204 identifiable. On the unwinnable stratum the reference itself is 0.287, below chance, as are the model, control-copy and scramble marks. On the identifiable stratum the model reaches 0.768 and a control-copy baseline 0.766, paired contrast $+0.002 [-0.026, +0.028]$, clustered over cell lines. Plotting the model alone would suggest it does better on higher-reference drugs. The arm marks are bare: the source carries no interval for a stratum mean, and this document draws only the clustered bootstrap, so nothing is drawn rather than an interval of another kind.

The third objection is whether the representation throws the drug away. Cell sentences keep only the rank order of genes and discard expression magnitude; if the drug lived there the following sections would attack the wrong thing. It is testable without any model, by asking whether true expression discriminates drugs better than rank does, and it does not.

The fourth is whether the scramble swaps in similar drugs. The swap-distance sweep bounds this rather than settling it, and it sits outside the within-plate rule, being an earlier run at `max_new_tokens=1200` scored on the unseen-drug tier of a different evaluation set, with swap partners not plate-restricted, and with $n = 40$ per bin wide enough to exclude a large effect but not to establish a null (§ A.3). Neither column has an interval excluding zero in any distance bin; in the single-cell column the far tail reads -0.019 across a $2.3 \times$ spread in swap distance, trend correlation $+0.051$. That column bounds a large

SEE § 4.11 gives that measurement

dissimilarity effect for the expression-frame model. The optimal-transport column cannot, since § 4.9 later shows that checkpoint clears zero under the canonical within-plate stratified test at the matched `max_new_tokens=1400` budget. Its apparent null is budget-dependent, and the stratified scramble replaced the sweep in the reported set.

THE CLAIM. On the calibrated metric the model is drug-blind in the expression frame, including on the identifiable stratum. The model is at chance overall (0.498 against a 0.576 within-well split-half precision reference), matched on the identifiable stratum by a zero-drug-information control-copy (0.768 against 0.766, paired contrast $+0.002 [-0.026, +0.028]$), and no more moved by a maximally dissimilar swap than by a similar one in the single-cell column of a non-canonical `max_new_tokens=1200` sweep outside the within-plate rule (§ A.3). The sweep's optimal-transport column does not carry that bound; § 4.9 later shows the same checkpoint is non-null under the canonical test at the matched 1400-token budget.

4.3 The drug is not missing from the computation

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■ EXPR

A model that ignores a drug it was told about can fail in two ways. Either the drug never enters its computation, an activation-side encoding failure that target- and input-side fixes might reach, or it enters and nothing reads it, a decoder failure only an objective-side fix could reach. Behaviour cannot tell them apart, since both predict the same scores. The weights can. A linear decode probe, read as a floor test and not as evidence of use, settles the first half.

§ 1.4 asks where useful prediction fails: is the drug absent from the data, lost from the model's representation, or encoded and underused by generation? The three sections that follow answer in one direction, and the answer is worth having before the instruments. Drug identity is written into the activations and stays written, more legibly with depth once batch structure is projected out, while generation consults it too weakly to account for the drug-blindness of § 4.2. The failure is not on the activation side. Neither half comes free: the presence result is close to guaranteed by construction, the low-gain reading rests on the weakest measurement in the chapter, and both are carried below with the discount attached.

Every behavioural measurement so far fits the last two readings equally. § 4.2 has answered the first where it matters here: the task is empirically discriminable for a substantial stratum at the tested pseudobulk budget, the reference rises with aggregation, 46.2% of conditions clear the declared identifiability bar, and on those the model is matched exactly by a baseline copying the untreated cell (0.768 against 0.766). The shortfall there is the model's; where in the model it lives needs the weights.

FRAME ■ **EXPR**
scores here are full-profile, § 4.1

IDENTIFIABLE STRATUM conditions whose own within-well split-half reference clears the pre-set bar

SCOPE LIMIT | carried into § 4.6 and chapter 5

The distance from a score to that stratum's reference is not available headroom. Across the three difficulty strata of § 4.2 the control-copy's own score tracks the reference almost step for step: 0.310 against a reference of 0.287 on the low-reference stratum, 0.540 against 0.689 on the marginal one, and 0.766 against 0.938 on the identifiable one. A predictor that reproduces the untreated cell and knows nothing whatever about any drug does best exactly where the ranking problem is easiest. What separates the strata on this criterion is largely drug-independent: within a comparison set every condition's untreated cells come from one plate-matched DMSO pool, so a control-copy scores high exactly where a condition's measured profile sits closer to that shared untreated state than its plate-mates' do. Most of the ranking these conditions are scored on can therefore be done without the drug, and the gap between a predictor and a high reference is not all of it room that better drug modelling would fill. Every score quoted on the identifiable stratum below inherits that reading, including the three target constructions of § 4.6.

THREE SUB-QUESTIONS, IN THE ORDER THEIR BLIND SPOTS CLOSE. Is drug identity present in the activations? Does the network use the region where it is written? And does it use *which* drug? Each instrument's blind spot is what the next must close; § 4.6 adds a fourth. Figure 4.7 maps all four, on scales that may not be read across.

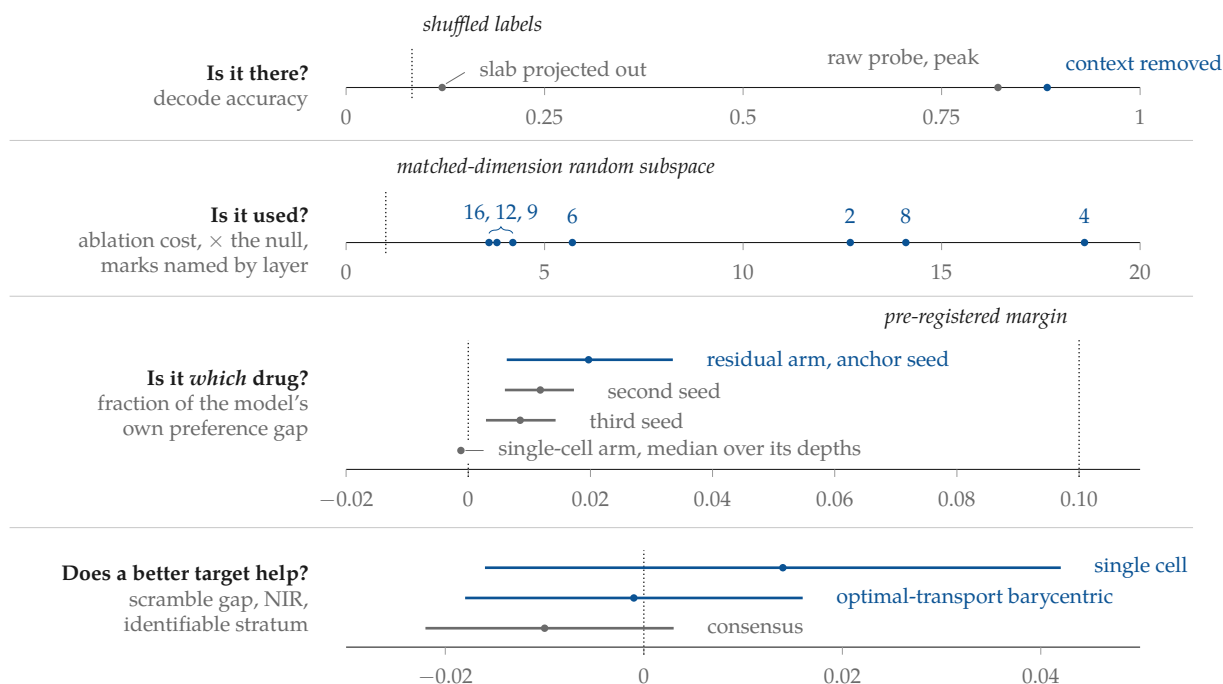


Figure 4.7. Each instrument answers a different question against a different comparator, and the four rules may not be read across. Presence is decode accuracy against a shuffled-label floor; use is ablation cost in multiples of a matched-dimension random subspace; identity is a fraction of the model's own clean preference gap, against the 10% margin pre-registered for calling an effect material; target value is a scramble gap in NIR on the identifiable stratum, against zero. Converting one rule into another is the inference this part exists to block. Bars are 95% intervals where the instrument returns one; the ablation rule carries none (Table 4.4, Table 4.5, Table 4.6).

THE PROBE, AND WHY ITS POSITIVE IS NEARLY FREE. The prompt names the drug in plain text (chapter 3), so recovering it from the activations establishes only that the model did not discard what it was handed. The informative outcome would have been a negative, and what the probe buys is the depth at which one would appear.

Forward passes only. Real tier-2 evaluation prompts are grouped by drug; for each, the residual-stream activation is captured at the last prompt position, where generation begins, at every layer. A cross-validated multinomial logistic-regression classifier learns which of the twelve drugs the prompt named, against the floor measured on shuffled labels ($1/12 \approx 8\%$). The twelve are a random draw from tier-2 drugs with at least forty scored cells each; that count sets the 8% floor and, in § 4.4, bounds the drug subspace at rank 11. Folds are stratified over prompts and not grouped by well, a limitation stated and not repaired, and the reason the accuracy is read as a presence test, not a generalisation estimate.

DECODE PROBE a classifier trained to name the drug from the residual stream at one layer

THE NEGATIVE NEVER ARRIVES. The probe reads 82% at layer 9 and 76% at layer 16, the deepest measured, against the $\approx 8\%$ floor. The between-drug to within-drug variance ratio collapses about three-fold across the final layers, so the drug is present throughout but geometrically de-emphasised in the layers nearest generation. That de-emphasis is in the raw geometry; once cell line, plate and dose are projected out, the series rises with depth instead, to 0.883 (88%) at the deepest layer. The survival carries the weight, not the peak, and that is the end of what a probe can say.

THE CLAIM. The drug is not missing from the computation. Its identity is linearly readable from the residual stream at every layer measured, at the position the first target token is emitted from. The raw probe reads 82% at its layer-9 peak and 76% at the deepest against an 8% chance floor; with cell line, plate and dose projected out it rises with depth instead, to 88% (§ 4.4). The prompt names the drug, so this is a floor test and its positive is close to free. It rules out that the drug field is lost before the decoder runs, an encoding failure in the activations. It does not rule out that the decoder ignores it. (§ 4.7 later uses *target encoding* for a different thing, how the target string is written, and nothing here bears on that.)

4.4 The drug region is used, at a bit player's weight

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■ EXPR

Knowing the drug is written somewhere says nothing about whether the network reads it. Intervening on the region requires localising it, purifying it of the batch structure it is confounded with, and proving it really holds the drug before any null about it can be believed. Generation is sensitive to removing the region, at a bit player's weight beside cell line.

The second instrument is built in four steps, each because the unguarded version of the same measurement misled us once already.

STEP ONE, WHERE THE DRUG LIVES. On 480 tier-2 prompts (twelve drugs, forty real cells each) at seven layers, the twelve per-drug mean activation vectors are centred on the global mean and stacked, and their singular value decomposition contributes its top ten orthonormal directions, V_{drug} . Twelve centred class means span at most eleven directions, so the slab keeps essentially the whole span, noisiest included, which for a removal claim is conservative. The layer grid is even (2, 4, 6, 8, 12, 16) plus layer 9, added after the raw probe peaked there. A cell-line subspace is built identically as the positive control.

STEP TWO, PURIFICATION. The raw drug slab is partly a batch slab. Drugs are confounded with plate and dose by the atlas's design, and probing dose from inside the drug subspace recovers it at 0.90–0.95 at every layer. The drug axis is therefore orthogonalised against a context subspace of cell line, plate and dose directions, giving V_{pure} . Every causal claim below is made on V_{pure} , never on the raw slab.

STEP THREE, THE REMOVAL GATE. Before any “ablating the drug changes nothing” result can be believed, the instrument must prove the slab actually contains the drug. Accuracy is measured before and after projecting the drug subspace out, and the fraction of above-chance signal removed must exceed 0.8; below that the null is unfalsifiable, a measurement of nothing. The gate passes at every layer. Accuracy falls from 0.41–0.82 to a constant 0.121 floor, removing 88–95% of the above-chance signal, so the nulls below are real properties of the model and not broken hooks.

STEP FOUR, THE CAUSAL MEASUREMENT IN LOGIT SPACE. The true sentence is teacher-forced and its per-token log-probabilities read off clean; a forward hook then projects the slab out at every position ($h \leftarrow h - (hV)V^\top$) and they are read again. The measure is the mean KL divergence from clean to ablated over the target tokens, on 60 prompts. Logit space repairs the project's first causal attempt, which injected a steering vector and generated; that vector was small against the residual-stream norm, its effect sat under the sampling-noise floor, and its “no effect” was uninterpretable. The ablation runs for V_{pure} (headline), a matched-dimension random subspace (the null), and the cell-line slab (the positive control). The prompt count is small and no interval is quoted; what intervals the probe arc has come from the identity test of § 4.5,

SEE the grouping ladder in § 4.2 is what makes cell line usable as a positive control

TEACHER FORCING the true sentence is scored token by token, never generated

CORRECTION the steering-vector attempt this replaces, § A.5

which is discounted there in turn, so no conclusion in this chapter rests on the probe arc alone.

Table 4.4. Drug-label information becomes more decodable with depth while the same slab carries little functional weight. All seven layers pass the removal gate (88–95% of above-chance drug signal removed); chance is 0.083. The last column is ablation cost divided by the matched-dimension random null on the same layer, a sign-and-magnitude reading and not an estimate with an interval. The null matches the slab's rank, not the variance it removes — the slab's subspace fraction is roughly six times the null's — so the ratio mixes drug content with that difference.

Layer	Decode (raw)	Decode (context removed)	Ablate pure drug (KL)	Random matched dim	Ratio
2	0.408	0.354	0.155	0.012	12.7×
4	0.585	0.429	1.563	0.084	18.6×
6	0.692	0.529	1.435	0.251	5.7×
8	0.779	0.569	0.596	0.042	14.1×
9	0.821	0.602	0.418	0.100	4.2×
12	0.754	0.873	0.133	0.035	3.8×
16	0.758	0.883	0.015	0.004	3.6×

Decode columns are accuracies over twelve drugs, answering different questions and not two estimates of one quantity. The raw column includes whatever cell line, plate and dose contribute; the other is the same probe after those directions are removed.

WHAT THE ABLATION SHOWS IS A DISSOCIATION. With cell line, plate and dose projected out, drug-label decodability rises with depth, from 0.354 at layer 2 to 0.883 at layer 16. Yet the purified slab's descriptive subspace fraction stays flat at ≈ 0.03 against ≈ 0.6 for cell line and ≈ 0.005 for the matched-dimension null, an activation-space summary and not a decomposition of biological drug-response variance, and ablating it costs 3.6–18.6× the matched-dimension random null. Figure 4.8 draws the dissociation whole: the rise, the flat fraction beneath it, and each layer's ablation cost beside its random null.

That is small beside the cell-line positive control, which the same ablation reads at 1.0–11.2× the drug's. Removing the drug subspace costs 9–37% of what removing cell line costs at layers 2–9, with layer 12 the single exception at 98% before layer 16 falls back to 25%. On subspace fraction cell line measures 15–22× the purified slab. Sensitivity is established at a bit player's weight, across all seven layers.

WHERE ABLATION STOPS. It cannot show that the slab's drug-specific content is what is read. The purified directions could carry generic strong-programme energy that merely co-varies with drug across the twelve. That needs a test which replaces what the slab says.

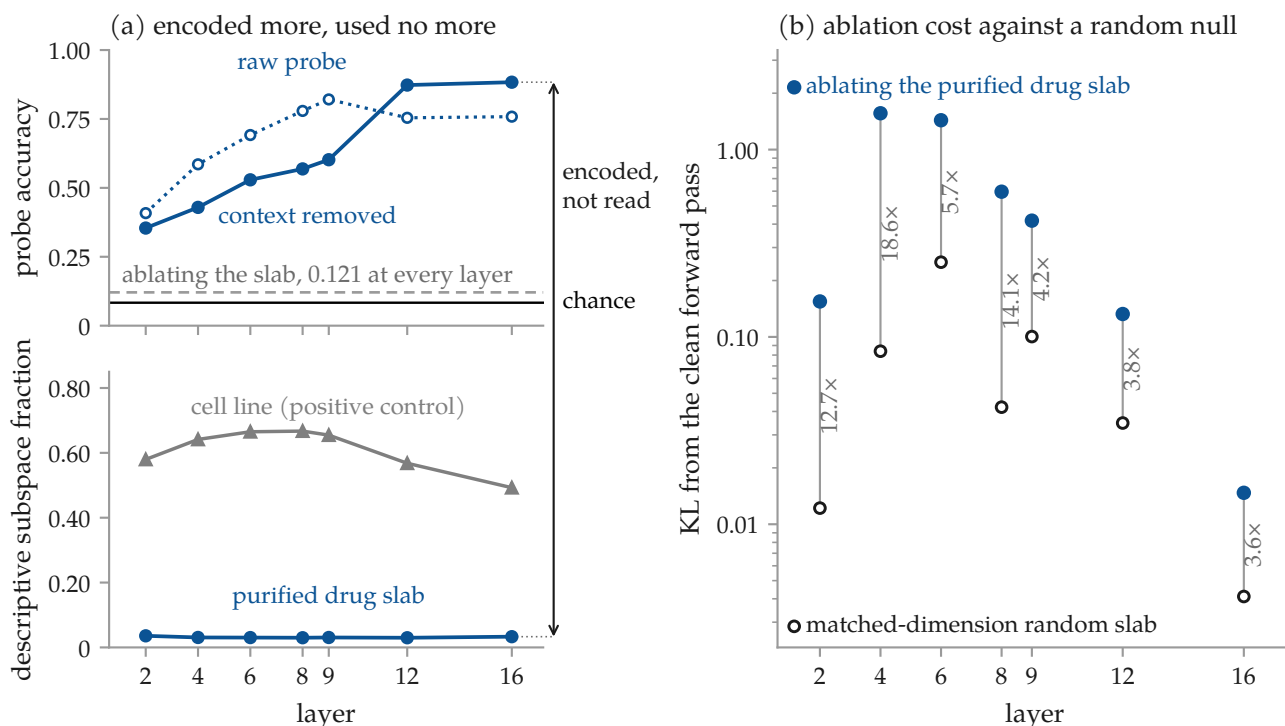


Figure 4.8. Drug-label information becomes more decodable with depth while interventions on the purified slab have limited effect. Layer is a true numeric axis in both panels, so the 9–12–16 spacing is drawn as it is. (a) Top: drug-label probe accuracy over *twelve* drug classes, so chance is $1/12 = 0.083$; the raw probe is drawn for contrast and the flat grey line is the same probe after the slab is projected out. Bottom: the descriptive subspace fraction of the purified drug slab against the cell-line positive control, on its own separate scale — it is an activation-space summary, not a decomposition of biological drug-response variance. The double-headed arrow at layer 16 spans the two, and is what *encoded, not read* names. (b) Mean KL from the clean forward pass when the purified drug slab is ablated, against a matched-dimension random subspace, $n = 60$ prompts per entry. Log axis because late-layer divergences are small for everything, and paired dots rather than bars: on a log axis a bar's length is measured from an arbitrary axis floor, whereas the vertical gap between a pair *is* their ratio, which is the number annotated beside each pair and the number in Table 4.4. **No intervals are drawn in either panel.** The stored dispersions are normal-approximation standard errors over the 60 prompts, not the thesis's one interval convention (95% bootstrap clustered over cell lines); as in Table 4.4 this is a sign-and-magnitude reading and not an estimate with an interval. These accuracies come from a different instrument from the twenty-four-class probe sweep run over the five arms of § 4.5, whose chance is $1/24$ on a different label set, and the two must not be read against each other. Whether drug *identity* is read requires § 4.5.

THE CLAIM. The purified drug-label slab carries decodable information and generation is sensitive to its removal, at a bit player's weight. Once cell line, plate and dose are projected out, decodability rises monotonically with depth, which the raw probe, peaking at layer 9, does not, while the descriptive subspace fraction stays flat at ≈ 0.03 , roughly 5% of the cell line's ≈ 0.6 , and is not a biological variance decomposition. Ablating the slab costs 9–37% of what ablating cell line costs at layers 2–9, with layer 12 the exception at 98%. Whether the slab's content is read, ablation cannot say.

4.5 Identity is read, at very low gain

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■ EXPR

Removal cannot distinguish a region whose content is read from one whose mere presence matters. The last question needs an instrument that replaces what the region says. Push the activations from the drug the prompt names towards another drug, and ask whether the model moves its preference towards that other drug's own held-out sentence. It moves, by a sliver, in one arm at one depth. The design is given at length because three earlier versions could have manufactured an effect.

VERDICT The generation decoder is sensitive to the drug-label activation signal at very low gain, in one training arm and at one rung of the displacement ladder.

The verdict first, because the instrument takes several pages to describe and the verdict is small. In one of the five training arms of Table A.3, at one depth of seven, injecting half the class-mean displacement from drug A to drug B inside the purified subspace ($\alpha = 0.5$, the rung the multiplicity correction is applied to) moves the model's preference towards drug B's own held-out sentence, against an identically constructed injection naming a third drug, by $+0.0197 [+0.0063, +0.0335]$ of the preference gap the model already has. One to two per cent, against a 10% margin pre-registered for calling an effect material. Two further seeds reproduce it, and of twenty-six headline tests it is the only survivor of multiplicity correction.

Read at face value that says the generation decoder does consult which drug the region names, at a gain far too small to account for anything in § 4.2. Read with the multiplicity discipline below it says less.

SCOPE LIMIT | carried into § 4.6 and chapter 5

This is the weakest load-bearing measurement in the chapter and it is carried as such. No argument downstream rests on it alone, and a reader who discards it entirely loses nothing that the drug-blindness result depends on. What the following section tests is a prediction of the *interpretation* of this result, not the result itself, and the inference there is written to run one way only for that reason.

FIRST, A CONTRAST SCORE. It is

$$\log P(g_B \mid \text{prompt}_A) - \log P(g_A \mid \text{prompt}_A), \quad (4.6)$$

where g_A and g_B are real cell sentences emitted under drugs A and B by held-out cells, each log-probability a per-token mean. Any offset uniform across positions, an inflated logsumexp normaliser included, then cancels exactly. It does *not* cancel in the stronger sense one might assume. Once g_A and g_B diverge the two teacher-forced sequences visit different states, so the normaliser's state-dependent part survives, unbounded here. The permuted arm controls for that, sharing anchor, construction and norm and differing only in naming the wrong drug. What survives the two-arm comparison is the model's preference between drug B's own held-out sentence and drug A's.

SECOND, ONE ANCHOR AND ONE NORM. Writing Π for the orthogonal projector onto the purified slab, the swap arm injects $\Pi(\mu_B - \mu_A)$, pointing at the drug whose sentence is scored; the permuted arm injects $\Pi(\mu_C - \mu_A)$ at identical norm, same anchor and construction, wrong drug. An isotropic random direction is kept only as a damage diagnostic, since it can disrupt the model in ways an observed class-mean displacement does not. The subtraction cancels the global mean exactly, and nothing is ablated, so the control has no removed term to trip over.

THE THREE FAILED DESIGNS TAUGHT THE CONSTRAINTS. A control drawn in the full 2048-dimensional stream puts $\approx 99.5\%$ of its energy where the intervention never acts, so it must live in the treatment's subspace. Ablate-then-add delivers (added – removed), so an uncentred class mean turns substitution into restoration, hence no ablation and an exactly centred displacement. A norm-matched random direction inflates the logsumexp normaliser (measured +2.75 nats), depressing every target whether or not it carries identity. The planted-world selftest validates the directional logic but cannot speak to the normaliser, since there each drug's sentence is a single token. Algebra in § A.5.

OUT OF SAMPLE, AND ON A COMPARABLE SCALE. Subspaces and class means are fit on one slice of cells; g_B comes from a third disjoint slice, never from the rows defining μ_B , else the injected vector and the scored sentence would share cells and the test would be partly circular. Because the five arms emit different target formats, every effect is divided by that model's own clean preference gap, measured with no intervention.

FOUR GUARDS CARRY THE INFERENCE. A drug-proxy screen and an injected-norm diagnostic are engineering, deferred to § A.5.

The first is the uncertainty model, a cluster bootstrap over ordered (A,B) pairs, because rows sharing a pair share the injected vector and the scored sentence. At the anchor seed's layer 12 that is 60 rows in 58 pair clusters over 24 anchor drugs, close to a row bootstrap. It guards against sharing within a pair, not across pairs sharing an anchor or target drug.

The second is the equivalence criterion, stricter than TOST. The interval must lie inside a pre-registered margin, set at 10% of the model's own clean preference gap, *and* contain zero. That conjunct runs one way only; it can withhold an equivalence verdict, never manufacture one. Under textbook TOST the headline +0.0197 [+0.0063, +0.0335] lies wholly inside the margin and would be returned as equivalent, read as a null; this rule returns a directional effect smaller than the margin.

The third is a per-layer headroom gate, excluding any layer where the cell-line positive control shows less than $5\times$ dynamic range, because a saturated instrument can support neither verdict. The fourth is a magnitude ladder, repeating the test at $\alpha \in \{0.5, 1, 2, 5, 10\}$ times the natural displacement, because a real identity effect should show a monotone dose–response. The ladder is the guard that bites.

CLUSTER BOOTSTRAP resamples whole (A,B) pairs, not rows, so a shared vector counts once

ONE SURVIVOR IN TWENTY-SIX TESTS. The design ran across the five training arms of Table A.3 and, for the two residual arms, at three further seeds. Of 26 headline tests, five arms by up to seven depths, exactly one survives Benjamini-Hochberg correction, the residual arm at layer 12 ($q = 0.026$, pooling the 26 across arms and not correcting within each as the pipeline stores them). That q rests on a bootstrap p , and the arm's probe file carries two draws of it for the same effect, identical to sixteen digits, at $p = 0.001$ and $p = 0.004$, either side of the 0.05/26 admitted at rank one; on the second draw nothing survives. The consensus arm at layer 2 ($p = 0.012$) does not ($q = 0.156$).

WHICH RUNG THE CORRECTION IS APPLIED TO MATTERS. The headline is the $\alpha = 0.5$ rung, half the natural displacement, so the ladder's first entry is that headline in raw nats (+0.0013) and not an independent point. At full displacement the effect is *larger* and noisier, +0.0265 [+0.0021, +0.0524] against +0.0197 [+0.0063, +0.0335], at $p = 0.033$ against $p = 0.001$ on the draw the correction uses. Each of $\alpha = 0.5, 1$ and 2 excludes zero; $\alpha = 5$ and 10 span it. In raw nats the rungs read +0.0013, +0.0017, +0.0044, +0.0061, +0.0046, rising to $\alpha = 5$ and turning over at $\alpha = 10$ (Figure 4.9, lower strip), consistent with a dose-response without establishing one. All five re-score the same 60 rows at scaled injection, no trend statistic is computed, and the pre-registered monotonicity holds over four rungs of five.

But applying the 26-test correction to the $\alpha = 1$ slice leaves *nothing* surviving, and applying it to $\alpha = 2$ promotes a different arm at a different depth (residual_holdout at layer 4, $q = 0.007$) while demoting this one (Figure 4.9). The slice decides not merely whether a survivor exists but which one, and below that the bootstrap draw decides whether one exists at all. That is the mark of a family dominated by noise, not of one containing a single robust effect. The multiplicity-corrected claim is therefore a statement about the $\alpha = 0.5$ slice and is not robust to that choice, which is why this section opened by discounting its own result instead of closing on it.

THE INSTRUMENT HAS BEEN SHOWN TO REGISTER A POSITIVE. The residual arm is independently known to respond to the drug its prompt names. On train, swapping the named drug and its mechanism for an orthogonal partner costs it +0.0898 in NIR, excluding zero clustered on cell line ([+0.0382, +0.1415]) and on drug ([+0.0291, +0.1506]), a behavioural result established in § 4.10. ➤ settled in § 4.10

Three things bound what it licenses. It is a training-split fact, and does not extend to held-out combinations. It is behavioural, so it says the arm responds to the drug its prompt names, not that the response is routed through the subspace this test injects into. And the two measurements are not on the same weights, the swap gap scored on the rebuilt target build and this probe on the earlier one (Table A.3), so drug-sensitivity and identity reading are attributed to residual-target training and not to any checkpoint. Narrower than a routing claim, and enough. The probe detects identity reading, replicably, in the arm behaviour calls drug-sensitive, and none at any depth in the arm that is behaviourally drug-blind. A null from an instrument never shown to register a positive is uninterpretable; this one has.

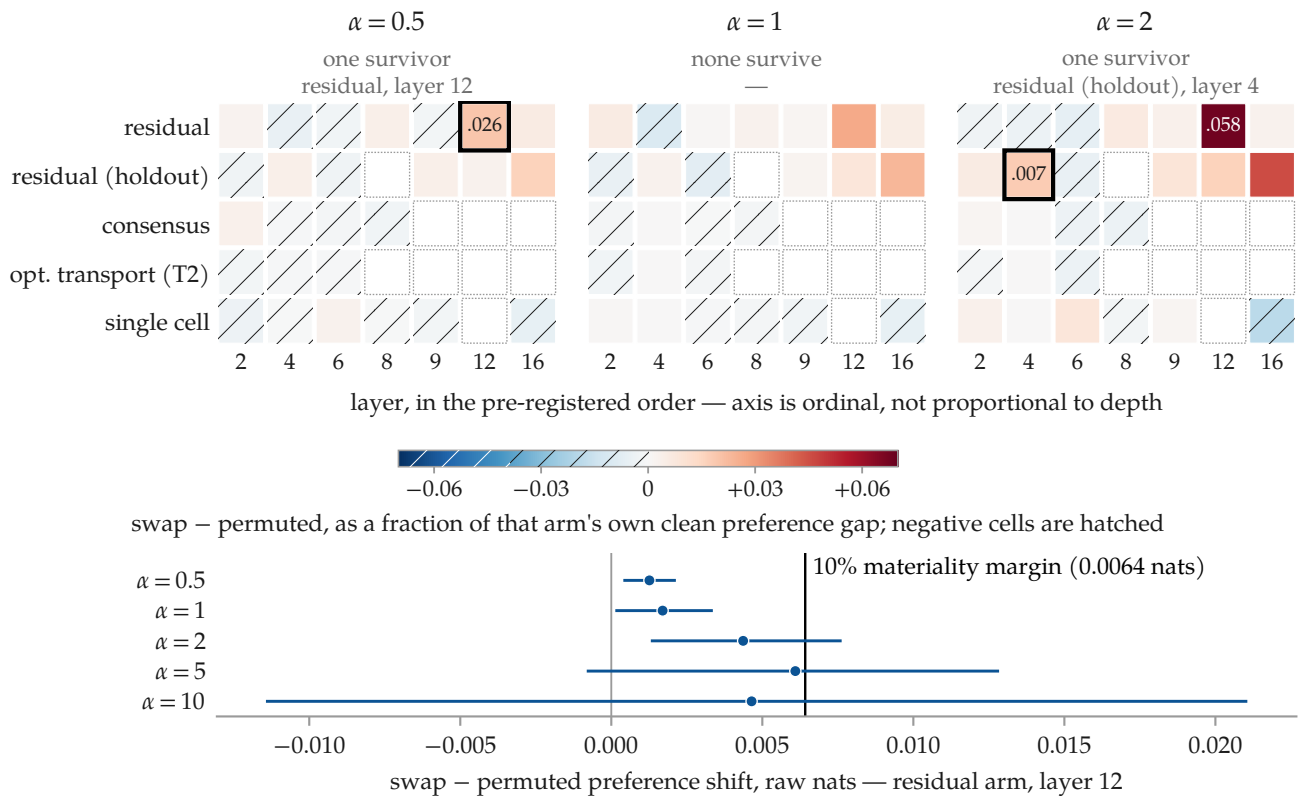


Figure 4.9. The one test that survives multiplicity correction changes arm and depth with the rung of the displacement ladder, and at one rung there is no survivor at all. Each panel is one slice of the ladder: the five training arms of Table A.3 (rows) by the seven probed depths (columns), $n = 26$ measured tests per slice, on one diverging scale shared across the three panels. Colour is the swap minus permuted effect as a fraction of that arm's own clean preference gap. The layer axis is ordinal, drawn at equal spacing, and is not proportional to depth. Nine of the 35 grid positions were never measured and are drawn as empty dotted outlines, not as zeros. Benjamini–Hochberg is pooled across arms within a slice, as the text applies it and not as the pipeline stores it; q is printed in the three cells that fall below 0.10 and ringed where $q < 0.05$. That criterion admits the residual arm at layer 12 at $\alpha = 0.5$ ($q = 0.026$, on the stored bootstrap draw the text discusses), nothing at $\alpha = 1$, and residual_holdout at layer 4 at $\alpha = 2$, which demotes layer 12 to $q = 0.058$; it is the only survivor among this chapter's twenty-six headline tests. Beneath, the same residual arm at layer 12 in raw nats across all five rungs, with the 10% pre-registered materiality margin as the vertical rule. Intervals throughout are 95% cluster bootstraps over ordered (A,B) pairs, 4000 resamples, which at this layer is 60 rows in 58 pair clusters over 24 anchor drugs. The three seed replicates are in Table 4.5 and are deliberately not drawn here.

THREE LIMITS BELONG WITH THE VERDICT. The effect appears at one depth of seven, and layer 4 is positive in a single seed and null in the other three. The single-cell arm has no layer-12 measurement, its cell-line positive control reaching $3.88\times$ there, under the $5\times$ gate; so the cleanest comparison, the same depth across two arms, does not exist, and the dissociation rests on that arm being null at all six depths it could measure. And the sibling residual_holdout arm does not reproduce the effect at the headline rung ($+0.0028$ and $+0.0049$ in the two seeds where it was measured); it surfaces instead at $\alpha = 2$, rung-dependence and not a second replication.

Read with the decodability series, the picture is label information that becomes more linearly decodable with depth once cell line, plate and dose are

Arm / seed	effect	95% CI	<i>p</i>
residual, seed 42	+0.0197	[+0.0063, +0.0335]	0.0010
residual, seed 43	+0.0118	[+0.0060, +0.0173]	0.00025
residual, seed 45	+0.0085	[+0.0029, +0.0143]	0.0015
single-cell arm, six depths	nothing detectable at any depth (median −0.0012)		

projected out, while the process that writes the prediction barely consults it. That is suggestive of the workspace account of § 2.7, an interpretation of the pattern and not a demonstration of the mechanism.

THE CLAIM. The generation decoder is sensitive to the drug-label activation signal at very low gain, in one training arm and at one rung of the displacement ladder. The anchor seed puts it at $+0.0197$ [$+0.0063$, $+0.0335$] of the model’s own preference gap, and $+0.0133$ averaged over the three seeds that measured it, one to two per cent, a fifth or less of the 10% margin pre-registered for calling an effect material. The multiplicity-corrected version of that holds for the $\alpha = 0.5$ rung on one of its two stored bootstrap draws, not for its neighbours. At $\alpha = 1$ nothing survives correction; at $\alpha = 2$ a different arm does. The claim is not that the decoder cannot read drug identity, and it is not the stronger mechanistic story earlier drafts told.

4.6 Denoising the target changes the target, not the decoder



If the defect is a decoder that barely consults the drug region whatever is written there, then improving what sits in that region should not help, and improving the training target is exactly that kind of intervention. It is also the repair a reader reaches for first, which is reason enough to test it. Three target constructions, one scoring, one stratum. None separates from a prompt naming the wrong drug.

The interpretation at the end of the last section is not established, and this section does not proceed as though it were. The inference runs one way only. A positive here would tell against the low-gain reading; a negative is evidence about target-side fixes, not confirmation of the mechanism. Because one failed construction proves nothing, two more were compared with it.

THREE POINTS, NOT A SWEEP. They are the original single-cell target, one optimal-transport barycentre and the consensus target. *Consensus* groups cells by (drug, cell line, dose) and averages them into one pseudobulk sentence, every original prompt retained so only the target changes. *Optimal transport* uses the full condition key carried by the source, (cell line, plate, drug, dose/treatment well), computing each condition separately, its treated cells coupled only to the shared DMSO controls from that condition's own (cell line, plate). Treatment conditions are never pooled.

The coupling π minimises $\sum_{ij} \pi_{ij} \|x_i - y_j\|^2$ under balanced uniform marginals, which impose mass conservation between the empirical control and treated measures, a modelling assumption and not an observed biological constraint. Sinkhorn iteration [29] solves it on the kernel $\Gamma = \exp(-C/\varepsilon)$, with C the pairwise squared-distance matrix. Couplings are computed in the released Tahoe scVI latent [31] ($n_{\text{latent}} = 10$), joined per cell by barcode and not re-encoded, since a re-encode produced a degraded latent (cell-line probe 0.11 versus 0.975). Targets are built without the scVI decoder, as convex combinations of that condition's observed treated cells, so they stay inside their convex hull, not necessarily on the data manifold.

The regularisation parameter locates the middle construction between two limits. As $\varepsilon \rightarrow \infty$ the coupling is uniform and the target is the treated mean (consensus); as $\varepsilon \rightarrow 0$ it approaches an unregularised sparse transport solution, not necessarily a map to one observed treated cell. Only three hand-picked constructions were trained and scored, so this is a three-point comparison and not a sweep of the family.

ONE SCORING, NAMED BEFORE THE RESULT. Every point was scored on the identifiable stratum, the conditions § 4.2 selected because their own within-well reference clears the pre-set 0.80 bar, under the random-partner scramble defined there. That is the stratum on which a zero-drug-information control-copy already matches the model, so the scope limit of § 4.3 applies to every row of Table 4.6.

An interval that covers zero earns its keep by what it excludes. On the identifiable stratum the single-cell arm's scramble gap is $+0.014 [-0.016, +0.042]$, against the 0.170 separating the model's 0.768 from that stratum's 0.938 reference. At the very top of the interval, naming the right drug instead of a wrong one would buy a quarter of what is there to be bought, and the point estimate puts it at 8% of that. Taking that distance at face value is the generous reading, for the reason declared in § 4.3. The interval does not show the effect is zero and no experiment here could; it puts it far below the room these conditions leave.

Table 4.6. Three target constructions, and no point separates from its random-partner scramble on the identifiable stratum ($n = 204$). Two things bound that negative. The consensus endpoint was stopped at roughly 60% of an epoch with its tier-1 NIR unscoreable, so the negative rests on the single-cell and optimal-transport endpoints. And the optimal-transport endpoint is not null in every frame. These rows use a random partner at 1200 generation tokens, while against the geometry-defined orth partner over all its conditions the same checkpoint moves from $+0.0144 [-0.0051, +0.0338]$ at 600 ($n = 250$) to $+0.0292 [+0.0144, +0.0441]$ at 1400 ($n = 248$), reported in § 4.9. Comparator, population and budget differ at once, and chapter 3 shows the budget is a claim-level variable, so the frames are set side by side and not put into ratio. This is a null under this scoring, not over the family of target constructions.

Target	ε	Δ_{scr}	Verdict
single cell (real treated cell)	≈ 0	$+0.014 [-0.016, +0.042]$	not separated
OT barycentric	0.5	$-0.001 [-0.018, +0.016]$	not separated
consensus (global mean)	$\rightarrow \infty$	$-0.010 [-0.022, +0.003]$	not separated

Δ_{scr} is the model's NIR minus its scramble's, so its null is zero and not chance. The three rows share a scoring, a stratum and a comparator; they are one comparison, not three independent experiments. All three intervals are *one-way* percentile cluster bootstraps over the 40 cell lines, not the two-way default.

TRAINING WAS NOT INERT, AND WHAT IT DID INSTEAD IS A RESULT. The consensus model stops echoing the control. On the identifiable stratum it scores 0.653 against a zero-drug-information control-copy at 0.766, where the single-cell model sits on the control at 0.768. Its predictions are not mode-collapsed either; there is real between-condition variation. But that variation is uncorrelated with real drug identity. The Mantel correlation between the model's between-drug distance structure and the true one averages $+0.03$ over cell-line and plate groups.

The failure this literature names first for a perturbation model that underperforms on well-calibrated metrics is mode collapse, where the prediction ignores the input and reverts to a mean, and the repair is to restore diversity in what the model emits [21]. This arm is not that. It moved off the control, kept the variation, and spent it on the wrong axis. Producing no structure and producing structure that does not track the drug are different failures with different repairs.

THE CLAIM. No point in the three-point comparison separates from its scramble on the identifiable stratum under the random-partner test. Two things bound that negative. The consensus arm was stopped mid-epoch, so the comparison rests on the single-cell and optimal-transport points, and the optimal-transport arm is not null in every frame: scored against the geometry-defined orth partner over all its conditions at the residual arm's generation budget, its gap excludes zero (§ 4.9). By the inference rule above that positive tells against the low-gain reading and not for it, and being in a different frame it does not settle the matter, but it is one

more reason the interpretation of § 4.5 is carried here as an interpretation and not a result. What the comparison establishes is that denoising the target changes what is in the target without changing what the decoder does with it under this scoring, and not that no target-side fix can work. Measuring how much of the target its tokens expose to the drug, which the next two sections do, is not the same as showing that the target encoding is the defect (§ 4.7). That construction changes five things at once.

4.7 The standard target hides the drug



A sentence lists genes in order of expression, so two drugs on the same cells give almost the same target string and the loss can barely tell them apart. Can the target be rewritten so that its tokens are the drug-specific part? The instrument counts how many top target positions carry a different gene for two drugs in one context. Almost none do before the rewrite; a clear majority do after, at an unchanged retrieval reference.

Encoding here means how the target string is written, not what the model’s internal states carry (§ 4.4). The loss weights every position equally, so the measurement is a count over the 200 target positions (Table 4.7).

Table 4.7. 83% of the standard target’s tokens are identical between any two drugs in the same context. The residual row is quoted at cell-line scope; at plate scope it reads 120.3 and 0.7467. The middle column is where the three encodings differ, and it does not on its own identify tokenisation as the binding constraint.

Representation	Top-200 tokens differing between drugs	Within-well split-half retrieval reference
full profile (the standard target)	34.6 / 200 (17%)	0.742
shift (treated – control)	111.4 (56%)	0.742
residual (drug-specific)	118.2 / 200 (59%)	0.7432

The two columns hold different quantities, a count out of 200 positions and a retrieval score, so rows compare within a column and never columns with each other. The third column pits disjoint cell subsets of one drug, in one cell line, on one plate, against each other. It is a precision reference, not a biological replicate ceiling, and it is near-constant across the three rows.

THE COUNT STOPS SHORT OF THE TOKENISATION CLAIM. The natural inference, that only a small fraction of the gradient carries drug identity, is a hypothesis about the optimisation; what is measured is the overlap of target gene sets, not a token-level loss or a gradient attribution. Gene positions are not loss-token positions under byte-pair encoding, and the residual target below alters aggregation, control referencing, generic subtraction, sign coding and sequence length at once. Separating those five factors needs a matched-budget factorial this thesis does not run.

THE COUNT IS NOT INHERITED FROM RESIDUAL SIMILARITY. A descriptive summary bins 212,528 exhaustive within-group drug pairs on the cosine between the two residuals. Mean differing top-200 genes across the near, orthogonal and opposite strata run 32.5, 34.6 and 35.8 for the full profile; 107.4, 111.3 and 114.4 for the shift; 118.4, 121.0 and 118.7 for the residual, with Spearman coefficients against residual cosine of -0.149 , -0.282 and $+0.019$. The residual target stays high in every stratum and lacks the monotone gradient the other two carry, which weakens the reading that the later swap-distance pattern is inherited trivially from target divergence without being an independent test of it. The pairs reuse drugs and groups, so no pair-level inferential claim is made.

DEFINITION The drug-specific residual

For each condition, a (drug, cell line, plate, dose) tuple with at least 40 treated and 20 plate-matched control cells,

$$r(d, c) = \underbrace{(\bar{y}_{\text{treated}} - \bar{y}_{\text{control}})}_{s(d, c)} - \underbrace{\frac{1}{|D_c \setminus \{d\}|} \sum_{d' \in D_c, d' \neq d} s(d', c)}_{\text{generic}(c)}. \quad (4.7)$$

The control subtraction removes the cell's own state, leaving the *shift*; the mean over drugs removes the *generic* drug programme every drug triggers. D_c is the training drugs sharing c 's plate and cell line, each entering once whatever its doses, and the condition's own drug is excluded. What remains is not what makes this drug different from all drugs. It is what makes it different *from the other training drugs that happen to share its plate and cell line*.

THE COMPARISON SET IS FIXED BY THE PLATING DESIGN. That last sentence is the price of estimating the generic locally, and mechanism is where it bites. § 4.12 finds mechanism-related drugs co-plated 0.404 of the time against 0.362 for a random draw, so for a co-plated mechanism class part of the class's own shared programme sits inside the generic and is subtracted out of every member's residual by construction.

That bias was tested. Same-mechanism plate-mates supply a mean 4.4% of a condition's generic, present for 83% of the 917 labelled mechanism-arm conditions, from Tahoe's *moa-fine* labels joined to the plate rosters the channel gate scored. The 155 conditions with *no* same-mechanism plate-mate, whose residuals retain the mechanism programme, give a gap of $+0.1145$ [$+0.0276$, $+0.2014$] against $+0.0737$ [$+0.0150$, $+0.1324$] for the 762 that have one, both against the same plate-matched null under the same two-way clustering. That is the direction the construction predicts, but not a corrected estimate. The intervals overlap across almost their whole range, the split is post hoc, and the cleaner stratum rests on 43 wells. It supports a statement about sign, not size.

SCOPE LIMIT | carried into § 4.12 and chapter 5

Two properties of this frame travel with every number measured in it. First, the generic is a plate-local mean, so where a mechanism class is co-plated the mechanism channel is later graded in a frame that has already removed some of

what it is asked to find, which makes an inconclusive verdict on mechanism the conservative reading and not a generous one. Second, the plate-scoped generic removes plate structure from the truths and, in expectation, from the predictions, but that is an expectation and not an identity. An individual residual may retain some plate structure, and the size of what remains was never measured. The absolute NIR values of this frame are read within it and never set beside the within-plate tables elsewhere in the chapter.

WHICH SUBTRACTION DOES THE WORK. Removing the control moves the between-drug token difference from 34.6 to 111.4; removing the generic adds a further 6.8 at cell-line scope and 8.9 at the plate scope this build uses. So the control subtraction alone buys 92% of the recoverable difference, 90% plate-scoped. The measurement is an overlap of gene sets on one atlas at $K = 200$ (6,602 conditions, `target_divergence.json`), so no margin is claimed for other atlases or other K . It squares with § 4.1, where what saturates $DE\text{-}\Delta r$ is reversion toward a central point, `revert_center` scoring 0.9999 without fitting anything.

The generic is not dispensable. It carries 39% of the shift's energy (§ 4.8), and subtracting it isolates the drug-specific remainder once the baseline is gone. The inclusion thresholds sit where the identifiability curve of § 4.2 flattens, the within-well split-half precision reference being 0.61 at 10 cells, 0.76 at 40 and 0.85 at 120.

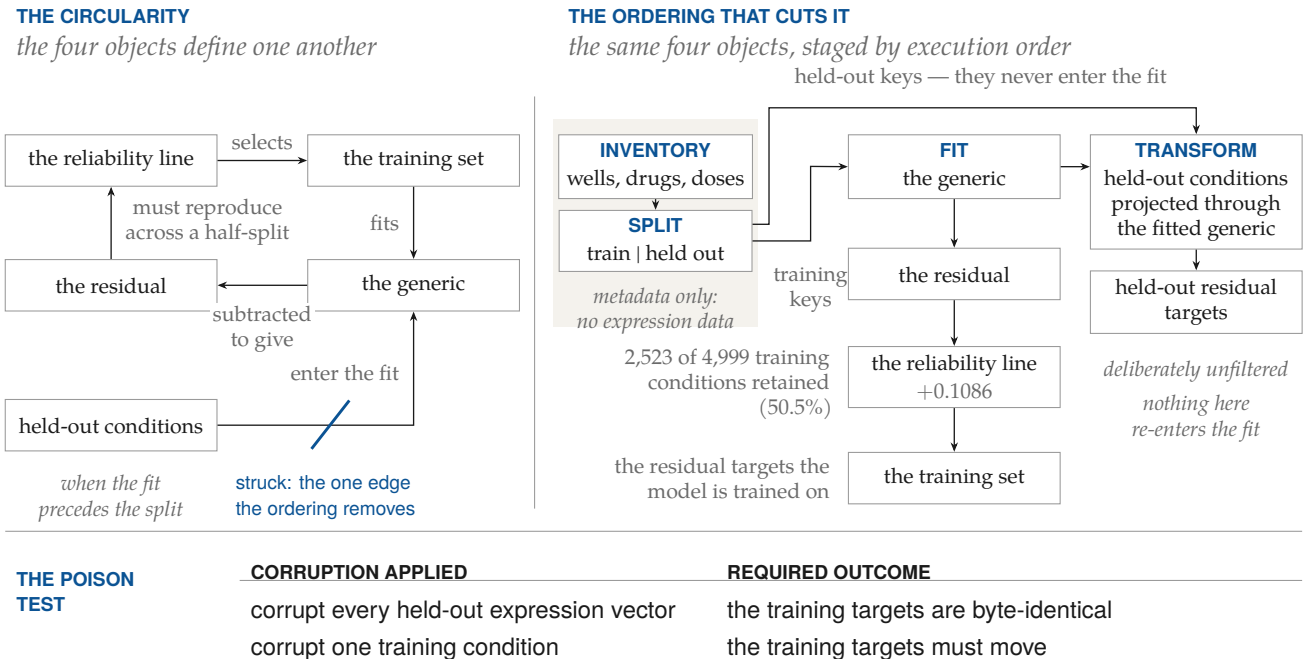
A DEFINITION THAT COULD HAVE BEEN CIRCULAR, AND WHAT BREAKS IT. Four properties of how the generic is estimated decide whether the remainder can be trusted. First, the shape of the problem. A condition enters training only if its residual reproduces across a half-split; a residual exists only once the generic has been subtracted; the generic is fitted on the training set; and the training set is what the reliability line selects. That is a loop (Figure 4.10, left), and an earlier build closed it, with generic and filter estimated over every condition and the holdout assigned afterwards, so each held-out target was defined partly by other held-out conditions.

The loop is cut by ordering. Inventory and split read metadata only, before any pseudobulk exists; fit hands the generic the training keys explicitly; transform projects held-out conditions through the fitted generic, which they never enter (Figure 4.10, right). Because the split is made on metadata alone, nothing downstream of it can have defined it. A poison test makes the ordering checkable in both directions. The other three properties fix choices the staging leaves open.

SECOND, THE SCOPE IS THE PLATE. The original argument for cell-line scope was reproducibility. 61.7% of conditions reproduce across a half-split at cell-line scope against 19.3% at plate scope (cosine threshold 0.2, $n = 6602$), on the reasoning that a plate holds too few drugs to estimate a mean. Both figures were computed with a shared control pseudobulk subtracted from both halves, which makes the halves correlate through their common control error, and the inflation favours cell-line scope precisely because the cell-line generic is itself less noisy. That alone retracts the retention argument. A disjoint-control

POISON TEST Corrupt the held-out side; the training targets must not move. Corrupt a training condition; they must.

CORRECTION the retention argument for cell-line scope is retracted here; the scope decision now rests on the two diagnostics below



THIRD, THE MEAN IS OVER DRUGS AND LEAVES THE CONDITION'S OWN DRUG OUT. Two things force that. A drug measured at five doses would otherwise carry five times the weight of a single-dose drug; and once the generic is train-only, a training condition's generic would contain its own drug while a held-out condition's would not, so the two splits would be scored against differently defined targets. The generic is therefore the mean over training drugs other than this one, within scope.

FOURTH, THE RELIABILITY LINE IS RESOLVED FROM THE DATA. The threshold was the largest single determinant of the training set, and it had been inherited. The inherited 0.2 came from an era when reliability was measured with shared controls, a different quantity on a different scale, under which it retained the 61.7% above against the 19.0% (948 of the 4,999 training candidates) it retains once each half is measured against its own control half. The two are not a matched pair, differing in population and scope as well as control handling, but between them a moderate bar had silently become a stringent one.

DEFINITION The reliability line

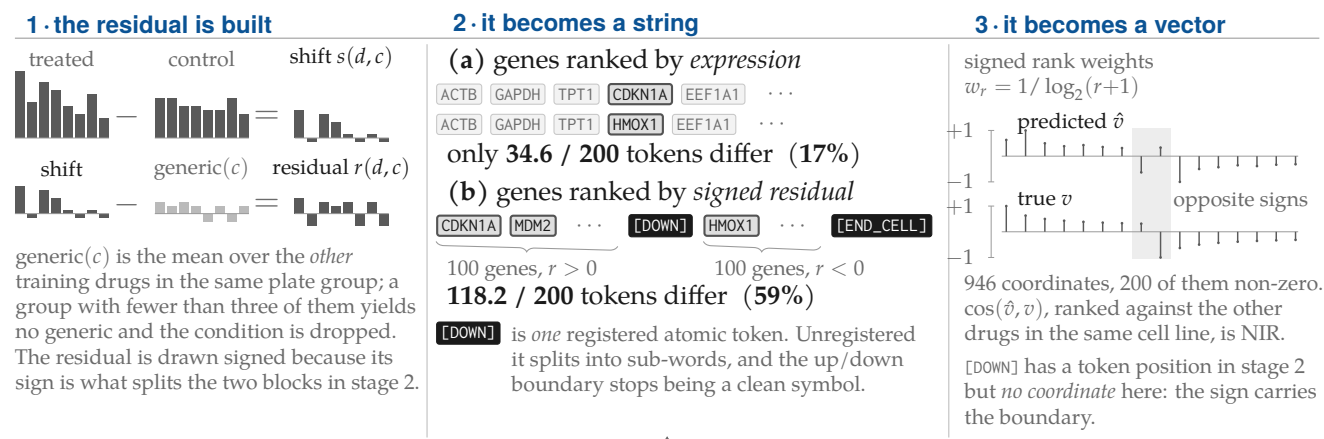
A condition is kept for training only if its residual reproduces across a half-split, $\cos(r_A, r_B)$ above a threshold, with each half measured against its own control half so that the two share nothing. No new threshold is picked by hand. The builder reports the whole retention curve against a null that pairs half *A* of one condition with half *B* of a *different* condition, same encoding, same dimensionality, same noise, no shared biology, and sets the line at that null's 95th percentile. A condition is kept when its two halves agree more than two unrelated conditions' halves do. At the measured null (95th percentile +0.109, resolving to +0.1086) this retains 2,523 of 4,999 training conditions (50.5%).

The inherited 0.2 survives as an argparse default in three downstream configs, `truth_repro_thr` in `re_v3.json` and `repro_thr` in `vardecomp_matched.json` and `channel_gate_v3.json`. It did not act as the training filter. Of the 300 scored training conditions, 181 sit below 0.2, the minimum is +0.109, and none falls at or below the resolved +0.1086. The held-out sets are deliberately unfiltered, since dropping held-out conditions for failing their own reproducibility bar would select the evaluation set on its own outcome; noise in retained training conditions costs learning efficiency, not validity.

The half-split agreement is a within-well sampling-precision criterion. It is not biological reproducibility, since 286 of 287 distinct (drug, dose) treatments in this cache occur in exactly one well. The retained fraction says how much drug-specific signal survives pseudobulk noise at this cell budget, not how much biology is real.

HOW THE RESIDUAL BECOMES A STRING, AND HOW IT IS SCORED. Figure 4.11 draws the path, from the two subtractions to the vector that is scored. The target is `<up genes> [DOWN] <down genes> [END_CELL]`, 100 genes per block, ordered by signed residual magnitude; sign is biology, so it gets a symbol. Predicted and true residuals are both mapped to a signed rank vector, top-100 up positive, top-100 down negative, weight $1 / \log_2(\text{rank} + 1)$; similarity is cosine. A generation supplies its ranks by token order within each block, and a token outside the

panel or already placed is skipped without consuming a rank, so a block fills to at most 100 genes. The sampled generations of one condition are averaged as vectors and the mean is scored once, not scored one by one and averaged. The score is NIR over other drugs in the same cell line. That set spans plates, in apparent violation of the same-plate rule of § 4.2, and stands only because the plate-scoped generic has already subtracted each plate's mean shift, in the expectation sense fenced off above.



and the within-well split-half retrieval reference is **unchanged**: 0.742 → 0.743

Figure 4.11. The information was always present; the standard target's tokens barely expose it — and the string the model writes is not the vector that scores it. *Stage 1.* The drug-specific residual is built by two subtractions. The control removes the cell's own state, leaving the *shift*; the mean over the other drugs in the same plate group removes the *generic* programme that every drug triggers, leaving what makes this drug different. The six gene profiles are illustrative shapes and carry no measurement — no axis is drawn — but the arithmetic within the figure is exact bar by bar, and the residual is drawn signed because its sign is what splits the two blocks in stage 2. *Stage 2.* That same measurement, written two ways. Ranking genes by expression (a) puts housekeeping genes first, so only 34.6 of the top 200 tokens differ between two drugs in the same context; ranking by signed residual (b) raises that to 118.2. (Divergence counts quoted at cell-line scope, as in Table 4.7; plate scope is similar, at 120.3.) Row (b) is the target string itself: a 100-gene up block, [DOWN], a 100-gene down block, [END_CELL]. *Stage 3.* What NIR ranks is not that string but a signed vector over the 946 panel genes, top- k up positive and top- k down negative. The stem heights are the exact weights $1 / \log_2(\text{rank} + 1)$, so the eight drawn per sign are 1.000, 0.631, 0.500, 0.431, 0.387, 0.356, 0.333 and 0.316; which gene sits at which coordinate is illustrative, and no cosine value is printed, because these two vectors are drawn rather than measured. Stage 2 and stage 3 are different objects and only stage 3 is scored. The bracket is the point of the figure — the within-well split-half retrieval reference is essentially identical under both encodings, so no information has been added. What changed is how much of it the token sequence exposes to the loss — a property of the encoding, which is not the same as showing the encoding is what binds (Table 4.7).

4.8 A per-drug constant does not exhaust the target



That share has not previously been computed for this task, and its frame is not the trained-on one. The stored generic averages over conditions and keeps the condition's own contribution, where the training frame averages over drugs and leaves the condition's own drug out. Self-inclusion scales a residual by $(m - 1)/m$ in a plate group of m conditions, so that frame difference depresses the share quoted here; it does not inflate it. The weighting difference has no established direction. A second bias runs the other way. A condition's pseudobulk sampling error enters its own shift whole and is averaged down in the group mean, so it lands in the residual term and inflates the estimated share at this cell budget (Table 3.1, and the reliability line above). The two oppose each other, the net is not established, and no noise-corrected share is estimated here.

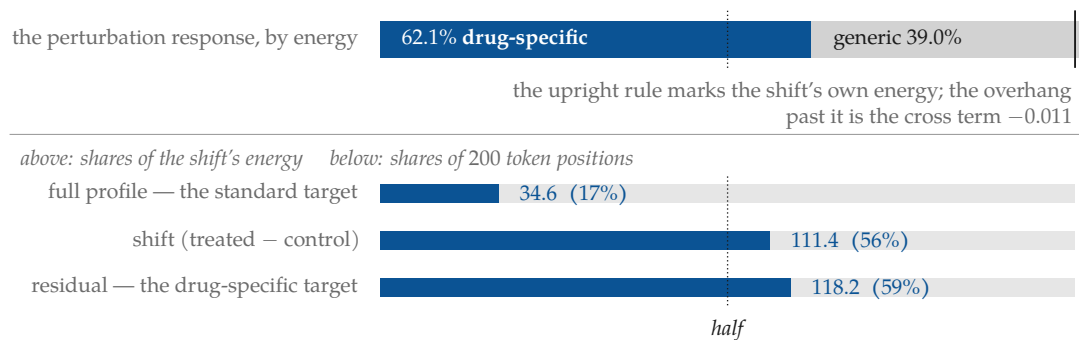


Figure 4.12. The drug is the majority of the perturbation response and almost none of the standard target's tokens. Above the rule: the shift's energy decomposes exactly into a drug-specific residual, a generic programme and a cross term, so 62.1% of what a perturbation does to a cell is specific to the compound. Below it: of the 200 positions a cell sentence spends on the standard target, only 34.6 carry a different gene for two drugs in the same context — the other 83% are the same string. Ranking the same measurement by signed residual raises that to 118.2, and at the plate scope this build trains in the residual row reads 120.3 (60%). The two halves of the figure are different quantities, an energy share and a count of token positions; they share the axis and the half-way rule and nothing else, and no difference between them is a number. What the picture states is an ordering: the signal is a majority of the response, the standard encoding exposes a small minority of it to the loss, and the residual encoding puts the exposed fraction back on the majority side (Table 4.7).

TWO COMPONENTS ASK FOR DIFFERENT WORK. Writing

$$r(d, c) = \beta(d) + \kappa(d, c) + \epsilon, \quad \mathcal{T} = \frac{\text{var } \beta}{\text{var } \beta + \text{var } \kappa}, \quad (4.8)$$

the per-drug constant $\beta(d)$, the drug *main effect*, is what a lookup table reproduces with no model at all. The remainder $\kappa(d, c)$ is the only component a conditional model could win on. This is DrEval's naive predictor [19] in transcriptomic space. Their NaiveMeanEffectsPredictor predicts $\hat{y}_{ij} = \mu_i^c + \mu_j^d - \mu$, cell-line and drug main effects with no interaction term, and $\text{generic}(c) + \beta(d)$ is that quantity lifted from a scalar potency to a 946-dimensional profile.

SEE § 4.11 builds that lookup and scores it.

Two warnings attach before any number is quoted. κ is written as a drug $\times$ cell-line interaction because the decomposition assumes one, but the estimator used does not identify such a term, so κ is called the *context-specific remainder* and $\mathcal{T}$ a disattenuated cross-context residual-similarity coefficient, not a variance share. And $1 - \mathcal{T}$ does not measure the headroom above the additive baseline. The cross-line pairs entering it may also differ in dose and in treatment well, and each residual is taken against a generic its own drug helped define, so it bounds that headroom from above at best.

DEFINITION The transfer coefficient

With 286 of 287 treatments in exactly one well there is no replicate from which to separate $\text{var } \kappa$ from $\text{var } \epsilon$, so the variance-components model is unidentified on its own. The cosine route buys identification through split-half reliability:

$$\hat{\tau} = \frac{\cos(r(d, c_1), r(d, c_2))}{\sqrt{\rho(d, c_1) \rho(d, c_2)}}, \quad \rho(d, c) = \frac{2 \cos(r_A, r_B)}{1 + \cos(r_A, r_B)}, \quad (4.9)$$

where r_A and r_B are disjoint halves of condition (d, c) and $\rho(d, c)$ is the Spearman–Brown reliability of that condition’s residual at its full cell count.

Dividing out the reliability is essential, since measurement noise alone depresses the raw cross-line cosine and would otherwise be read as interaction. Two cautions ride with it. Spearman–Brown is derived for correlations between parallel halves, and what licenses it on a cosine here is the planted-truth validation in chapter A. And the reliability entering it is a within-well split-half precision, optimistic relative to true biological reproducibility. That optimism sits in the denominator, where an over-large ρ makes $\hat{\tau}$ too small, so $1 - \hat{\tau}$ is, if anything, *overstated*. An earlier draft had that direction the wrong way round.

THE CONTROL THAT HAD TO BE REPLACED. The first run compared $\mathcal{T}$ against a negative control of different drugs on the same plate, which returned $+0.485$ against $\mathcal{T} = 0.511$, and was correctly declared void. That control differs from the estimand in two ways at once, different drug *and* same cell line, and plate structure surviving a cell-line-scoped generic inflates same-plate comparisons while leaving cross-line comparisons untouched. The $+0.485$ is voided, not matched against the separate -0.017 plate-scoped diagnostic. Only a structure-matched control, breaking the drug match alone, bounds the estimand.

CORRECTION the same-plate negative control is voided, not matched; the structure-matched control replaces it

Table 4.8. The five checks accompanying the transfer-coefficient estimate, and what each one bounds. At $\kappa = 0$ the *uncorrected* cross-line cosine is only 0.734, which read naively reports 27% interaction where the truth is zero, while the disattenuated estimator recovers the truth to within 0.001. That recovery licenses the reference column of Table 4.9.

Check	What it bounds
disjoint control halves	shared-control inflation: $+0.242$ at cell-line scope, -0.000 at plate
structure-matched negative control	the estimand itself; returns -0.008
simulated $\kappa=0$ null	estimator bias: returns 1.001 where the truth is 1.000
dose strata on molar doses	a reference scale: the same drug and line at another dose
leave-one-drug-out generic	train/held-out target asymmetry (§ 4.7)

An earlier cache carried the well identifier in the dose field, so what was labelled dose transfer compared physical wells, not concentrations.

Table 4.9. The transfer coefficient, estimated on real molar doses with dyadic cluster-robust intervals. The matched negative control sits on zero. The same-plate control that produced the earlier false alarm reads +0.485 under a cell-line-scoped generic. $\mathcal{T}$ is a cross-context similarity coefficient, not a variance share, and $1 - \mathcal{T}$ must not be read as a drug $\times$ cell-line interaction share.

Quantity	Estimate	95% interval	Reference
$\mathcal{T}$ (cross-context, plate-scoped generic)	0.553	[0.514, 0.593]	simulated $\kappa=0$ null = 1.001
similarity scale NOT shared across contexts, $1 - \mathcal{T}$	44.7%	[40.7%, 48.6%]	mixes cell line, dose, well and estimator
$\mathcal{T}$ across dose (same drug, same line)	0.707	[0.632, 0.782]	reference scale
structure-matched negative control	−0.008	spanning zero	must be ≈ 0

WHAT THE COEFFICIENT IS, PRECISELY. $\mathcal{T}$ is a descriptive sensitivity analysis and not a load-bearing estimate, and the reason sits in its configuration. The scoped calculation retains conditions at a reproducibility threshold of 0.2, and it carries every caution above at once: a *self-including* generic, pairs free to differ in *dose* and in treatment well, and an optimistic reliability in the denominator. That last one pushes $\hat{\mathcal{T}}$ down, so $1 - \mathcal{T}$ errs high and bounds from above what a per-drug constant leaves on the table, without measuring it.

DOSE IS THE Milder PERTURBATION. Holding drug and cell line fixed and changing only the molar concentration, transfer is 0.707 [0.632, 0.782] on 363 pairs against 0.553 [0.514, 0.593] pooled cross-context; the intervals do not overlap. That contrast is between one clean axis and a mixture, since the cross-context arm differs in well and dose as well as cell line, but it is the closest this section comes to a single-axis comparison.

The same ordering served as the weaker of the two scope diagnostics. At cell-line scope it inverts. The dose arm falls to 0.4846, a different quantity from the voided same-plate control, which coincidentally reads 0.4853, while the cross-line coefficient holds at 0.511. That is the wrong way round for dose against cell line, but the expectation is a prior, so it is used as a plausibility check. The second diagnostic needs no prior. At plate scope the shared-control and split-control builds of $\mathcal{T}$ agree to −0.000, while at cell-line scope they differ by +0.242, plate structure leaking through a cell-line-scoped generic. That is what the scope decision rests on, and it is why the cell-line-scoped build cannot corroborate when it returns a similar $\mathcal{T} = 0.511$; that build is the one declared void, its own same-plate negative control reading +0.485.

WHETHER ANYTHING COULD LEARN THE REMAINDER. If a cell line modulated drug responses in a consistent direction, that direction would be estimable from the line's own cells. Writing $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$, the test compares $\cos(\kappa(d, c), \kappa(d', c))$ for different drugs in the same cell line against the same quantity across different cell lines, with $\hat{\beta}$ estimated leave-one-cell-line-out; estimated over all lines it would contain $r(d, c)$ itself, and subtracting a mean

containing its own term induces a $-1/(m-1)$ correlation that biases the signal arm downward by construction. Both arms are disattenuated by κ 's own split-half reliability.

Averaged over lines, no consistent per-line direction shows up. Under the plate-scoped generic the same-cell-line arm reads -0.0149 $[-0.0251, -0.0050]$ and the different-cell-line arm $+0.0003$ $[-0.0071, +0.0081]$, an excess within line of -0.0153 for which no interval was computed. The intervals come from a cell-line-clustered bootstrap over 38 and 42 clusters, on 4,000 same-line and 3,884 cross-line pairs over 1,282 conditions. Neither arm shows the positive within-line excess a per-line direction would require, and the negative sign of the same-line arm is not read as evidence: same-line pairs that also share a plate share a generic, and that common subtraction pushes their cosines negative by construction, where cross-line pairs share no generic at all. It is the construction the different-drug same-plate diagnostic reads at -0.017 above. A cell-line-scoped counterpart was not computed in the canonical run, and the row an earlier draft carried for it is withdrawn.

Two negative results do not add up to irreducibility: this one, and a direct test of whether mechanism-matched drugs deviate from their per-drug constants in the same way within a cell line, which returned nothing detectable. The second bounds its own power severely. κ 's split-half reliability in this atlas is 0.195 at the 0.05 reliability floor (0.324 at the stricter 0.20 floor), and same-mechanism drugs are so heavily co-plated that only 3–7% of matched pairs survive a different-plate requirement (1% at the stricter floor). Resolving κ 's structure would take more cells per condition or a plating design that crosses mechanism with plate.

THE CLAIM. There is a conditional target that a per-drug constant cannot express. Two tests looked for learnable structure in it and found none, and neither settles whether there is any. κ 's split-half reliability is 0.195 at the floor the mechanism test runs at, and only 3–7% of mechanism-matched pairs survive a different-plate requirement, so its structure is unresolved here rather than shown to be zero. Its size is quoted as what it is: $\mathcal{T} = 0.553$ $[0.514, 0.593]$ is a disattenuated cross-context similarity coefficient, so 44.7% $[40.7\%, 48.6\%]$ of *that similarity scale* remains unshared across the compared contexts. It is not a variance share and not a drug $\times$ cell-line interaction, since the estimator pools pairs differing in cell line, dose, well and target estimator, so it bounds what a per-drug constant leaves on the table without establishing that a model could reach any of it.

4.9 Re-encoding produces prompt sensitivity



Every target change tried so far left the model indifferent to the drug named in its prompt. Does training on the residual change that? The instrument is the stratified scramble, the same prompt with another drug's name and mechanism substituted, read as a ladder of strata because the swapped-in drug is itself a drug the model knows and so is not a neutral comparison. The model moves. It is not the only target change that makes it move.

On the 300 trained conditions of the residual-frame evaluation the model beats the geometry-defined orth scrambled partner by $+0.0898$ [$+0.0291$, $+0.1506$], clustered on the drug and the treatment well (`re_v3.json`). That is the first non-null drug effect in this project, first in the order the experiments were run, and not the only target construction that produces one.

WHY THE COMPARISON SET IS NOT EXCHANGEABLE, AND WHAT FOLLOWS. The comparator is not a null. Its own score is a monotone function of which drug it was handed, which makes it an active treatment; § 4.10 measures the size of that. Two consequences fix the reading protocol here. The series is quoted as a *gradient*, since a metric artefact would not order itself by swap dissimilarity; and where a single magnitude is needed it is the orth one, the least contaminated comparison available and not an independently established null, that stratum having been defined using true residual geometry.

SCRAMBLE The same prompt with the drug's name and mechanism replaced by another drug's, in three strata by residual similarity.

WHY THE STRATA ARE CUT BY RESPONSE GEOMETRY AND NOT BY MECHANISM. A scramble that swaps in a drug with a similar true response is a weak test, and the annotation that looks like it should prevent that does not. In the characterisation behind Table 3.1, two drugs sharing a `moa-fine` annotation sit a mean distance 2.322 apart against 2.377 for two drugs that do not, 1,244 same-mechanism against 17,563 different-mechanism pairs, a ratio of 0.977. Same-mechanism drugs in Tahoe are just over two per cent closer to each other than arbitrary pairs are, so “different mechanism” does not imply “different response”. The strata are therefore cut by measured residual cosine within the cell line. (Mechanism does carry signal in the residual frame, § 4.12, a different frame from the full profiles measured here.)

THE GRADIENT, ON THE EARLIER TARGET BUILD. The stratified scramble of § 4.2 gave a gap that grew with swap dissimilarity (Figure 4.13), $+0.0351$ [$+0.0061$, $+0.0656$], $+0.0716$ [$+0.0371$, $+0.1059$] and $+0.1429$ [$+0.1112$, $+0.1787$] across near, orth and opposite, each differenced against its own stratum-specific partner. These are the residual arm's 1400-token run; the two control arms beside them were scored at the 600-token cap of chapter 3, where their stratum gaps run from -0.021 to $+0.026$ and were read at the time as flat and null. That reading holds for the single-cell arm at both budgets. It does not hold for the optimal-transport arm, whose 600-token gaps are themselves monotone in swap dissimilarity.

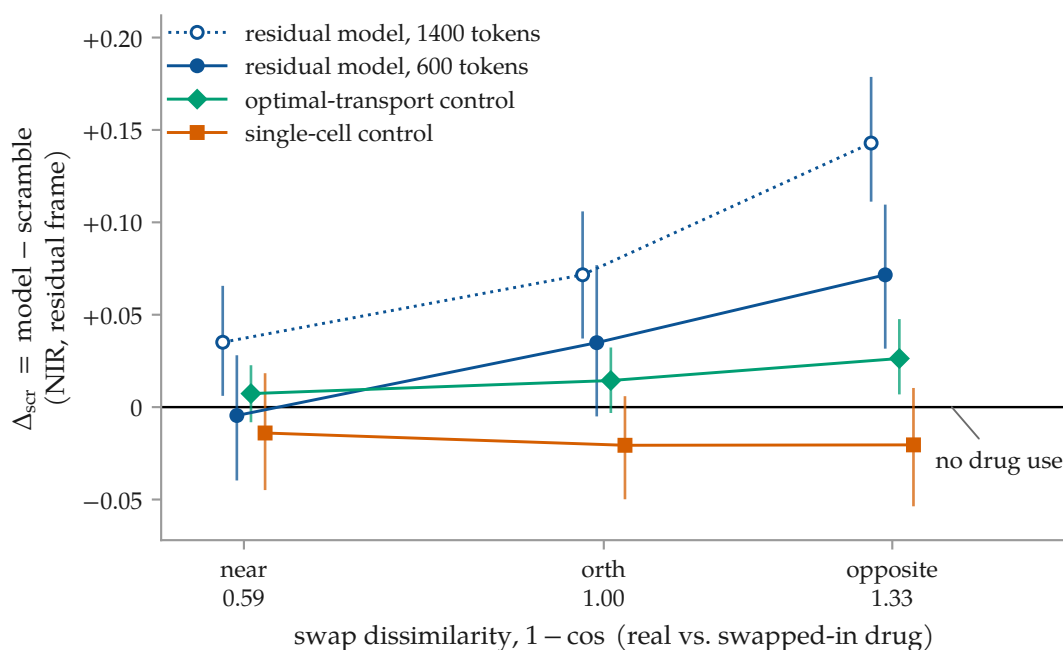


Figure 4.13. On the earlier target build, the residual model’s advantage over its own scramble grew with how dissimilar the swapped-in drug’s residual was; the single-cell control scored through the identical pipeline stayed flat and null, while the optimal-transport control did not. Residual frame throughout. The horizontal axis is measured swap dissimilarity, $1 - \cos$ between the real and the swapped-in drug’s residual, byte-identical across all four runs, so the three stratum positions are exact; the connecting lines are guides between three measured strata and not a sampled continuum, and the four series are nudged sideways so that their intervals do not overlap. The three solid series are at a matched 600-token budget; the residual model’s 1400-token run is the dotted overlay, since the budget alone roughly doubles the gap on the two strata where it is positive at 600 ($+0.0349 \rightarrow +0.0716$ on orth, $+0.0716 \rightarrow +0.1429$ on opposite). Intervals are 95% bootstrap clustered over cell lines, $n = 250$ conditions per point. The near stratum is the weakest test by construction, since where the swapped-in drug’s true residual is similar, an unchanged prediction is correct.

THREE DIAGNOSTICS THAT NEED NO SCRAMBLE. All three agreed on that build, and the residual model led on all three. The correlation between a condition’s reproducibility and model NIR was $+0.144$, the model doing better where the drug effect is real, against -0.117 and -0.216 for the controls. Prediction diversity was highest for the residual model. And $\cos(\text{prediction}, \text{own truth})$ exceeded $\cos(\text{prediction}, \text{other drugs})$ for all three arms, the residual model’s margin, $+0.084$, being roughly twice the optimal-transport arm’s $+0.049$ and three times the single-cell arm’s $+0.026$.

ONE MEASUREMENT LESSON BEFORE THE WITHDRAWAL. The 600-token cap roughly halved the effect at every stratum where it was positive there, so $+0.0716$ appears twice in this section, once as the 600-token opposite gap and once as the 1400-token orth gap, and the two are not the same quantity. Re-run at 1400 tokens, the single-cell control stays flat, $-0.0128 [-0.0458, +0.0202]$ against its own orth partner. The optimal-transport control does not.

✕ **CORRECTION 1** | “OT achieved nothing”, § 4.6

At 600 tokens the optimal-transport arm’s orth gap was $+0.0144$ $[-0.0051, +0.0338]$ and spanned zero; at 1400 it is $+0.0292$ $[+0.0144, +0.0441]$ and does not. The token budget was suppressing a real effect in that arm, the same lesson the residual arm taught, applied to a control previously read as inert. The 600-token evidence pointed the same way and was read too cautiously; that arm’s stratum gaps there were $+0.0073$, $+0.0144$ and $+0.0263$, monotone in swap dissimilarity like the residual arm’s, with only opposite excluding zero. The consequence reaches backwards. The optimal-transport target is one point in the three-point comparison of § 4.6, and that negative holds only at the budget and comparator it was measured with. At a matched budget and against the geometry-defined orth partner, whose true residual is on average orthogonal to the target’s ($\cos = +0.035$), the optimal-transport point is not null. The single-cell point is the one whose interval still spans zero. The geometry-based selection does not independently establish a null in either case.

THE COMPARISON THAT COSTS THIS SECTION ITS STRONGEST PHRASING. The two arms’ pooled gaps against their own orth partners overlap almost entirely (`ctrl1400_summary.json`), so re-encoding the target produces prompt sensitivity but is *not* the only target change that does.

Three limits stop that comparison being read as an ordering. The arms are scored in different frames, so only each arm’s gap against its own comparator is comparable; they are separate evaluations over different condition sets, so this is an arm-level and not a paired comparison; and those sets differ in split composition by more than the arms differ from each other. The residual figure pools the three splits of Table 4.10, whose own gaps are $+0.0898$, $+0.0332$ and -0.0315 , while the optimal-transport evaluation carries no split labels at all. A spread of that size across splits, against a 0.0069 difference between the two pooled numbers, means the near-equality does not order the arms in either direction.

THE SLOPE CARRIES ITS OWN NULL. Regressing each prediction’s NIR on the cosine between the named drug’s truth and the target’s truth gives a slope whose null is zero for a model that ignores the drug it is told. Its points are the three scrambled partner prompts and nothing else, the target-drug prompt never among them, and § 4.10 fits it on all three splits. The fit is pooled least squares over every point in a split with a single intercept, three points per condition, no condition-level intercepts absorbed, so the zero null is exact within a condition while the pooled slope also carries a between-condition component. What it closes is the freedom to pick a comparator, since it spends all three strata rather than one.

THE CLAIM. Re-encoding the target produced the project’s first non-null drug effect. On trained conditions the model beats the geometry-defined orth partner by $+0.0898$ $[+0.0291, +0.1506]$ and beats chance outright, with no comparator arm at all, by $+0.0981$ $[+0.0407, +0.1554]$, both clustered on the drug and the treatment well, the drug being the axis the claim

rides on (§ 4.10). The comparator-free slope of retrieval against the named drug's truth is $+0.3728 [+0.2797, +0.4660]$, and zero is its null for a drug-blind model, so no choice of comparator manufactures it, though it is fitted over the scrambled partner prompts alone, with the target-drug prompt never a point on the line. It is first in sequence and not uniquely responsible: at a matched 1400-token budget the optimal-transport target clears zero against its own orth partner too ($+0.0292 [+0.0144, +0.0441]$ over 248 conditions that carry no split labels, against the residual arm's $+0.0361 [+0.0159, +0.0564]$ pooled over 1,394 that mix trained and held-out splits), on a separate evaluation in a different frame, so the two are not ordered against each other. The model has been shown to respond to the drug; whether that response is any use is the next question.

4.10 Responding to the drug is not predicting it



A model that moves when the drug token moves has cleared a low bar, because reproducing a training average requires that much too. Everything scored so far came from conditions the model trained on, so what has been shown is that the token is consulted, not that anything transferable was learned. Separating the two needs a holdout, and the atlas's own randomisation decides which one means anything: whole treatment wells. Prompt sensitivity survives that test. Whether the residual itself transfers comes back unsettled.

The physical layout of chapter 3 forces the choice. A (drug, cell line) holdout, the field's standard Leave-Pairs-Out split and the one an earlier version of this work used, leaves the held-out condition's treated well sitting in the training set through its other ≈ 48 cell lines. Measured against that holdout, 124 of the 255 treated wells for which split information exists placed a held-out condition and a training condition in the same well. That estimand measures within-atlas matrix completion, not independent generalisation, and no processing of the targets changes it.

Holding out whole wells is the only assignment-respecting choice the design offers, and it answers a narrower question, generalisation to an unseen treatment well of a seen drug. Fixed-dose cross-context transfer, present in one line and absent in another, is structurally unidentifiable in this atlas.

WHAT EACH SPLIT HOLDS. train takes every condition of the remaining wells, 4,999 before the reliability line and 2,523[?] **AUDIT: TALLY AGAINST THE GATE'S RETAINED COUNT** after it. unseen_combo holds 36 treated wells out whole (904 conditions, 900 after inventory filters), and it never takes a drug's last training well; without that guard a two-well drug could silently become an unseen drug while sitting in the unseen-combination arm, and the two arms would measure the same thing. unseen_drug holds every condition of 10 held-out drugs (714 conditions, 711 after inventory filters). Zero wells contribute conditions to both sides.

SEE § 4.11: what a held-out gap has to beat.

unseen_drug is a control and not a third result. Tahoe fine-tuning never sees the token, so a large effect there would indict the comparator, not report a biological finding. “Unseen” means held out from fine-tuning with mechanism supplied, not unknown to the pipeline; the Pythia-1b backbone was pretrained on natural language and the prompt still carries the Mechanism: field.

Two warnings ride with the design. DrEval reports, citing [23], that apparently good performance on pair-holdout splits can be reached simply by predicting each drug’s average response across the training cell lines, which is why the baseline ladder of § 4.11 is not optional. And the evaluation draws under a per-split quota, because a uniform draw reproduces the pool’s proportions and starves the split the experiment exists to test; in a first run, 250 constructed held-out combinations yielded only 33 scored conditions. Under the quota the evaluation scores 1,394 conditions in all, 300 trained and 199 unseen-drug, the rest unseen combinations, almost the whole pool of 900 that arm affords, while unseen_drug remains a quota of the 711 available.

THE FIRST RESULT OF THIS DESIGN IS ABOUT THE COMPARATOR. It is a finding and not an erratum, and it reframes every gap quoted so far. Across the three swap strata the scrambled model’s NIR is 0.550, 0.501 and 0.423 as the named partner’s true residual moves from aligned ($\cos = +0.22$) through orthogonal (0.00) to anti-aligned (-0.20) with the target. The near and opposite scores differ from chance in opposite directions; the geometry-defined orth partner scores 0.501 and is empirically near chance on these conditions. That makes orth the least contaminated available comparator, not an independently established null, since its stratum was defined using true residual geometry.

The consequence is structural. A gap measured against such an arm decomposes as

$$\Delta_{\text{scr}} = \underbrace{(\text{NIR}_{\text{model}} - 0.5)}_{\text{how much naming the truth helps}} + \underbrace{(0.5 - \text{NIR}_{\text{scramble}})}_{\text{how much naming the lie hurts}}, \quad (4.10)$$

and only the first term is wanted. The second is not a rounding error. Pooled over all 1,394 scored conditions the model’s own advantage over chance is $+0.0367$ and the opposite comparator’s deficit is $+0.0772$, so 68% of the $+0.1139$ gap that arm reports is the comparator failing and not the model knowing. Against orth the second term is approximately zero here, so every gap in Table 4.10 is measured against that partner, its magnitude conditional on the observed near-chance score.

The diagnostic is free and not specific to this experiment. Score the comparator arm against chance; if its score moves with which label it was handed, the gap it defines is not a measurement of the model. A swap drawn to be maximally dissimilar, the obvious way to build a strong negative control, is exactly the swap that maximises the second term.

Table 4.10. Whole-well holdout on the corrected evaluation, with the truth verified against the build's own fit digest and the full held-out population scored. The two right-hand columns carry the SAME point estimate and differ only in what they treat as the unit of generalisation. On `unseen_combo` the gap excludes zero clustered on cell line and does not clustered on drug, which is why that row is reported as not established. The `unseen_drug` row corroborates but does not prove: its orth-partner gap is negative while its opposite gap is +0.0379.

Split	<i>n</i> (lines; drugs; wells)	model NIR	Δ_{scr} , clustered on <i>cell line</i>	the same, clustered on <i>drug</i>
train	300 (39; 76; 136)	0.598 [0.549, 0.647]	+0.0898 [+0.0382, +0.1415]	+0.0898 [+0.0291, +0.1506]
unseen_combo	895 [?] (42; 34; 36)	0.533 [0.492, 0.575]	+0.0332 [+0.0069, +0.0594]	+0.0332 [−0.0031, +0.0695]
unseen_drug	199 (40; 10; 28)	0.459 [0.404, 0.514]	−0.0315 [−0.0836, +0.0206]	−0.0315 [−0.1049, +0.0419]

Every gap printed here is measured against the geometry-defined orth partner, whose own score is empirically near chance on these conditions; opposite figures never appear as a column. Model NIR intervals test against the chance value 0.5 with no comparator. Intervals are two-way cluster-robust over cell lines and wells, with a Student *t* reference on $\min(G_{\text{line}}, G_{\text{well}}) - 1$ degrees of freedom, which is why the well counts are printed. The `unseen_drug` arm spans 28 wells, and $t_{27} = 2.05$ against 1.96 widens its interval by five per cent (chapter 3).

THE CONTROL BEHAVES AS THE DIAGNOSIS PREDICTS. The unseen-drug gap against opposite is +0.0379 [−0.0083, +0.0841], positive where the orth-partner estimate on the same conditions and predictions, −0.0315 [−0.0836, +0.0206], is negative and covers zero as a control should. That point estimate is the `unseen_combo` gap with the sign reversed, so the row is read as uninformative and not as a confirmed null. A cluster-robust variance is asymptotic in the number of clusters, not conditions, and this arm spans 40 cell lines but only 28 treatment wells, so its two-way variance is carried on 27 degrees of freedom. A cell-line-only cluster bootstrap puts the opposite gap clear of zero at [+0.0044, +0.0727]; adding wells as the second clustering axis returns it to zero, at the normal quantile as well as at $t_{27} = 2.05$. What decides the row is the unit of clustering.

THE HELD-OUT ROW SETTLES PROMPT SENSITIVITY ONLY. On `unseen_combo` the gap clears zero clustered on cell line and well but not clustered on drug and well (Figure 4.14a, right), so it survives only one of the two units the claim needs. Drug is the axis a claim about transferring a *drug*-specific residual has to generalise across, so generalisation to an unseen treatment well of a seen drug is not established.

SCOPE LIMIT | carried into § 4.11 and chapter 5

The precision matters: not established is not the same as shown to be zero. The drug-clustered interval reaches +0.0695, so an effect of up to roughly +0.07 remains compatible with these data at the 895[?] **AUDIT: ARM SIZE, TWO RECORDS DISAGREE** conditions this arm affords, spread over 34 drugs, a limit of power at the size the holdout gives, reported as one. On `unseen_drug`, $n = 199$ leaves a half-width of 0.055 around the model's own NIR and an 80%-power detection limit of 0.079

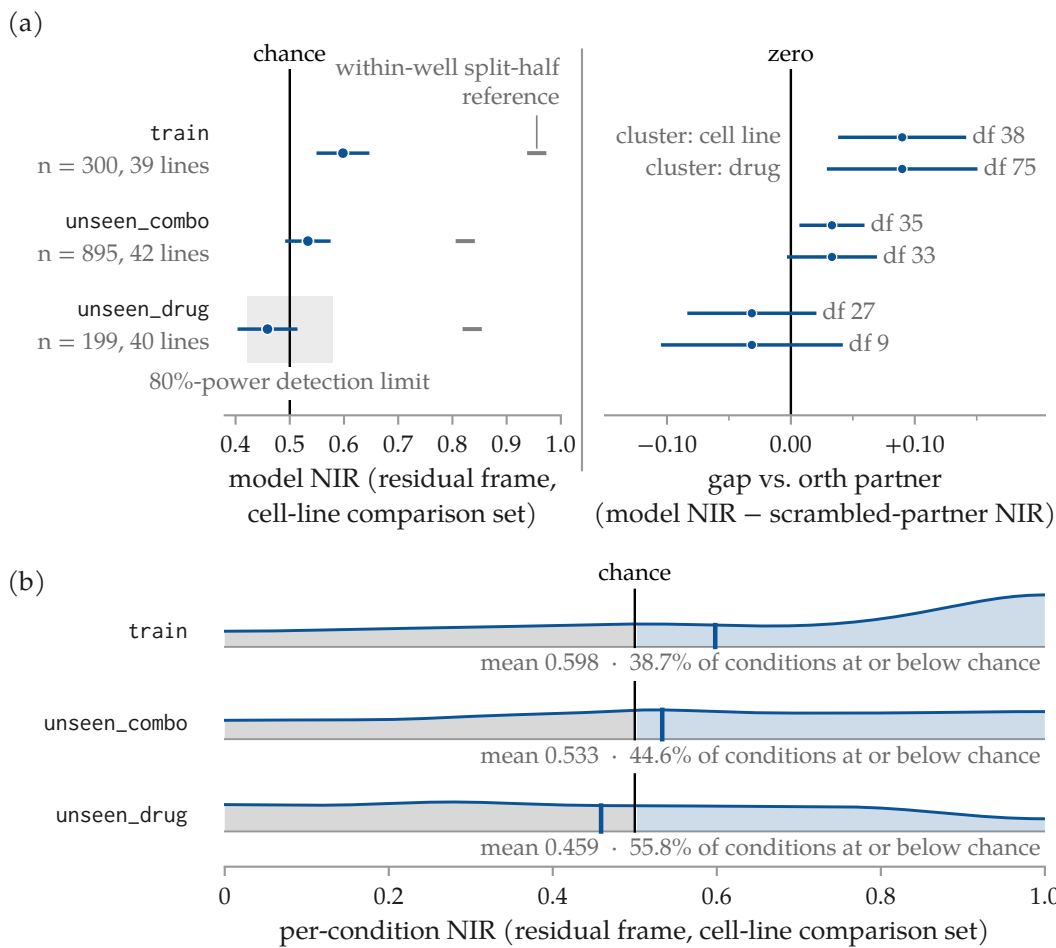


Figure 4.14. The residual-frame model clears chance on trained conditions only; the held-out-combination gap excludes zero clustered on cell line but not clustered on drug, and the unseen-drug gap flips sign. Residual frame throughout, and NIR ranks each prediction against the other drugs measured in the same cell line. (a) Left, model NIR against the chance value 0.5 (solid rule), with 95% two-way cluster-robust intervals; $n = 300$, 895[?] and 199 conditions on train, unseen_combo and unseen_drug. The grey dash on each row is that split's raw within-well split-half precision reference, a measurement-precision figure computed from two halves of the same well and not a replicate experiment; it carries no interval and is drawn as a bare marker, and only train is filtered at the reliability line. Right, on its own axis, the gap against the geometry-defined orth partner. Each row has one point estimate, drawn twice with two intervals because the verdict depends on which unit of generalisation is clustered: cell-line-clustered above, drug-clustered below, each labelled with the Student t degrees of freedom it is carried on. The grey band on the unseen_drug row is that arm's 80%-power detection limit around chance, ± 0.079 , the smallest departure the arm could detect and not a confidence region. (b) The per-condition NIR distributions behind those three means, as reflected-boundary kernel densities on a shared axis, each with its mean marked and the region at or below chance shaded. The pooled 0.537 is a mean over nearly flat distributions rather than a location: even on train, 38.7% of conditions sit at or below chance, against 44.6% and 55.8% on the two held-out splits.

against chance, shaded on that row in Figure 4.14a. Neither row licenses a statement that the effect is absent.

The three splits are not scored on comparable populations either, and the difference is one of selection. Their raw within-well split-half precision references, the grey dashes of Figure 4.14a, are 0.956, 0.824 and 0.836, a spread of 0.132; train is filtered at a reliability line the two held-out splits are not, and imposing that line on all three brings the references to 0.955, 0.961 and 0.960, a spread of 0.006. The held-out shortfall therefore cannot be softened by appeal to a lower noise floor.

× **CORRECTION 2** | a training-exposure or memorisation advantage

The difference between the trained and held-out estimates is $+0.0567 [-0.0014, +0.1148]$ with cluster-robust $p = 0.0556$. An unclustered permutation returns $p = 0.0142$, quoted only for contrast, because ignoring the clustering this chapter mandates everywhere else flatters the result. It does not clear the conventional threshold, and the two arms differ in three ways at once in any case, in whether the condition was trained on, in which drugs they contain, and in how their scored populations were selected. Restricting the comparison to the 30 drugs present in both arms reverses the sign, with the trained arm sitting *behind* the held-out one, $-0.0606 [-0.1506, +0.0295]$, $p = 0.1797$, on 79 trained and 750 held-out conditions, and the pooled difference is a composition effect: 83.8% of held-out-combination conditions sit on those shared drugs against 26.3% of trained ones. The claim of a training-exposure or memorisation advantage is therefore withdrawn. None is established, the drug-matched estimate points the other way, and the two arms are scored on populations selected by different rules.

TWO INSTRUMENTS NO COMPARATOR DEFECT CAN TOUCH. First, the slope test. Regressing each prediction's NIR on the cosine between the named drug's truth and the target's truth gives $+0.373 [+0.280, +0.466]$ on train, $+0.336 [+0.223, +0.450]$ on unseen_combo and $+0.342 [+0.215, +0.469]$ on unseen_drug, clustered on the drug. All three exclude zero, and all three still exclude it clustered on the cell line ($[+0.272, +0.474]$, $[+0.241, +0.432]$ and $[+0.284, +0.400]$). Its points are the three *scrambled* partner prompts and nothing else, exactly three per condition, so 900, 2,685 and 597 points on the three splits. On unseen_drug those partners are overwhelmingly drugs the model *was* trained on, so the slope shows that the model responds to a named trained drug in a scoring context whose own drug was held out, and not that it knows the held-out drug.

Second, the model's own score against chance, with no comparator at all. It clears 0.5 on train, by $+0.0981 [+0.0407, +0.1554]$ clustered on drug and well, and is not distinguishable from chance on either held-out split (Table 4.10).

A POSITIVE SLOPE BESIDE A CHANCE-LEVEL SCORE IS NOT A CONTRADICTION. The model has a coarse sense of what a named trained drug does, and not of what it does in this well, and NIR asks the second question, ranking against the ≈ 40 other drugs in the same cell line. That is the drug main effect without the part of the residual that is not shared across contexts, which § 4.8 measures

at 44.7% ($1 - \mathcal{T}$, with $\mathcal{T} = 0.553 [0.514, 0.593]$), a figure that mixes cell line, dose, well and estimator. It is also why the lookup wins in the next section: `drug_lookup` is that main effect, estimated straight from data far more precisely than the model learned it. The condition-level picture agrees. The cosine between prediction and own truth on trained conditions averages $+0.0389$ (sd 0.1119 , median $+0.0220$), spanning $[-0.3046, +0.3976]$ with 33.7% clearing $|0.1|$.

ONE ASYMMETRY ATTACHES TO EVERY HELD-OUT NUMBER IN THIS CHAPTER. The evaluation's truths are not refitted. The generic is reconstructed from the build's own fit inventory and verified against the build's `fit_digest`, and a mismatch aborts the run. The reliability line selects the *training* split, so train is by construction the reliability-surviving population, and the two held-out splits are not filtered at all. In the unseen-combination arm 499 conditions sit below the $+0.1086$ line and 74 have a negative split-half cosine; in the unseen-drug arm, of 199 conditions, 88 sit below the line and 18 are negative. This is deliberate, since filtering the evaluation set on its own outcome is the selection this chapter removed, but it leaves the comparison conservative in the direction that matters. The held-out arms are scored on their full, noisier populations while the trained arm is not.

THE CLAIM. Re-encoding the target made the model respond to the combined drug-and-mechanism prompt. That much is established and survives every stress: on trained conditions it holds clustered on the cell line and on the drug, it holds against chance with no comparator at all, and the comparator-free slope excludes zero on all three splits under both clusterings ($+0.373 [+0.280, +0.466]$, $+0.336 [+0.223, +0.450]$, $+0.342 [+0.215, +0.469]$).

Transfer to held-out combinations is not established. The gap there is $+0.0332$. Clustered on cell line and well it excludes zero, $+0.0332 [+0.0069, +0.0594]$; clustered on drug and well it does not, $+0.0332 [-0.0031, +0.0695]$. It is the drug axis that the claim has to survive. The split is also weaker than its name: 30 of its 34 drugs and 39 of its 42 cell lines appear in training, and 82% of its conditions recombine a seen drug with a seen cell line. The effect is a majority tilt rather than a uniform one, positive for 20 of 34 drugs (sign test $p = 0.39$).

What the slope on the *unseen-drug* split does and does not add has to be stated exactly. At $+0.342 [+0.215, +0.469]$ it shows that the prediction still tracks the drug *named in the prompt* when the condition's own drug was held out. But the drugs named in that regression are the three scrambled partners, which on this split are overwhelmingly training drugs, and the real target-drug prompt contributes no point to it. The slope is evidence that the model responds to named trained drugs in an unseen-drug scoring context, and not evidence that it knows the held-out drugs.

4.11 A lookup table wins

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▲ RESID

What does a drug-response prediction have to beat before it is worth having? The instrument is a ladder of predictors scored through one pipeline, the lowest rung of which needs no model at all, only a table of what each drug did elsewhere. That table wins.

VERDICT The model reads the drug and recovers a small fraction of its residual.

A model can respond to the drug in its prompt and still predict badly, so the pre-registered bar was never “beats chance”. Residual targets are plausibly memorisable, and a predictor that looks up what this drug did elsewhere needs no model. Two objections to § 4.2’s drug-blindness verdict come first, and neither needs a checkpoint.

IS THERE ANYTHING TO KNOW?. A ladder of mean-shift baselines, each predicting the average shift of a progressively narrower group, answers that. It is scored in the expression frame on $DE-\Delta r$, so its rows may not be set beside any NIR row below. On tier 2 the global mean scores 0.056; cell line raises it about two-and-a-half-fold to 0.142; mechanism does not move it (0.052); both together are no better than cell line alone (0.120 against 0.142). Tiers 1 and 3 agree. The informative conditioning variable here is the cellular context, not the drug class, the same asymmetry § 4.4 finds inside the model’s activations. Mechanism does carry signal once the generic programme is subtracted (§ 4.12); in full-profile space the generic response swamps it.

LOOKUP Predicts this drug’s residual as its mean residual in the *other* cell lines.

DOES THE ENCODING KEEP THE DRUG?. A sentence keeps only the rank order of genes. If the drug lived in the discarded magnitudes, this chapter’s diagnosis would be attacking the wrong thing. It does not. At single-cell resolution, cosine, Pearson and Spearman similarities on true normalised expression all sit within 0.04 of chance (0.529–0.531 [0.515, 0.543]), the same near-chance regime as rank. What recovers the signal is aggregation. At fifteen cells the control-referenced shift cosine reaches 0.793 [0.775, 0.809]. So the re-encoding of § 4.9 cannot be read as restoring magnitudes; it keeps rank throughout and changes only what is ranked. The same measurement also bounds what a single cell can deliver. The single-cell same-drug versus different-drug gap is only +0.006 [+0.003, +0.008], so these instruments do not resolve drug identity from one observed cell at this sequencing depth and 946-gene panel.

WHAT MAKES THE LADDER FAIR. Each baseline in Table 4.11, set out against chance and the reference in Figure 4.15, predicts the drug-specific residual through one pipeline, on the same truths, encoding and comparison set. Two disciplines keep it honest. The lookups are fitted from training conditions only; fitted from the full cache, a lookup scoring a held-out condition would read residuals the model was never shown, an oracle and not a baseline, so that construction is excluded. And `drug_lookup_1` averages a chosen cell line’s conditions, because 62% of drugs are multi-dose and a single condition would confound “different cell line” with “different dose”. `moa_lookup` is not an oracle

in any split either. Its partner pool is restricted to *training* conditions in the same cell line, so a held-out condition never enters the prediction made for it. The unrestricted-pool defect audited in § 4.12 was in the channel gate; the restriction was already in force here, so the $n = 218$ row is on the restricted side of that repair.

DEFINITION Coverage of the reference

A predictor's *coverage* is the fraction of the distance from chance to the within-well split-half precision reference that it closes,

$$(\text{NIR}_{\text{arm}} - 0.50) / (\text{NIR}_{\text{ref}} - 0.50).$$

Coverage above 100% therefore means the arm scores above the reference, a statement about the reference and not a headroom estimate. Reference and coverage are computed on the same support as the arm they describe.

Table 4.11. A training-only per-drug lookup is the bar to beat. Reference, model and drug-agnostic arms use all 1,394 conditions; the drug lookups exist on a common support of 1,192, where model and reference score 0.549 and 0.857. There, model minus drug_lookup is -0.3631 $[-0.3982, -0.3280]$; the lookup scores 0.913 pooled, 0.980 on trained conditions and 0.890 on held-out wells, where its same-well advantage is absent. moa_lookup has narrower support ($n = 218$) and is not separated from the model ($+0.0436$ $[-0.0480, +0.1352]$); its pooled row must not be compared with rows on other supports.

Arm	What it knows	NIR	Note
within-well split-half reference	disjoint cell halves, one well	0.854	precision, not a replicate; 0.857 on the lookups' support
drug_lookup (training-only)	drug's mean residual, other lines	0.913	the bar to beat; $n = 1,192$
drug_lookup_1	the same, from one other cell line	0.822	still beats the model; same support
moa_lookup ($n = 218$)	drugs sharing this drug's MoA	0.552	narrower support; not separated
model	control cell + drug	0.537	0.549 on the lookups' $n = 1,192$
control_copy	nothing (copies the control)	0.506	drug-agnostic
generic	nothing (predicts the average)	0.500	the zero vector
random	nothing	0.501	floor

Rows on different supports are not comparable. Chance is 0.50 in the ▲_{RESID} frame under the exchangeability condition audited in § 4.1.

THE PRE-REGISTERED MIDDLE BRANCH FIRED. Of the three branches written before the result, the middle one fired. The model reads the drug and recovers only a small fraction of its residual; transfer to held-out combinations is not established (§ 4.10). Whether its sensitivity to the prompt reduces to the Mechanism: field is not settled here. On the 218 conditions where that compari-

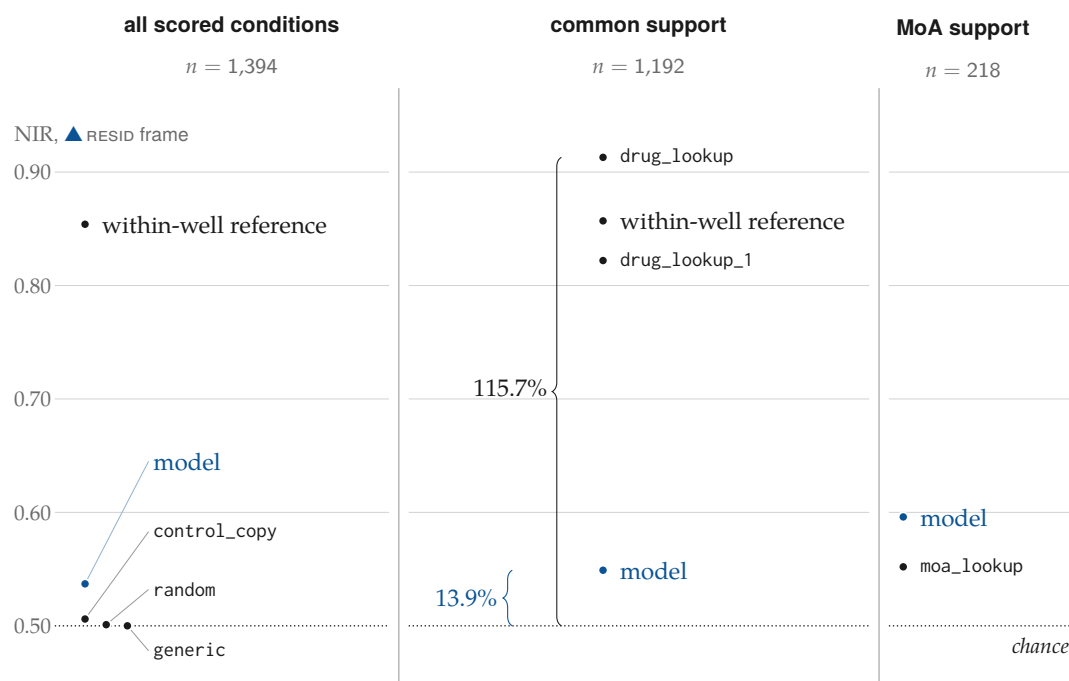


Figure 4.15. A predictor that needs no model, no prompt and no checkpoint scores above the reference the model is measured against, and the three columns may not be read across. Marks are NIR in the ▲ RESID frame, where chance is 0.50. Each column is scored on its own support ($n = 1,394, 1,192$ and 218), so a mark in one may not be set beside a mark in another; even the reference differs between the first two. On the common support the model covers 13.9% of the distance from chance to the within-well split-half precision reference, and a training-only per-drug lookup covers 115.7% of it, which is a property of the reference and not headroom (Table 4.11). On the MoA support the model scores 0.596 against moa_lookup's 0.552, a $+0.0436$ the interval does not separate from zero. No interval is drawn: the source gives them on differences between arms, not on the level of one arm, and those are in Table 4.11. *Reading the labels:* monospace names an arm exactly as Table 4.11 names it, roman names the reference in prose, the accent marks the model, and there is no fourth style. A label a rule would strike or a neighbour would crowd is lifted clear on a hairline; the dot carries the value. Horizontal position inside a column means nothing, and arms whose scores nearly coincide are dodged sideways so each keeps its own dot.

son exists, $\text{model} - \text{moa_lookup} = +0.0436 [-0.0480, +0.1352]$, clustered on cell line and well, an interval spanning zero.

THIS IS DR EVAL'S FINDING, IN TRANSCRIPTOMIC SPACE. The finding travels furthest of anything here, needing no checkpoint, no prompt and no cell-sentence encoding. DrEval reports that deep learning models barely outperform a naive predictor of mean drug and cell-line effects [19], and our drug_lookup is the $\beta(d)$ term of that additive model, measured where the generic(c) term has already been subtracted. On a 946-gene transcriptomic response, not a scalar potency, that drug main effect reaches 0.913, above the 0.857 reference itself. DrEval's inherited warning [23], that good pair-holdout performance is reachable by predicting each drug's training average, is here the hypothesis. It holds.

SEE § 4.8 measures what that constant leaves out

THE MISSING HEAD-TO-HEAD, AND WHY THE BAR SURVIVES IT. A third objection is that this chapter’s model is the wrong one. A conditional model built for perturbation response [9, 8, 10] was never run on this split, so the head-to-head a reader wants is missing, and that is a real gap. But the ladder is architecture-free, so a competitor can be dropped in and scored, and what it has to clear for a seen drug in a new context is `drug_lookup`, 0.913 pooled, or 0.890 on the held-out split. Not 0.500. For an unseen drug no lookup exists, so that regime needs a different bar and gets one in § 4.12.

SCOPE LIMIT | carried into § 4.12 and chapter 5

The two regimes must not share a number. `drug_lookup` is the bar for a *seen* drug in a new context. No lookup exists for a drug that never appeared in training, so nothing here sets a bar for that regime. Neither may rows on different supports be combined: 1,394, 1,192 and 218 select conditions by different rules, and a distance to the reference is comparable within a support and not between supports.

WHY THE LOOKUP BEATING THE REFERENCE IS NOT HEADROOM. On the common support the lookup scores *above* the reference, $\text{drug_lookup} - \text{reference} = +0.056$, tempting to read as “the modelling target has vanished”. Three reasons, all favouring the lookup, block that inference. The reference is scored against a half-sample truth and every baseline against the full-sample truth; `drug_lookup` averages ≈ 38 conditions against that single noisy half; and NIR near 0.91 compresses most of the cosine range into a few thousandths. The coverage figures are themselves distances to a split-half reference chapter 3 calls optimistic; that optimism sits in the denominator and if anything makes both coverage figures too small, while the half-sample truth of the first reason pushes the other way. The net magnitude is unknown and the ordering carries the claim.

AVERAGING IS NOT WHY THE LOOKUP BEATS THE MODEL. Of those three reasons only the averaging could also deflate the comparison with the model, and `drug_lookup_1` tests it by drawing on one other cell line. It scores 0.822, and paired on the same support the model trails it by -0.2723 [-0.3074 , -0.2372], clustered two-way on cell line and well. This drug’s mean residual in one other cell line beats a 1B-parameter model handed this condition’s own control cells. Neither gap is an average over a mixed population (Figure 4.16): condition by condition, the model scores above `drug_lookup` on 85 of the 1,192 and above `drug_lookup_1` on 214, 7.1% and 18.0%, counts over conditions carrying no interval.

THE BAR IS NOT EQUALLY CLEAN ON BOTH SIDES OF THE SPLIT. The pooled 0.913 covers two populations. On trained conditions the lookup scores 0.980. On held-out wells, where it cannot borrow the target’s own well, it scores 0.890 against the model’s 0.533 on the 893[?] held-out-well conditions where both are scored, a count this thesis quotes two ways and has under audit; the paired gap there is -0.357 [-0.403 , -0.311]. Nothing in the argument turns on the exact size of that population. The ordering holds on both sides, and the held-out

side, where the well advantage is absent, is the side a claim about transfer has to survive.

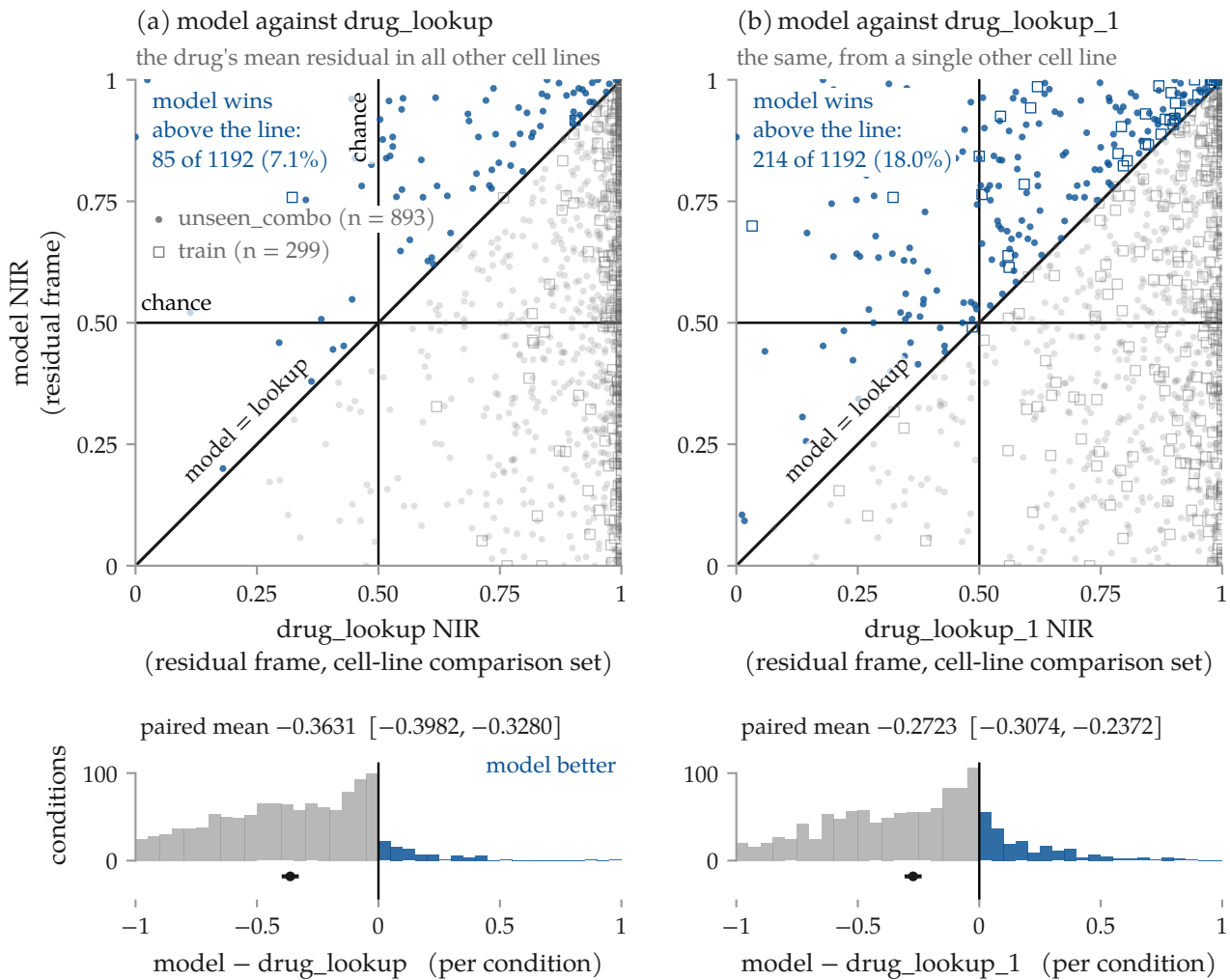


Figure 4.16. The headline gap is a tilt of nearly the whole population and not a mean over a mixed one: the model scores above `drug_lookup` on 7.1% of conditions, and above the single-cell-line lookup on 18.0%. Every mark is one of the 1,192 conditions of the common support — those on which a training-only lookup exists — and both axes are NIR in the **▲** `RESID` frame, comparison set other drugs in the same cell line. On this support the model scores 0.549 and the within-well split-half precision reference 0.857; the 1,394-condition figures quoted elsewhere (0.537 and 0.854) are a different population and are not combined with these. (a) Against `drug_lookup`, the drug’s mean residual in all other cell lines, 0.913 pooled: the model is above the identity line on 85 of 1,192 conditions, 7.1%. (b) Against `drug_lookup_1`, the same quantity from a single other cell line, 0.822 pooled: 214 of 1,192, 18.0%. Marker shape gives the split, 299 train and 893[?] unseen_combo; no unseen_drug condition appears, because no training lookup exists for a drug held out of training, and that regime is therefore not addressed by this figure at all. The lower panels are the per-condition paired difference on a shared count axis, with the paired mean drawn below the counts as a point and bar, the bar a 95% bootstrap interval clustered two-way on cell line and well: $-0.3631 [-0.3982, -0.3280]$ in (a) and $-0.2723 [-0.3074, -0.2372]$ in (b). The win counts are counts over conditions and carry no interval; ties (23 and 28 conditions) count as not a win. The 0.50 rules are chance on both axes, labelled once in (a).

THE CLAIM. The model reads the drug and recovers a small fraction of its residual. The honest summary is the paired gap on common support ($n = 1,192$), $-0.3631 [-0.3982, -0.3280]$. The model covers 13.9% of the distance from chance to the within-well split-half precision reference where a training-only lookup covers all of it and more. Those are common-support figures, model 0.549 against reference 0.857; over all 1,394 scored conditions, not all of which admit a lookup, the model is 0.537 against a reference of 0.854, and the two populations are not combined. The pooled gap does combine the two sides of the split, on which the lookup is not equally clean. On held-out conditions alone it is $-0.357 [-0.403, -0.311]$, and the lookup's 0.913 is 0.980 on trained conditions against 0.890 on held-out ones. That the lookup exceeds the reference is a property of the reference, not a headroom estimate; the reference is a single noisy half-sample while the lookup averages many cell lines, and NIR compresses near its top. Nor is averaging what beats the model. Restricted to a *single* other cell line the lookup scores 0.822 and the model still trails it by $-0.2723 [-0.3074, -0.2372]$. This is DrEval's finding replicated in transcriptomic space, the drug main effect at 0.913 where a 1B-parameter model reaches 0.549. Being a bar and not a result is what makes it reusable, and any method proposed for seen-drug transfer on this atlas has to clear it, not merely chance.

4.12 Same-target retrieval is the strongest observed lead

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▲ RESID

The bar just set exists only for a drug that appeared in training. A drug that never did can be reached at all only through some property it shares with drugs that were seen, a protein target, a mechanism, a chemical shape. The instrument predicts each drug from its annotated neighbours in the same cell line, against a null matched on everything but the channel itself. One channel carries a lead, on a very short roster of drugs.

VERDICT Same-target retrieval is the strongest observed lead on its own restricted support; because the three channel arms differ in support, drug roster, partner count and annotation coverage, that is a descriptive result and not an identified biological ranking.

For a channel X, a drug's residual is predicted as the mean residual of the other drugs in the same cell line sharing property X with it, the construction of `moa_lookup` (§ 4.11) generalised from mechanism to any annotated property.

THE NULL HAS TO DIFFER IN EXACTLY ONE RESPECT. The gate's estimand requires the channel arm and its null to differ in the channel and in nothing else. Every channel is paired with a plate-matched random null carrying the same number of partner drugs, drawn from the same cell line, with the same split across the target's own plate and other plates. Mechanism- and target-related drugs are co-plated more often than chance, so a null matched on count alone would differ in plate membership too, two differences where the estimand allows one.

Where a plate-matched null spans too few cell lines the channel is reported UNTESTABLE. “We could not build the control” must never be written as “closed”. Coverage is printed first as well, because a channel annotated on few drugs was never really tested and cannot be closed by a null result.

THE PARTNER POOL WAS WRONG, AND EVERY VERDICT HERE IS FROM THE RE-RUN.

The gate was intended to draw partners only from drugs that were in training, which is what makes it a fair contest in every split. An audit of its own implementation, not of the already-restricted lookup arms of § 4.11, found the pool spanning every retained condition in the cell line. The restriction is now in the code. Of the 4,131[?] retained conditions only 2,523[?] are training, so nearly two in five the old pool could draw on were ineligible. Every verdict here is from the corrected run, which moved the target gap down, the mechanism gap up, and chemistry’s down by half.

CORRECTION the channel gate’s partner pool, restricted and re-run

Table 4.12. Coverage first, then the gap. The drugs behind each channel are stated before any verdict, because a null result on a channel never really tested is not a closure. The plate-matched null is the estimand-correct control; the count-matched column shows sensitivity to plate membership, not a causal share. The gate asks the plate-matched lower bound to clear a pre-set margin of 0.03 clustered on cell line and well. On that axis, and only there, protein target clears it (+0.0745) while mechanism (+0.0279) and chemistry (+0.0035) do not. All three exclude zero, so the two that fail are inconclusive rather than equivalent to their nulls, and neither may be written as closed.

Channel	Drugs (an-not./scored)	<i>n</i>	vs plate-matched null	(vs count-matched)	Verdict
protein target	264 / 18	712	+0.1351 [+0.0745, +0.1958]	+0.1498	live, cell-line axis [†]
mechanism (MoA)	180 / 26	921	+0.0805 [+0.0279, +0.1332]	+0.0623	inconclusive
chemical structure	377 / 94	4131 [?]	+0.0261 [+0.0035, +0.0487]	+0.0122	inconclusive

[†]The axis is written into the target verdict because “live” is not usable without it. Clustered on the drug, the target interval falls short of this margin against both estimand-correct nulls and clears it only against the weaker count-matched one.

The counts are annotations in Tahoe’s drug metadata and not testable coverage here; within this cache only 18, 26 and 94 of the 94 scored drugs received a channel arm at all.

THE POPULATION THE QUESTION IS ACTUALLY ABOUT. The repair restricted where the *predictors* come from, not which conditions are *scored*. The targets are all retained conditions, spanning 94 drugs, of which 14 are held out of training entirely, the 10 of the `unseen_drug` quota plus four appearing only in held-out wells. The target arm covers 18 drugs, **two** of them held out. Restrict the targets to held-out drugs and, clustered two-way on cell line and well as everything else here is, the target gap is +0.1197 [+0.0499, +0.1894] on those two; mechanism falls from +0.0805 to +0.0439 [−0.0846, +0.1723] on four and chemistry to +0.0378 [−0.0029, +0.0785] on fourteen, both spanning zero (Figure 4.17, third block). Only the target channel survives, and it survives on two drugs.

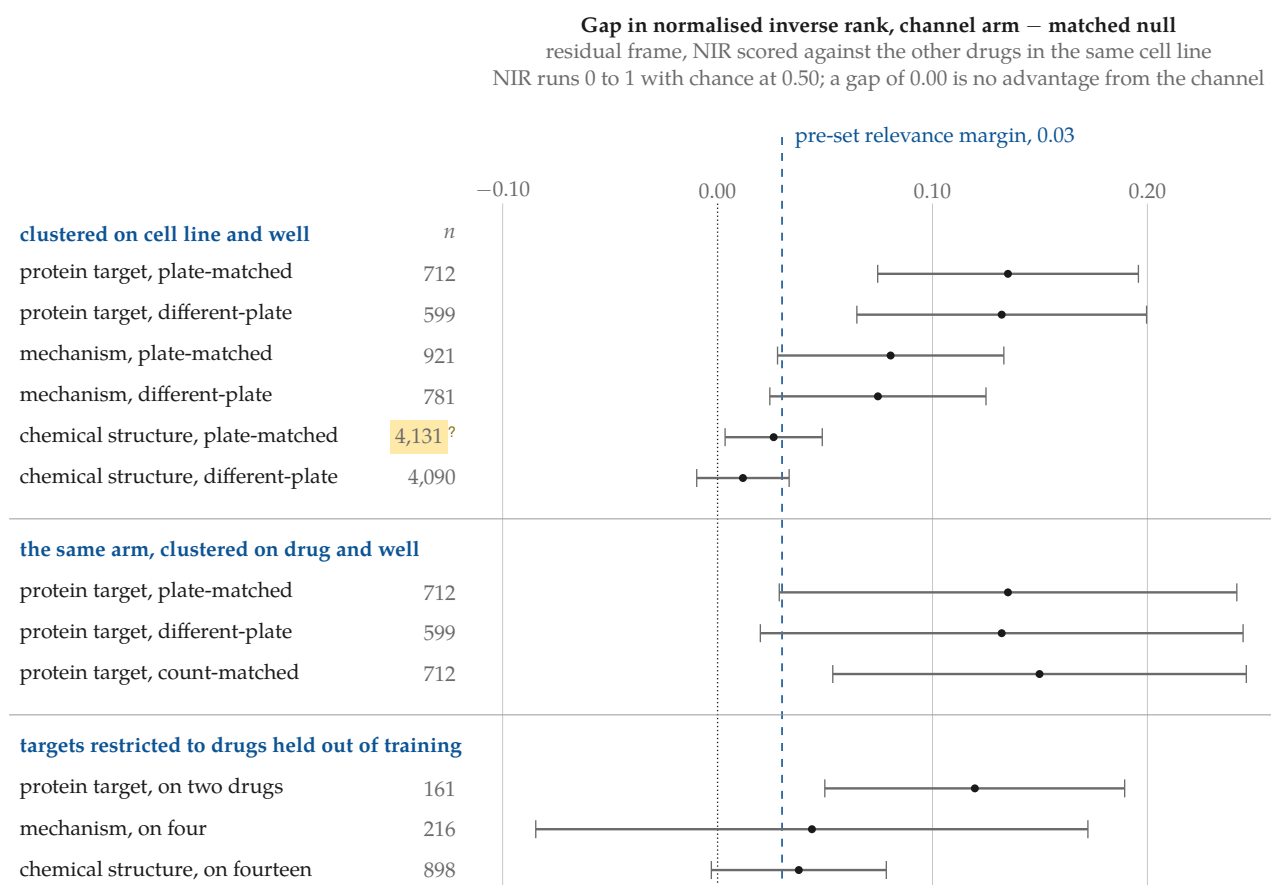


Figure 4.17. The gate's verdict is decided by what the interval is conditioned on, not by where the point estimate falls. Every row is residual-frame NIR scored against the other drugs in the same cell line: the arm predicts a drug's residual from its annotated neighbours there, and its matched null draws the same number of partner drugs from that cell line. The dot is the estimate, the bar its 95% two-way cluster-robust interval, the dashed line the pre-set relevance margin of 0.03 that the gate asks the *lower* bound to clear. The count beside each row label is the number of conditions that row scores, not of drugs; the first block's plate-matched counts are the n of Table 4.12. The second block re-clusters the same target arm on the drug, the unit a claim about unseen drugs has to generalise across, leaving the point estimates untouched; the lower bound now falls short against both estimand-correct nulls and only the weaker count-matched one clears it. The third restricts the targets to drugs held out of training. Rows may be read within a block and not across blocks.

WHAT EIGHTEEN DRUGS ARE MADE OF. The target arm's 18 annotated drugs are almonertinib, amsacrine, cepharanthine, clonidine, elagolix, gemfibrozil, goserelin, idarubicin, lumateperone, mitoxantrone, neratinib, norepinephrine, olanzapine, rapamycin, relugolix, temsirolimus, tofacitinib and tofacitinib citrate. Their mean partner count is 1.19. Of the arm's 712 predictions, 579 come from a *single* partner drug and 133 from two, against 2.46 partners for mechanism and 5.00 for chemical similarity. And sometimes that partner is an unusually close relative. Tofacitinib and Tofacitinib (citrate), the free base and the citrate salt of one drug, are scored in the same 20 cell lines and have one partner each; so are rapamycin and its ester prodrug temsirolimus, in 22 lines. Neratinib and almonertinib pair the same way in 20 lines, and those two are chemically distinct, so the pattern is not only duplicate chemistry. The channel establishes that a same-target neighbour predicts the drug. That a

target *annotation* generalises across a class is the stronger statement; testing it would mean dropping the nearest analogue from the pool, and that was not run.

THE TWO DRUGS THE VERDICT RESTS ON. They are **amsacrine** and **relugolix**, and because the duplicate-chemistry pattern above would otherwise be the obvious objection to them, their pools are worth naming. Amsacrine is annotated to TOP2A and its partners are doxorubicin, epirubicin, idarubicin and mitoxantrone; relugolix is annotated to GNRHR and its partners are elagolix and goserelin. Neither pool contains a salt, prodrug or stereoisomer of the drug it predicts, so this verdict is not the tofacitinib case in another guise — though amsacrine’s pool is three anthracyclines and a close structural analogue of them, which is a drug class and not a duplicate. The support is 161 conditions over 39 cell lines and 5 treated wells, 70 of them amsacrine in two wells and 91 relugolix in three; the five wells are the smaller cluster count and are where this interval’s t_4 reference comes from.

SCOPE LIMIT | carried into § 4.13 and chapter 5

What this section measures is *channel availability* — whether the information a channel carries is present in the atlas at all — and not unseen-drug predictive performance, which two drugs cannot establish for the target channel and which mechanism does not survive at all. Every verdict here, including those printed in Table 4.12, is to be read in those terms. The bound is on similarity transfer through each channel and not on any model, which could combine channels or learn a non-smooth structure-to-response map that a nearest-neighbour average cannot express, though at a few hundred drugs the lookup is effectively the bound.

SEE chapter 3: why the smaller cluster count sets the reference distribution

WHY THE CLUSTERING AXIS DECIDES THIS SECTION. Two conditions from one treated well share a physical assignment and a batch; two from one cell line share the biology. Every interval on a mean in this chapter is therefore two-way cluster-robust,

$$V = V_{g_1} + V_{g_2} - V_{\text{intersection}}, \quad (4.11)$$

where g_1 and g_2 are the cell line and the treated well, whose intersection is the single condition, so $V_{\text{intersection}}$ is the ordinary independent-sampling variance (chapter 3). Such a variance is asymptotic in the *number of clusters* and not in the number of conditions, so the reference distribution is Student’s t on $\min(G_{g_1}, G_{g_2}) - 1$. Wells sit inside drugs rather than crossing them, so on the drug axis the intersection is the well and not the condition; retaining the crossed form there subtracts less and widens the interval, and the drug-axis figures in this chapter are the demanding ones by construction.

AND ON THIS SECTION’S OWN ARM IT CHANGES THE VERDICT. The gate clusters on cell line and treated well, right for a statement about this experiment. But the claim the channel is asked to support, that an *unseen drug* is reachable through its target annotation, generalises over drugs, and only 18 annotated drugs carry the target arm. Against the plate-matched null the target gap is +0.1351 [+0.0287, +0.2416] clustered on drug and well (df = 17; Figure 4.17,

second block against the first), whose lower bound falls 0.0013 short of the margin; against the different-plate null it is +0.1322 [+0.0199, +0.2445]. Both exclude zero; neither clears 0.03. Only the count-matched null clears it here, +0.1498 [+0.0536, +0.2460], and that is the control already set aside as weaker. Mechanism, which excludes zero on the cell-line axis, spans it on the drug axis in all three constructions. The honest summary is therefore narrower than *live*, and the axis is chosen by the claim, not by which interval is narrower.

SENSITIVITY TO PLATE MATCHING, MEASURED AND NOT ARGUED AWAY. Mechanism partners are co-plated with the target 0.404 of the time against 0.362 for a random draw; elsewhere the diagnostic is negligible, protein target 0.368 against 0.366 and chemical similarity 0.370 against 0.367. Under a third construction drawing all partners from other plates, target and mechanism keep intervals clear of zero and chemistry's spans it (Table 4.13). Absolute NIR there is 0.638, 0.575 and 0.499 against nulls of 0.506, 0.500 and 0.487, so removing the shared plate does not push the arms below chance. It barely moves target or mechanism, two and seven per cent off their plate-matched gaps, while chemistry attenuates again. But it also changes support and available partners, so the attenuation does not identify co-plating as its cause.

THE ORDERING IS DESCRIPTIVE, AND ONE OF ITS TERMS IS A FLOOR. Target, then mechanism, then chemistry, under every construction and again on held-out targets. The target and chemistry intervals do not overlap under any of the three nulls; each adjacent pair's do, and on the drug axis even the two ends overlap. These arms differ in support, drug set, partner count and annotation coverage, so the order does not identify a biological ranking. One correction runs in a known direction. The residual frame subtracts a plate-local generic, and § 4.7 shows that for a co-plated mechanism class part of that class's own shared programme goes with it, so mechanism's middle position is a floor and not an estimate. Below target the two failures differ. Mechanism's lower bound falls 0.0021 short, a knife-edge that is not a negative result in either direction; chemistry's interval is the *narrowest* of the three, 0.045 wide against 0.121 for target, but straddles the margin. Neither is a closure.

A PRE-REGISTRATION THIS OVERTURNS. Two design documents concluded that unseen-drug generalisation was out of reach, and both inferred it from the chemical-structure channel alone, Morgan fingerprints and a molecular language model. An earlier drug-knowledge specification had instead identified protein targets as "the key feature", since transcriptional response is driven by which protein a drug hits, something structure encodes only indirectly. The same-target lead shows that generalisation from chemistry to every channel was not licensed. The gate agrees with the specification about which lead to pursue; with arms on different supports it does not validate it as a hierarchy. Tahoe's own metadata carries protein-target annotations for 264 drugs (280 distinct symbols, 550 edges), which no analysis in this project had read.

THREE CAVEATS BOUND THE TARGET VERDICT. The first is annotation coverage. The arm covers 18 of the 94 scored drugs. The second is scale. A shared property buys a fraction of what the drug itself buys; the target arm sits +0.14

above chance where a same-drug lookup for a seen drug sits +0.41 (§ 4.11, a different condition set, so a comparison of scale and not a paired one). The third is overlap. Whether target adds anything over MoA is unsettled here; settling it would need a leave-one-mechanism-out split, since a held-out drug's mechanism class is generally still represented in training.

THE CLAIM. Same-target retrieval is the strongest observed lead on its own restricted support; because the three channel arms differ in support, drug roster, partner count and annotation coverage, that is a descriptive result and not an identified biological ranking. The target arm excludes zero on every construction under both clusterings, and on held-out targets, the population the unseen-drug question is actually about, the gap is +0.1197 [+0.0499, +0.1894] on two drugs, clustered on cell line and well as everything else here is. On the full arm it clears the pre-set relevance margin of 0.03 on the cell-line axis under all three constructions; clustered on the drug, the axis a statement about unseen drugs rides on, it clears the margin only against the weaker count-matched null and falls short against both estimand-correct ones. The arm rests on 18 annotated drugs, two of them held out. Mechanism excludes zero on the cell-line axis, falls 0.0021 short of the margin there, spans zero on the drug axis, and spans zero again on held-out targets (+0.0439 [-0.0846, +0.1723]); it cannot carry an unseen-drug claim in any conditioning, and the frame understates rather than flatters it. Chemistry's interval is the narrowest of the three but straddles the margin, so it does not locate the effect wholly either side of that threshold; it does rule out a large effect, and under the different-plate construction it no longer clears zero, though support and partner availability change with plate separation, so that attenuation does not identify co-plating as its cause. None of the three may be written as closed, and none as live without naming the axis. Similarity transfer through the target channel is measurable and is the lead to pursue first; whether it is large enough to matter is unresolved at 18 drugs, and mechanism is unresolved and not excluded, graded in a frame that subtracts part of its own signal before scoring it.

4.13 Which part of the prompt is read remains unresolved

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■ EXPR / ▲ RESID

The previous section asked whether the information an unseen drug would need is present in the atlas at all. The complementary question is whether the model reads what it is already handed, and which part of it, the name or the mechanism line beside it. The instrument swaps one span of the prompt at a time and leaves every other byte identical. It does not settle the question, and what follows is why, and what would settle it.

The chapter has established that the prediction responds to that information as a block; it has not established *which* part of it is read. Saying so precisely

matters, because the next step writes itself. Append the protein targets to the prompt, retrain, and see whether unseen-drug performance rises. That arm assumes the bottleneck is information availability. Everything in this chapter points the other way, but the direct evidence is weaker than earlier drafts claimed.

THE INSTRUMENT, AND THE ONE GUARD IT NEEDS. The scramble has until now replaced the drug name and the mechanism together, so a non-null gap establishes sensitivity to the combined prompt without attributing it to either span. A field decomposition swaps exactly one span to the opposite partner, the name with the mechanism kept, the mechanism with the name kept, and both together. One guard is essential. If the swap partner happens to share the original's mechanism, swapping only the mechanism produces a prompt identical to the model's own and the arm scores exactly zero by construction, which read naively is indistinguishable from "the model ignores the mechanism", the very claim under test. Such conditions are dropped, which is why the mechanism arm carries fewer conditions.

WHAT THE ARMS SAID, AND WHY IT DOES NOT STAND. On the earlier target build the decomposed arms suggested the prompt sensitivity is carried almost entirely by the name span. Swapping only the drug name gave Δ_{scr} of $+0.0809$ [$+0.0592$, $+0.1010$] ($n = 570$); swapping only the mechanism gave $+0.0091$ [-0.0174 , $+0.0383$] ($n = 325$); both together gave $+0.1024$ [$+0.0827$, $+0.1236$]. All were scored against the opposite comparator that § 4.10 shows inflates every gap read against it. They *were* re-run under the corrected evaluation (Table 4.13, Panel B), but the defect survives it, being a property of the swap partner and not of the target build. The claim that the model "reads the name and not the mechanism" is therefore withdrawn pending comparators whose own scores are empirically near chance. What the current run establishes is sensitivity to the combined prompt, with the mechanism field's contribution unresolved.

WHY THE TWO PANELS CANNOT BE ADDED TOGETHER. Panel A says a channel carries information in the atlas; Panel B asks whether the model uses what it is handed. Neither answers the other's question, and an ordering that reverses when the comparator is removed is not an attribution.

CORRECTION

the name-over-mechanism reading, withdrawn pending neutral comparators

THE PRACTICAL CONSEQUENCE, NARROWER THAN EARLIER DRAFTS DREW. The target-prompt arm is not retired by measurement. It is untested. What argues for building it carefully is the shape of the failure above. The model's deficit is not confined to missing information about unseen drugs, because on seen drugs it covers 13.9% of the distance from chance to the within-well reference, against a training-only lookup that covers all of it and more (§ 4.11). What argues for building it at all is the one thing a lookup cannot do, handle a drug it has never seen, the regime where § 4.12 finds measurable transfer through the protein-target channel, on 18 drugs.

THE OPPORTUNITY IS NOT THE PART THAT FAILS TO TRANSFER. § 4.8 puts $1 - \mathcal{T}$ at 44.7% [40.7%, 48.6%], and it is tempting to call that the model's headroom. It is not. That figure is a shortfall in cross-context similarity, mixing cell line,

Table 4.13. Availability does not establish use. Panel A shows target and mechanism retrieval surviving the different-plate null while chemistry does not; only target clears the relevance margin, and only on the cell-line axis, on an arm covering 18 drugs (§ 4.12). Intervals are two-way cluster-robust over cell line and treated well. Panel B cannot attribute the combined-prompt effect between name and mechanism, because its two partners lie on opposite sides of chance (0.434 and 0.520), inflating one gap and suppressing the other. Comparator-free margins reverse the apparent ordering (+0.0367 for name, +0.0506 for mechanism), so the split is unresolved pending comparators whose own scores are empirically near chance. Identical mechanism swaps are omitted, which is why that row has fewer conditions.

Panel A: channel availability — metadata-neighbour retrieval			
Channel	Drugs	<i>n</i>	Gap vs plate-matched random null
	(annot./scored)		
protein target	264 / 18	712	+0.1351 [+0.0745, +0.1958]
mechanism (MoA)	180 / 26	921	+0.0805 [+0.0279, +0.1332]
chemical structure	377 / 94	4131 [?]	+0.0261 [+0.0035, +0.0487]
the same three, against the stricter DIFFERENT-PLATE null			
protein target		599	+0.1322 [+0.0648, +0.1996]
mechanism (MoA)		781	+0.0746 [+0.0243, +0.1249]
chemical structure		4090	+0.0118 [-0.0097, +0.0333]
Panel B: field use — one span swapped, every other byte identical			
Span swapped	Δ_{scr}	<i>n</i>	Comparator direction (partner NIR)
drug name, mechanism kept	+0.1026 [+0.0625, +0.1427]	1394	inflated (0.434)
mechanism, drug name kept	+0.0303 [-0.0078, +0.0685]	738	suppressed (0.520)
both spans (the standard arm)	+0.1139 [+0.0781, +0.1496]	1394	inflated (0.423)

Panel A is repeated from Table 4.12 so the two panels can be read against each other; it is the same measurement, not a second one.

dose, well and estimator; it is neither a variance share nor an interaction. With 286 of 287 treatments in exactly one well there is no replicate from which to separate a context-specific remainder from measurement noise, which is why the coefficient is a disattenuated cross-line cosine and not a fitted variance decomposition. Whether any of that shortfall is learnable structure and not noise is separated nowhere here. The opportunity is the unseen drug, and whether the model can be made to take it is not answered in this chapter.

SEE § 4.8 for the estimator and its five checks

4.14 The investigation in one table

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■ EXPR / ▲ RESID

The results are distributed across two measurement frames, and both put chance in the same place. That coincidence is the trap this chapter has to leave its reader safe from. What follows is a ledger of the principal comparison arms, frame by frame, then the results that are estimates and belong outside it.

The expression frame of Table 4.14 scores a predicted cell profile against other drugs on the same plate. The residual frame scores a predicted residual against other drugs in the same cell line, after the generic programme has been subtracted.

THE SAME CHANCE VALUE, TWO DIFFERENT REFERENCES. Both use 0.50 as chance under the exchangeability condition audited in § 4.1. Figure 4.18 puts each frame on its own ruler, scaled so that chance and that frame’s own within-well precision reference fall at the same two places: the 0.537 in one frame and the 0.498 in the other are not two attempts at one task, and the difference between them is neither an improvement nor a decline. Read each against its own reference, the expression-frame model does not clear chance at all, and the residual-frame model clears it on trained conditions only, 0.598 [0.549, 0.647] there against 0.533 [0.492, 0.575] and 0.459 [0.404, 0.514] on the held-out splits, neither distinguishable from 0.50 (§ 4.10). The pooled 0.537 is three-quarters held-out condition, and no arm is quoted with an interval at the pooled level, so it is quoted as a mean and not as a clearance.

AND THE MARGIN BETWEEN THEM HAS A SUPPORT. What separates the frames is a margin § 4.11 sizes at 13.9% of the distance from chance to the reference, computed on the $n = 1,192$ common support where the model scores 0.549 against a reference of 0.857, and not on the $n = 1,394$ whose 0.537 and 0.854 are quoted above.

SCOPE LIMIT | carried into chapter 5

No number in one frame may be set beside a number in the other. Chance is 0.50 in both, which is exactly what makes the mistake easy to make; the references above chance are 0.854 and 0.576, and every distance quoted in this chapter is a distance to its own frame’s reference. Within the residual frame the same applies to supports: 1,394, 1,192 and 218 select conditions by different rules, and rows on different supports are not comparable.

WHAT THE LEDGER DELIBERATELY LEAVES OUT. Table 4.14 omits the near scramble, a deliberately weak test; `moa_lookup`, whose $n = 218$ support is not comparable with the principal rows; the residual-frame `control_copy` and `random` negatives, which serve the calibration diagnostic; and the field-attribution arms, whose non-neutral comparators leave attribution unresolved. The expression-frame `control-copy` stays because it carries the identifiable-stratum comparison, and `orth` and `opposite` because their contrast exposes how comparator score depends on partner geometry.

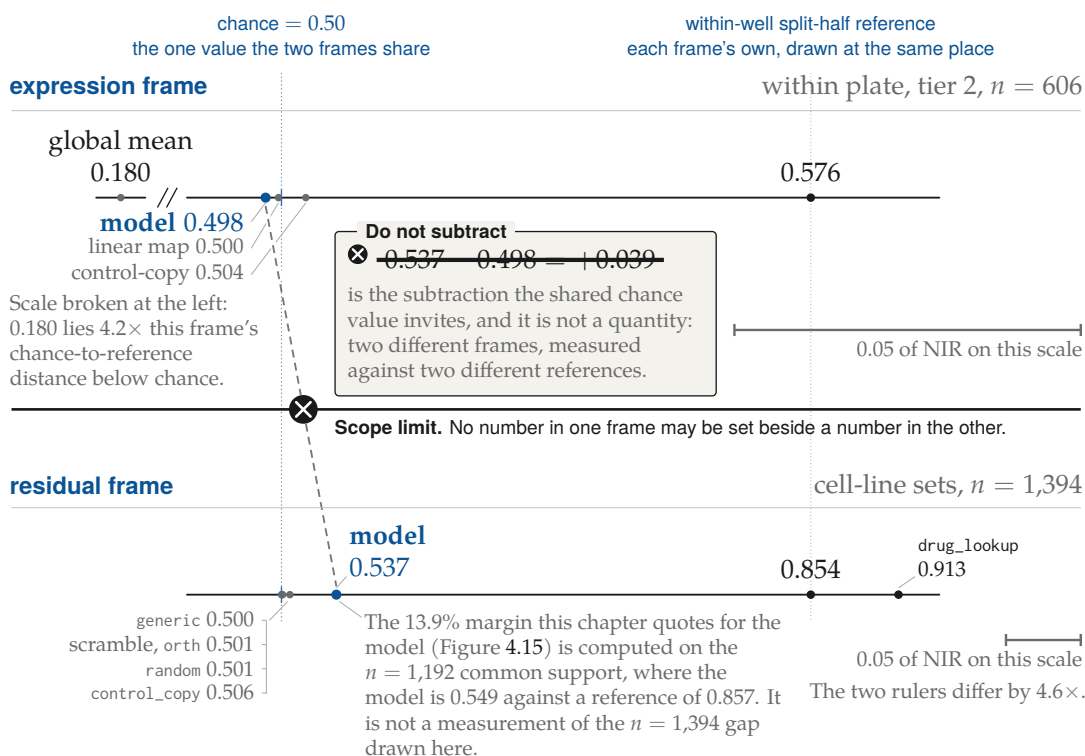


Figure 4.18. The two frames share a chance value and nothing else, and the shared value is exactly what makes the mistake easy. Each scale is stretched so that chance falls at the same place in both and so that each frame's own within-well split-half precision reference falls at the same place as the other's; nothing else is aligned, so the two frames carry different rulers — the same 0.05 of NIR is $4.6\times$ longer above than below — and that differential stretch is the figure. *Upper:* the expression frame, within plate, tier 2, $n = 606$, where the model scores 0.498 against a reference of 0.576 and is matched by a drug-agnostic linear map (0.500) and a control-copy (0.504); the scale is broken at the left because the global mean, 0.180, lies $4.2\times$ that frame's chance-to-reference distance below chance. *Lower:* the residual frame, cell-line sets, $n = 1,394$, where the model scores 0.537 against a reference of 0.854, a training-only per-drug lookup reaches 0.913 above that reference, and generic, orth, random and control_copy all fall within 0.006 of chance, one mark at this scale. Both references are within-well split-half precision figures — two disjoint halves of one treated well — and not biological replicates (chapter 3). Chance is 0.50 in both under the exchangeability condition audited in § 4.1, and because it is the one value the two frames share it is exactly what makes the struck-out subtraction in the middle of the figure easy to perform. Every value is a pooled mean drawn as a bare marker; no arm shown here is quoted with an interval at the pooled level, so none is drawn (Table 4.14). The 13.9% margin annotated below is computed on the $n = 1,192$ common support and is not a measurement of the $n = 1,394$ gap drawn here.

FIVE RESULTS ARE ESTIMATES, NOT PREDICTOR ARMS. They are stated outside the ledger because nothing in it is their comparison set.

- **The metric audit.** Of the five metrics graded, NIR is the only one whose DRF is positive, $+0.635$ [$+0.604$, $+0.663$] across plates, $+0.545$ [$+0.516$, $+0.569$] within plate, $+0.704$ [$+0.667$, $+0.736$] in the residual frame; the other four are negative and exclude zero. Three, panel- τ , spearman_expr and DE- Δr , are inverted rather than merely uninformative, scoring an

Table 4.14. The principal comparison arms on the calibrated metric, by frame. Chance is 0.50 in both; reference rows are within-well split-half precision figures, not biological replicates (chapter 3). The final expression rows show the 204 conditions clearing the pre-set 0.80 identifiability threshold, where a drug-agnostic control still matches the model. Of the scramble strata, *near* (0.550) and *opposite* (0.423) differ from chance in opposite directions while *orth* (0.501) is empirically near chance here; because the strata are defined using true residual geometry, *orth* is the least contaminated available comparator and not an independently established null (§ 4.10).

Frame	Arm	NIR	What it knows
expression (within plate, tier 2, $n = 606$)	within-well reference	0.576	two disjoint halves of one treated well; precision, not a replicate
	control-copy	0.504	nothing; it reproduces the untreated cell
	linear map	0.500	the control cell, no drug
	model	0.498	control cell + drug
	global mean	0.180	nothing
	model , identifiable stratum	0.768	the same model, on the 204 conditions whose own reference clears 0.80
	control-copy, same stratum	0.766	nothing, and it matches the model there
residual (cell-line sets, $n = 1394$)	within-well reference	0.854	two disjoint halves of one treated well; not a replicate, and on the lookups' support, where it is 0.857, the lookup exceeds it
	drug lookup (training-only)	0.913	this drug's residual from other cell lines, on the $n = 1,192$ conditions admitting a lookup, where the model scores 0.549
	drug lookup, one line	0.822	the same, from one other cell line, same support
	model	0.537	control cell + drug
	scramble, <i>orth</i> partner	0.501	geometry-defined by residual cosine to the target ($\cos \approx 0$); its own score is <i>empirically</i> near chance here
	scramble, <i>opposite</i> partner	0.423	an active treatment, not a null
	generic response	0.500	nothing (the zero vector, by construction)

uninformative baseline above the positive control; *weighted_r2* sits closest to zero and its sign tracks cell count, so it carries no directional claim (§ 4.1).

- **Representation and readout.** Drug identity is decodable from the activations, and the generation readout consults it at one to two per cent of the preference gap, a fifth or less of the pre-registered material margin, in

one training arm and at one rung of the displacement ladder; the chapter's weakest load-bearing result (§ 4.5).

- **Prompt sensitivity.** Re-encoding the target as the drug-specific residual gives $+0.0898$ [$+0.0291$, $+0.1506$] on trained conditions, clustered on the drug. A budget-matched optimal-transport control clears zero too, at $+0.0292$ [$+0.0144$, $+0.0441$], so the re-encoding is first in the sequence and not uniquely responsible (§ 4.9).
- **Transfer.** $\mathcal{T} = 0.553$ [0.514 , 0.593], a cross-context similarity coefficient and not a variance share (§ 4.8).
- **The channels.** Same-target retrieval is the strongest observed lead on its restricted support; only that arm clears the relevance margin, on the cell-line axis on all three constructions and not on the drug axis under either estimand-correct null, on 18 annotated drugs of which two are held out. The arms differ in support, roster, partner count and annotation coverage, so this is not an identified biological ranking (§ 4.12).

The distance between the model and the least contaminated available scramble is the prompt-sensitivity effect this work recovered; the distance between the model and the lookup is what it did not.

■ EXPR / ▲ RESID

A limitation is worth writing down when it changes what a number licenses.

DRUG COVERAGE. A study of 106 drugs can establish that a model uses the drug it is handed; it cannot describe how drugs in general behave. The residual analysis uses 620,564 cells covering 106 drugs across 50 cell lines, roughly a tenth of the atlas’s drug set, ample for testing whether the model uses the drug and for measuring transfer but not a population estimate. Claims about the *distribution* of drug behaviours belong to the broader characterisation, which used 6,628 conditions.

RELIABILITY LINE
The half-split cosine threshold above which a condition's residual counts as reproducible.

That calculation’s own filter is the inherited $\cos > 0.2$ line, where the training set uses 0.109, so the two estimates bracket the trained-on population without reproducing it. The generation evaluation mismatches differently again, applying 0.109 to train and leaving the held-out splits unfiltered. That one is load-bearing and is taken up below.

THE INTERVALS OMIT GENERATION VARIANCE. The intervals here are cluster-robust with each condition’s score treated as fixed: two-way over *cell line* and *treated well* on a $\min(G_{\text{line}}, G_{\text{well}}) - 1$ Student *t* reference for the headline gaps, reported again clustered on the *drug*; multiway dyadic for the transfer coefficient; a one-way clustered bootstrap, unit named, in a few tables.

They capture between-cluster variance and leave out the variance from sampling the model’s predictions. With four sampled generations per condition at temperature 0.8 in the residual-frame arms and eight in the expression-frame ones (chapter 3), that source is not negligible, and no across-seed replication of the generation arms was run, deferred or planned. Every interval quoted from a generation arm covers the sampling of conditions, cell lines and wells, and stops there.

One measurement *was* repeated across seeds, recorded so it is not mistaken for the replication that was not run. The activation probe of § 4.5 returned +0.0197, +0.0118 and +0.0085 on three seeds, each excluding zero. Its spread does not transfer to NIR and does not establish a scale for generation variance. Those runs share a checkpoint and score teacher-forced log-probabilities, so their seed moves the prompts, the partners and the bootstrap, and leaves both the weights and the sampled prediction untouched. They are scored on 60, 240 and 240 prompts in that order, so the largest of the three effects is also the smallest-*n* one.

NO TRAINING RUN WAS REPEATED. The seeds above move prompts, partners and resampling. None of them moves the weights. Every arm here was trained once, from one base checkpoint under identical hyperparameters (Table A.3), so each dissociation this thesis rests on — the residual re-encoding against the standard target, and the optimal-transport target against both — compares one training run with one training run. A second run of the same arm under a different data order would land on a different checkpoint and score differently, and the size of that difference was never measured here. It is therefore not covered by any interval in this document: the intervals quantify sampling over conditions, cell lines and wells, and a between-arm gap of the size reported would survive or fail a training replication that nobody has run. The ten-epoch diagnostic varies the horizon and not the seed, so it bounds undertraining and not run-to-run spread. What this costs is specific rather than general: the *direction* of the re-encoding effect is corroborated by the token-divergence measurement, which needs no model at all, but the *magnitude* of every arm-to-arm difference is an n-of-one comparison.

A SWING THAT LOOKED LIKE GENERATION VARIANCE AND WAS NOT. Unseeded generation is the kind of defect that surfaces as an unstable number, and once it seemed to.

✕ **CORRECTION 3** | seed instability in the held-out-drug control

Two superseded evaluations of the held-out-drug control returned values of Δ_{scr} far enough apart to be recorded as evidence of seed instability. Only one of the pair survives in the results archive, so the size of the swing is no longer quotable. The diagnosis was wrong. It named the trigger and missed the cause.

Both of the superseded figures are gaps against the opposite swap, which § 4.10 shows is not a null; it carries a term for how far the *partner* drug was pushed the wrong way, which changes with the draw. On the canonical evaluation the same control reads +0.0379 against opposite and −0.0315 [−0.0836, +0.0206] against the neutral stratum, clustered two-way on cell line and well.

SEE § 4.10, where a neutral stratum replaces opposite as the comparator

Two mechanisms were at work, and both are now addressed. `do_sample` advanced from global torch state, so the two arms of a contrast were drawn from different points of one stream; each call now seeds its own generator from (condition, arm, batch), and the comparator has been corrected. The general caution stands regardless; this swing is no longer evidence for it.

PSEUDOBULK RESOLUTION. Residuals are condition-level pseudobulks over ≥ 40 treated cells, so single-cell heterogeneity of the response is not modelled in this frame. Earlier project results show why: at single-cell resolution the drug moves a cell less than noise does (SNR ≈ 0.75 , median 0 differentially expressed genes at FDR $q < 0.05$).

CROSS-PLATE ABSOLUTES. Residual-space scores use comparison sets grouped by cell line, not by plate. The *difference* Δ_{scr} is leak-immune, since both arms share the same control cell. The absolute NIR values are a different matter.

§ 4.7 argues the grouping is admissible because the plate-scoped generic has already subtracted each plate’s mean response from truths and predictions alike. But it removes plate structure in expectation only, and the generic is estimated from a small plate group, so some may survive in an individual residual. The rescore that would size what survives, regrouping the generation-free arms within plate, needs no GPU time and was not run.

What exists instead is one-sided. Table 5.1 bounds it from above. In the expression frame with no generic subtracted, holding the plate constant moves a drug-agnostic **mean baseline** from 0.399 to 0.161 and the precision reference from 0.625 to 0.575 — the yardstick falls with the baseline (Figure 5.1), so the leak is not an offset one side of a comparison could be corrected by. Residual-frame absolutes should be read within their own frame and never set beside a within-plate number.

FRAME ■ EXPR the plate-leak audit

FRAME COVERAGE. The drug-specific residual is $\|r\|/\|s\| = 0.786$ of the shift against 0.620 for the generic programme and batch, and the two are essentially orthogonal, the shift’s energy decomposing into residual 0.621, generic 0.390 and a cross term of −0.011, summing to 1.000. The frame covers 62.1% of the shift’s *energy*, a real but not total share, and every result here should be read within that bound. The generic 39.0% is drug-agnostic and trivially matched, part of what a drug-agnostic predictor supplies on absolute-accuracy metrics, though the generic programme is not what saturates DE- Δr ; reversion toward a central point is (§ 4.1).

THE SPLITS ARE SCORED ON DIFFERENT POPULATIONS. The within-well split-half precision reference differs sharply across the three splits as scored, 0.956 on train, 0.824 on unseen_combo and 0.836 on unseen_drug. The obvious reading is that held-out conditions carry less recoverable signal. That reading is wrong.

Table 5.1. Expression-frame NIR, the leak audit. A drug-agnostic mean baseline more than doubles its score when the plate is free to vary. The precision reference falls once the plate is held constant, because batch structure was inflating it too.

Predictor (drug-agnostic)	cross-plate	within plate
mean baseline	0.399	0.161
control-copy	0.551	0.510
precision reference (within-well split-half)	0.625	0.575
fraction of drugs at chance	43%	49%

A wider diagnostic run than the $n = 606$ tier-2 benchmark the blindness verdict is stated on, at $n = 1,163$ cross-plate and $n = 998$ within plate. Holding the plate constant costs every condition with no same-plate comparison set, so the within-plate column is a subset of the cross-plate one rather than the same conditions rescored.

Its within-plate figures therefore sit slightly off Table 4.14, control-copy 0.510 against 0.504, mean baseline 0.161 against 0.180, precision reference 0.575 against 0.576.

It is the selection asymmetry noted above. `train` is drawn from a pool already filtered at the 0.109 line, and none of its 300 scored conditions falls at or below it, while the held-out splits are unfiltered and still contain conditions whose two halves disagree outright. Impose the training line on all three and the references agree at 0.955, 0.961 and 0.960, a spread of 0.006 where there had been 0.132 (Figure 5.2). Matched on reproducibility, a held-out condition is as recoverable as a trained one.

Those conditions carry less signal is therefore not a reason to discount a held-out shortfall, and this thesis nowhere uses it as one. What survives is narrower and binding. The matched-population rescore covered the precision reference only. The model's own arms were never rescored, so what the held-out shortfall becomes once the populations agree is not reported here, and the training-exposure contrast of § 4.10 inherits the mismatch.

SCOPE LIMIT | discharges the limit carried from § 4.11

Three selection rules are in force at once, and no result computed under one of them may be set beside a result computed under another. A raw score read across the three splits is not answering the same question on each, because the splits were filtered by different rules. An absolute NIR computed with the comparison set grouped by cell line is not comparable with one computed within plate. And a per-drug lookup is the bar for a drug that was *seen* in training and is being asked about a new context: no lookup exists for a drug that never appeared in training, so that regime keeps its own bar, set in § 4.12.

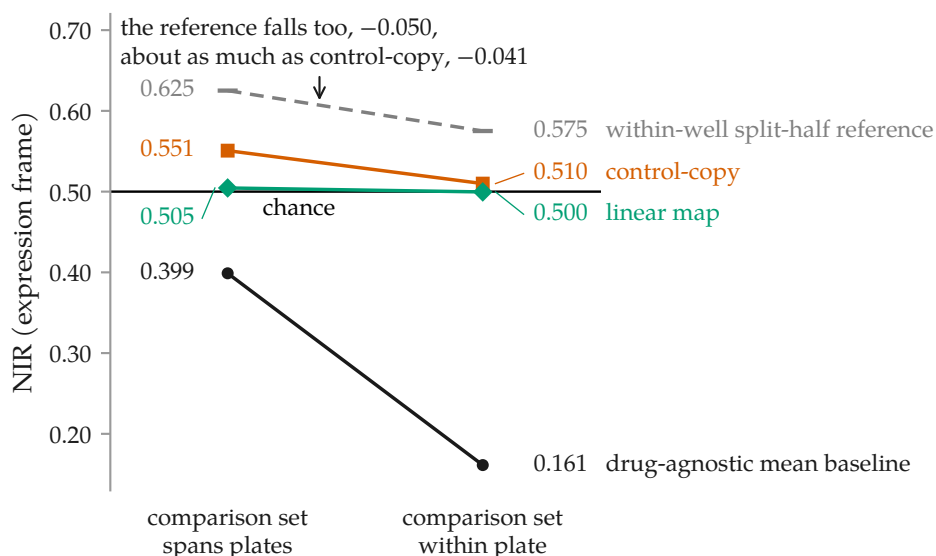


Figure 5.1. The batch shortcut inflates the precision reference as well as the baseline, so it is not an offset that could be subtracted from one side of a comparison. Expression-frame NIR, tier 2 unseen drugs, drug-agnostic arms only: the same audit as Table 5.1, with the linear map read from the same block of the same two records. Forcing the comparison set within plate collapses a mean baseline that carries no drug information whatsoever, 0.399 to 0.161, and leaves control-copy close to where it was, 0.551 to 0.510 — but it also lowers the within-well split-half precision reference, 0.625 to 0.575, by about as much as it lowers control-copy. That reference is the yardstick these expression-frame numbers are read against, so the leak has no one side to be corrected on. The two columns are not the same conditions: holding the plate constant costs every condition with no same-plate comparison set, so the within-plate column is a *subset* — $n = 998$ conditions, 46 drugs, 41 cell lines — of the cross-plate column's $n = 1,163$, 50 drugs and 43 cell lines, and each segment joins two scorings of overlapping but unequal populations rather than one set of conditions rescored. The marks are bare because neither record carries an interval for these quantities, each entry being a value and its n , and this document draws only 95% bootstrap intervals clustered over cell lines. The rank-frame variants stored beside these values are a different frame and are not drawn, and none of the values here may be set beside a residual-frame number.

5.2 What the unseen-drug arm licenses

Two defects in the unseen-drug arm must be stated. A third, the comparator, was found and corrected, and § 4.10 reports the arm against the neutral stratum throughout.

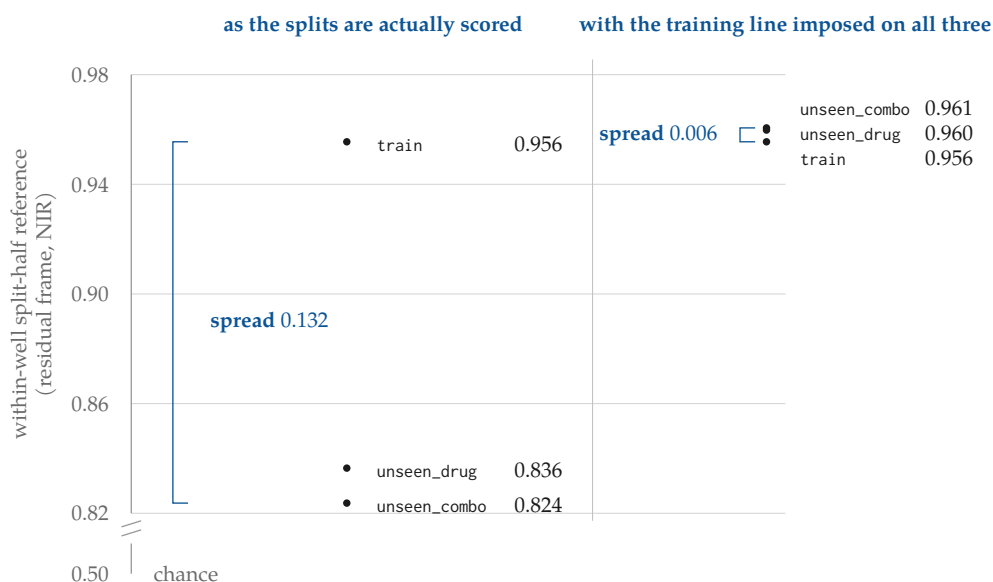


Figure 5.2. The three splits do not differ in recoverable signal; they differ in which conditions were allowed into them. The vertical axis is the within-well split-half precision reference — one half of a condition’s cells scored against the other half’s truth, in the residual frame, on the NIR scale — and it is *truncated* to 0.82–0.98, a sixth of the scale; chance is 0.50, marked below the axis break and far under everything drawn. Both panels share that truncated scale, so the two spans may be read against each other. Left, the reference as each split is actually scored, 0.956 on train against 0.824 and 0.836 on the held-out splits, a spread of 0.132. Right, the same reference with the resolved +0.1086 reliability line imposed on all three, 0.956, 0.961 and 0.960, a spread of 0.006; at this scale the three sit inside a bracket two millimetres tall, so they are listed beside the cluster in their own order rather than labelled in place. The left-hand spread is a selection artefact. The markers are bare: no interval is carried for this reference in the source record, and none is drawn rather than invented.

THE DESIGN LEAKS THE DRUG’S CLASS. Holding out a drug does not remove everything the model knows about it. The prompt contains Mechanism: {moa}, so a held-out drug still arrives with its class label. The correct design is *leave-one-mechanism-out*.

The leak is measured, and the measurement is not clean. Scrambling the Mechanism field alone moves the model’s NIR by +0.0303 [−0.0078, +0.0685] ($n = 738$, clustered on drug and well), an interval containing zero. That understates the channel; it does not bound it, because this row’s swap partner scores 0.520, above chance, and a comparator above chance suppresses the gap (Table 4.13, Panel B). A mechanism-only lookup is the same measurement from the other side and equally inconclusive. Pooled over the $n = 218$ conditions on which both are scored, model − moa_lookup = +0.044 [−0.048, +0.135], clustered two-way on cell line and well, spans zero, so the comparison settles nothing in either direction.

On unseen_drug, the split where it would matter, the mechanism lookup is ahead of the model by 0.099 on 44 conditions; in the contrast’s reported direction, model minus lookup is −0.099 [−0.222, +0.008], clustered on cell line, spanning zero and far too wide to settle anything. That measurement has no power at exactly the point where the design defect bites. It is not evidence

that the leak is doing work. The design is wrong, the arm cannot say what the wrongness cost, and its proper use is as an instrument control.

IT IS UNDERPOWERED BY CONSTRUCTION. The half-width on the unseen_drug gap is ± 0.052 clustered two-way on cell line and well, and ± 0.073 clustered on the drug; the smallest effect detectable at 80% power is ≈ 0.079 . The largest effect the mechanism channel could deliver is the distance from chance to moa_lookup, which scores 0.552 pooled across splits against a chance value of 0.500, so ≈ 0.052 . The arm *cannot* detect the largest effect it could plausibly be measuring (Figure 5.3).

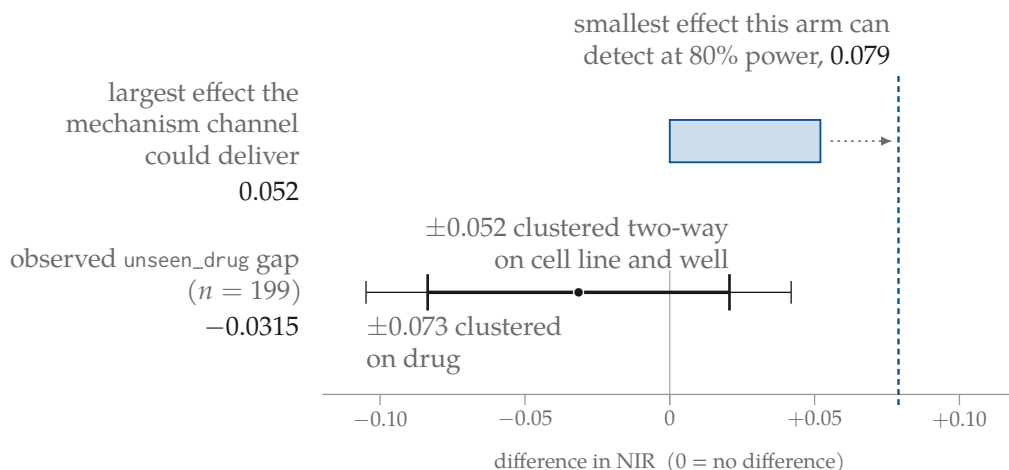


Figure 5.3. The bar ends before the threshold begins. Every quantity here is a difference in NIR on one scale; no absolute NIR is plotted. The dashed rule is the smallest effect the unseen_drug arm can detect at 80% power, 0.079, on the $n = 199$ conditions that arm scores. The bar is the largest effect the mechanism channel could deliver — the distance from chance to the mechanism-only lookup, which scores 0.552 pooled across splits against a chance value of 0.500, so ≈ 0.052 — and it stops short of the rule. Beneath it is the arm’s own estimate: model NIR minus the geometry-defined orth scramble comparator on the same conditions, -0.0315 , drawn with the thick span the ± 0.052 half-width clustered two-way on cell line and well and the thin span the ± 0.073 half-width clustered on the drug. Both cover zero, and the half-width on the tighter of the two is itself the size of the bar. The arm is licensed as an instrument control and is not evidence that unseen-drug transfer is zero: an effect of the largest size the mechanism channel could deliver would sit inside that same interval, so what it supplies is the absence of a contradiction rather than a confirmed null.

That cuts both ways, but asymmetrically. As a *control* the arm is not falsified, since the estimate $-0.0315 [-0.1049, +0.0419]$, clustered on drug, covers zero. It does not sit *at* zero, being negative and of the same order as the largest effect the arm could plausibly be measuring, so what it supplies is the absence of a contradiction. That is why the arm validates the comparator in § 4.10 but cannot argue that unseen-drug transfer is zero. “Unseen” is narrow in a second way too. Pythia-1b’s pretraining is natural-language text in which drug names occur, and the prompt still supplies Mechanism.

SCOPE LIMIT | carried into chapter 6

The `unseen_drug` arm is licensed as an instrument control. It is not evidence that unseen-drug transfer is zero. Its estimate covers zero, but an effect of the largest size the mechanism channel could deliver would sit inside that same interval, so what it supplies is the absence of a contradiction rather than a confirmed null. “Unseen” means held out from Tahoe fine-tuning and nothing wider than that.

5.3 What would settle it

A limitation earns its place when something could be run to remove it.

THE ARM TO BUILD CAREFULLY. The obvious continuation of § 4.12 is a channel-conditioned prompt that appends protein targets to the instruction, retrains, and evaluates on a leave-one-mechanism-out split. We do not recommend building it eagerly.

§ 4.13 does not resolve *which* span of the prompt is read. Swapping the drug *name* alone moves the prediction by +0.1026 [+0.0625, +0.1427] ($n = 1,394$); swapping the *mechanism* field alone moves it by +0.0303 [−0.0078, +0.0685] ($n = 738$), both clustered on drug and well. Both swap to an opposite-signature partner, which is not a null, so each gap carries a comparator term running in opposite directions across the rows, inflating the name arm, whose partner scores 0.434, and suppressing the mechanism arm, whose partner scores 0.520. Read as each row’s distance from chance the ordering inverts (+0.0367 on the name row against +0.0506 on the mechanism subset), so the split between the two spans is unresolved.

Pending a re-run against the neutral partner, the case against a target-conditioned prompt rests not on demonstrated mechanism-blindness but on the size of the readout deficit. On the common support the model covers 13.9% of the distance from chance to the within-well bar that a context-blind lookup covers entirely, and adding a field to a prompt does not touch that deficit.

Worth building instead is an intervention that makes the drug *causal* rather than merely present, a learned per-drug modulation of the transformation from control to treated, so the drug parameterises the computation instead of sitting in the residual stream as an input the readout can round away. That is a research architecture, outside the scope of what remained here.

TWO CHANNEL QUESTIONS THAT REMAIN OPEN. Target and mechanism overlap, so whether target adds anything *over* mechanism is unresolved, and it matters: the prompt already exposes mechanism, and whether the model reads that field is itself unsettled. Second, the gate of § 4.12 measures *similarity transfer*, an unseen drug resembling a seen drug with the same target. It does not test a *compositional rule*, that a named target’s own pathway moves, which would apply to any drug whose target is named, including one with no similar drug in training. That claim is the stronger one, it does not reduce to a lookup, and it is untested.

Both sat behind a defect in the gate itself. The partner pool was not restricted to training drugs, so a held-out drug’s response could be predicted from a partner that was also held out; the gate has since been re-run with

the restriction in place. The repair narrowed what may be claimed, and the clustering axis narrows it again. The target, mechanism and chemistry arms differ in support, drug set, partner count and annotation coverage, so their estimates are descriptive and do not identify a biological ranking.

CORRECTION the gate's partner pool, restricted and re-run in § 4.12

Protein target clears the relevance margin on the cell-line axis, misses it on the drug axis against both estimand-correct nulls, clearing it there only against the weaker count-matched one, and rests on 18 annotated drugs of which two are held out; mechanism misses its lower bound by 0.0021 on the cell-line axis and spans zero on the drug axis; chemistry's interval straddles the margin on both. Chemistry attenuates under the different-plate construction, but that construction also changes support and partner availability, so the attenuation cannot be assigned causally to co-plating. On its restricted support, same-target retrieval is the strongest observed lead, and stands only as a direction worth testing. The binding limitation is coverage: 18 drugs cannot settle it, and the experiment that would is a target-annotated panel broad enough to cluster on.

UNSEEN CELL LINES. A cell line never seen with any drug is a different generalisation axis from the held-out wells tested here, and one the plating design does not compromise in the same way. It asks whether the control cell's state is used *compositionally* with the drug.

The transfer coefficient scopes it only loosely. 44.7% [40.7%, 48.6%] of the similarity scale $\mathcal{T}$ is measured on is not shared across the compared contexts, and no structure was recovered in that unshared part by tests that do not settle it. That figure is not a drug $\times$ cell-line interaction share, since it mixes cell line, dose, well and estimator (§ 4.8), so it bounds what a per-drug constant leaves on the table without establishing that a new cell line is the axis the remainder lies along.

SEE § 4.8: $\mathcal{T}$, and what its complement does not measure

No consistent per-line direction was found, so the achievable target on a new cell line plausibly reduces to recalling the drug's per-drug constant, which a training-only lookup already does at 0.890 on held-out wells, the nearest measured analogue of a new context (0.913 pooled, inflated on trained conditions by the well advantage). The axis is less promising than it appeared, though the argument is weaker than a coefficient-based one.

OBJECTIVES THAT WERE CONSIDERED AND REJECTED. Three training objectives were designed and then set aside on their own mechanics, recorded so they are not re-litigated.

- **Reinforcement learning with a contrastive reward.** Its per-sequence reward keeps the token-dilution ratio out of the gradient, attacking a bottleneck the residual encoding has already lifted (34.6 $\rightarrow$ 118.2 differing tokens at cell-line scope, 120.3 at the plate scope the trained build uses). Its reward optimises similarity to the true residual, which drug_lookup already achieves at 0.913 pooled, 0.890 on held-out wells. Its best case is convergence to the lookup.
- **A differentiable profile-level contrastive loss.** The natural construction, $\text{score}(g) = \sum_i P(\text{token}_i = g) w(i)$, presumes gene $\approx$ token, but 86.9% of the 946 panel genes share a first sub-token under this tokeniser (mean 3.254

BPE tokens per gene, only 8 single-token). Under teacher forcing the target *is* the drug-specific residual, so the true prefix identifies the drug within a few symbols and the gradient vanishes at the readout solution the loss exists to improve on.

- **Loss re-weighting toward differentially expressed genes.** A re-weighted cross-entropy against a single target contains no negatives, so nothing forces one drug’s prediction to differ from another’s. It leaves the discrimination failure where it was, and that failure is what the measurements establish, whether or not the readout is also blind to direction specifically, which § 4.5 declines to establish.

ONE MODEL SIZE, AND THE SMALLEST ONE. Every result here comes from a 1B-parameter backbone. C2S-Scale releases checkpoints up to Gemma-2 at 27B, and we tested none of them, so the most predictable question an examiner can ask, whether it works if the model is bigger, has no answer in this thesis.

The constraint this work points to is not capacity. The target tokens are overwhelmingly shared across drugs — only 17% of them differ between two drugs in the same cell line under the standard target, against 59% under the residual encoding at cell-line scope and 60% at the plate scope the trained build uses (§ 4.9) — so very little of the training signal can carry drug identity. That step is a hypothesis about the optimisation; no token-level loss or gradient attribution was computed, so no fraction of the gradient is claimed. A larger model receives the same diluted signal from the same objective, and re-encoding lifted the effect while leaving capacity untouched, which points away from capacity without establishing what it points towards. But the re-encoding changed five things at once, aggregation, control referencing, sign coding and length among them, and a budget-matched optimal-transport target lifts the effect too, so § 4.9 declines to identify the tokenisation as the constraint that binds.

The genuinely different argument for scale is *prior knowledge*. A 27B model plausibly knows, as a fact about the world, that a named drug inhibits a named kinase, which a 1B model may lack. § 4.13 leaves unresolved whether the model uses the *mechanism* field it already has, at $+0.0303 [-0.0078, +0.0685]$ ($n = 738$) with a comparator above chance suppressing the gap, and real pharmacological knowledge is exactly the case where that might change. Testing it needs only a low-rank adaptation of a mid-sized checkpoint. We did not run it, and it is the first thing we would run with more time.

ONE EPOCH, AND A THIN SLICE OF THE ATLAS. Model size is pre-empted above at some length. Data scale is the other half of that question and it is not the same one. Fine-tuning saw 620,564 cells, 600,000 treated and 20,564 control, drawn from the roughly one hundred million Tahoe-100M holds, covering 106 drugs of its $\approx 1,100$, for one epoch at one learning rate (Table A.3). Under a hundredth of the atlas was used, and no arm saw any cell twice.

The diluted-signal argument above is a claim about the objective and survives this: a model trained on ten times the cells receives the same target strings, of which only 17% of positions differ between two drugs in one cell line (§ 4.9), so more data of the same kind does not make the drug easier to see

in the loss. The ten-epoch diagnostic separately rules out undertraining in the optimisation sense, since validation loss is best at the first epoch and worse at every epoch reported after it. Neither argument rules out the possibility that a drug needs to be seen in more cellular contexts than this sample provides before its response becomes learnable, and that possibility is not established either way. It is a cheaper experiment than a larger backbone and it is not one we ran.

THE STARTING CHECKPOINT WAS NEVER PREPARED FOR THIS TASK. We fine-tune from C2S-Scale-Pythia-1b-pt, whose cell-sentence pretraining corpus is atlas data alone, CELLxGENE and the Human Cell Atlas, with no perturbation corpus of any kind, and whose released capabilities do not include perturbation prediction.

That is why the held-out splits are as clean as they are. A drug withheld here entered neither the fine-tuning data nor the generic that defines its target, the split being assigned from metadata before any expression was aggregated, so no Tahoe perturbation response for it can have been memorised. But the failure we report is then a failure of *this* starting point, and the fair follow-up is whether a checkpoint already tuned for perturbation prediction would behave differently.

Within the C2S-Scale line that question cannot be answered from what is public. The perturbation-specialised model there is a Gemma-3 4B checkpoint that was not released. Outside that line it is answerable, and we did not answer it. Tahoe-x1 [13] is a perturbation-trained foundation model on this same atlas, public before this thesis was submitted. It is not a cell-sentence model, so it does not isolate the representation variable, but it is the obvious external check on whether a readout failure of this kind survives perturbation-specific pretraining, and it was not run here. That is a gap in this work, not a gap in the public record.

Every arm starts from the same base and differs only in the variable under test, with one material exception, the consensus arm, stopped at roughly 60% of an epoch with its tier-1 NIR unscorable (Table A.3). The *comparisons* between arms are therefore unaffected even where the absolute level is not, and that arm's null is quoted with the qualification wherever it is used.

GENERALITY. Everything here is one backbone on one atlas in one representation. Whether the readout-specificity failure belongs to cell-sentence models specifically, or to autoregressive conditional generation over transcriptomes generally, is open. The mechanistic protocol of § 4.4 to § 4.5 (purified subspace, removal gate, positive control, matched-norm swap) transfers unchanged to any model with an accessible residual stream.

■ EXPR / ▲ RESID

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

The expression-target checkpoint makes no useful condition-specific predictions. Residual encoding reveals real sensitivity to trained drug names, but the checkpoint recovers less than a simple lookup and establishes no transfer to new drug-context combinations or unseen drugs.

SCOPE LIMIT | carried out of the thesis; see chapter 5

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6.1 Only one of five metrics survived its own calibration

A negative result is worth no more than the ruler that produced it, and the field's default ruler failed a test it should have passed. A zero-information predictor placing every gene at the middle rank scores $DE-\Delta r = 0.999913$ $[0.999908, 0.999919]$, above both the two-cell reference and the model. Dynamic Range Fraction therefore grades a metric by whether it rewards a real replicate more than that arm.

Only NIR of the five is positive, returning $+0.635$ $[+0.604, +0.663]$ across plates over 25 cell lines and $+0.545$ $[+0.516, +0.569]$ within plate over 27, each at Holm $p = 0.002$. Three of the other four invert; the weighted R^2 variant lies nearest zero with its sign tracking cell count, so no directional claim is made for it. A residual-frame calibration returns $+0.704$ $[+0.667, +0.736]$ over 42 cell lines against a zero-drug-information control. All are percentile cluster bootstraps without small-sample correction, clustered on cell line after an earlier artifact clustered on drug rows; plate stays crossed with cell line, so they are robust on that axis alone.

SEE § 4.1 builds the audit and reports all five arms

CORRECTION corrected in § 4.1

What travels is the procedure. A perturbation metric must first show positive dynamic range against a real replicate and an uninformative predictor; the lookup is a second such bar, the removal gate a third.

WHERE THIS LEAVES THE CALIBRATION DISPUTE. Miller et al. [20] argue that miscalibrated metrics can manufacture apparent failures of deep perturbation models (§ 2.4). This thesis confirms the point, applies it, and still reaches a different verdict. Their rank-based NIR survives the audit and the checkpoint is at chance on it, which bounds their conclusion's extension from genetic perturbation to these chemical predictions without overturning it.

6.2 In the expression frame the checkpoint is drug-blind

On the calibrated metric the checkpoint fails useful condition-specific prediction. Within plate on unseen drugs it scores 0.498 against 0.504 for a control-copy that reproduces the untreated cell and 0.576 for a within-well split-half precision reference.

THE NULL IS NOT A FAILURE TO MEASURE. A null can always be blamed on the wells, so the aggregate was re-scored where the wells are demonstrably adequate. On the 204 conditions whose own reference clears the pre-set 0.80 identifiability bar and averages 0.938, the model reaches 0.768 and control-copy 0.766, a paired difference of $+0.002$ $[-0.026, +0.028]$. Split-half retrieval does not prove a drug's response is predictable from an untreated cell, but at the tested design and cell budget it rules out a failure to measure distinct treated populations.

THREE INSTRUMENTS THAT FAIL DIFFERENTLY AGREE. Changing the drug and mechanism while holding the cell fixed moves the prediction by $+0.014$ $[-0.016, +0.042]$ on the identifiable subset. Output invariance, which asks whether changing the drug token changes the generated text at all and not whether it changes it correctly, returns 0.000 $[-0.019, +0.016]$ in

top-100 Kendall τ , bounded at ≤ 0.016 against $\tau = 0.198$ between two generations from a byte-identical prompt; the Jaccard version is tighter, $+0.000 [-0.008, +0.008]$. Forced choice, grading a prediction against its own condition's truth and against another condition's, returns an own-truth win rate of 0.48 against a chance 0.50 fixed by construction, where the precision reference reaches 0.67–0.83. Their weaknesses do not overlap, which is why the agreement carries weight. No drug effect is shown to be zero by any of them, and any effect they miss is too small to support the intended prediction task.

6.3 The drug is not absent; it is barely consulted

The third question of § 1.4 asks where that null comes from: the drug absent from the data, absent from the model's internal representation, or present in the representation and not consulted when the model generates. The re-scoring above rules out the first at the tested design and cell budget. Behaviour cannot separate the other two, since both predict the same scores, so the weights were read instead.

THE LABEL IS NOT LOST BEFORE GENERATION. After cell line, plate and dose are projected out, drug identity becomes more linearly decodable with depth, reaching 0.88 at layer 16 against an 8% chance floor. A later probe suite asked whether that signal is read; its activation-swap interpretations did not survive their controls. What survives supports only low-gain reading of a decodable drug label, in one training arm and not another, surviving pooled multiplicity correction at one displacement rung only and on one of two stored bootstrap draws; no conclusion rests on it alone.

CORRECTION the swap arm and its failed controls are in § 4.5

6.4 Re-encoding produces the first trained-drug effect

If the drug is in the data and in the activations, the next suspect is the training target. Standard cell-sentence targets for two drugs in the same context share 82.7% of their top-200 genes, a property of the targets and not a token-level attribution of the gradient. Residual encoding raises the differing fraction to 59% at cell-line scope and 60% at the training plate scope.

Against the geometry-defined orth partner, trained conditions score $+0.0898$, with intervals excluding zero clustered on cell line and well, $+0.0898 [+0.0382, +0.1415]$, and on drug and well, $+0.0898 [+0.0291, +0.1506]$. That is the project's first measurable trained-drug effect.

THE COMPARATOR IS PART OF THE MEASUREMENT. A prompt replacement is not automatically a null. The scrambled arm itself scores 0.550, 0.501 and 0.423 as the partner moves from aligned through orthogonal to anti-aligned, so a dissimilar lie inflates the apparent benefit of the correct name. The orth stratum, with the comparator's own score reported, avoids that inflation without manufacturing an independent null.

THE GAIN IDENTIFIES A PACKAGE, NOT A MECHANISM. Residual encoding changes aggregation, control referencing, generic subtraction, sign coding and sequence length together, and separating them needs a factorial comparison that was not run. Nor is it uniquely responsible. At the matched 1,400-token budget the optimal-transport arm also clears zero against its orthogonal comparator, $+0.0292$ [$+0.0144$, $+0.0441$], where its 600-token estimate spanned zero. A generation cap can manufacture a null, so any claim about the residual advantage must retain the budget comparison.

ORDINARY UNDERTRAINING WAS EXAMINED SEPARATELY. Validation cross-entropy on the ten-epoch residual diagnostic is best after epoch 1 (1.9199) and ends at 2.2568 while training loss keeps falling. The schedule was stretched, not continued, and no checkpoint from it was scored with NIR; extending this optimisation does not improve its held-out objective, and no claim is made that every schedule must fail.

6.5 The trained-drug effect establishes no transfer

One question is whether the prediction moves when the drug named in the prompt changes. The other is whether the prediction is right, moving *toward* the response of that drug. The trained-condition effect establishes the first for drugs represented in fine-tuning, and it does not establish biological transfer.

Conditions in held-out treated wells score $+0.0332$, excluding zero clustered on cell line and well, $+0.0332$ [$+0.0069$, $+0.0594$], but not on drug and well, $+0.0332$ [-0.0031 , $+0.0695$]. Drug is the axis a transferred residual must generalise over. That split is largely recombination, with 30 of its 34 drugs and 39 of its 42 cell lines in training and 82% of its conditions recombining a training drug with a training cell line. On genuinely unseen drugs the gap is -0.0315 [-0.0836 , $+0.0206$], both axes spanning zero. The splits differ in reliability selection, so their point estimates are not a clean decay with novelty.

THE PARTNER SLOPE MEASURES THE TRAINED DRUGS. It is positive on every split, but fitted only over scrambled partner prompts, most of which are themselves trained drugs on the unseen-drug split, and the true target drug is not a point on that line. Held-out predictions sit near chance when ranked against other drugs in the same cell line. The map is coarse and trained-drug-to-residual, not a transferable response model.

6.6 Nothing built here beat the lookup

A per-drug lookup needs no checkpoint, no prompt and no cell sentence. It predicts a drug's residual as its mean residual in other cell lines, fitted from training conditions only.

On the 1,192 conditions where both are defined it scores 0.913, the model 0.549, the within-well reference 0.857. It reaches 0.980 on trained conditions, 0.890 on held-out ones, 0.822 restricted to one other cell line. Its excess over the split-half reference reflects a many-line average against a noisy half-sample on a metric compressed near its upper limit, not performance beyond biological

SEE § 4.11 scores the full ladder through one pipeline

headroom. Nothing conditional built here beats direct retrieval of the drug's mean residual.

A BAR, NOT A FULL ACCOUNT. That average is not a full account of the drug's effect. A disattenuated cross-context transfer coefficient is $\mathcal{T} = 0.553$ [0.514, 0.593]; a fully shared simulated null returns 1.001 and a structure-matched negative control -0.008 $[-0.036, +0.020]$. About 45% of the similarity scale on which $\mathcal{T}$ is measured is not shared across compared contexts. The lookup's 0.913 therefore does not mean drug response is nearly constant; it means NIR recognises the shared per-drug component and compresses distinctions near its upper end.

SCOPE LIMIT | carried out of the thesis; see chapter 5

The unshared part is explicitly *not* a drug-by-cell-line interaction or a variance share. The estimator mixes cell line, dose, well and target estimation, was run with `same_dose_only` and `loo_generic` off, and still leaves roughly 0.3 unshared when dose varies within one cell line. Reading $1 - \mathcal{T}$ as the learnable opportunity is equally wrong. It is an upper bound on what the constant leaves unshared under this estimator, not evidence a model can predict that remainder.

Two tests looked for learnable structure in that remainder and found none detectable. Within-line consistency is -0.0149 $[-0.0251, -0.0050]$ against $+0.0003$ $[-0.0071, +0.0081]$ across lines, and a mechanism-matched context test is null. Neither makes the remainder irreducible. That second quantity has low split-half reliability and mechanism is heavily confounded with plate, so resolving it needs more cells per condition or a design crossing mechanism with plate.

DR EVAL'S LESSON, WITH A VECTOR IN PLACE OF A SCALAR. DrEval finds deep drug-sensitivity models often add little beyond mean drug and cell-line effects [19]; here the predicted object is a 946-gene vector, not scalar potency, and a per-drug mean again dominates. Simple baselines are not thereby biologically complete. A conditional architecture earns its complexity only by beating the retrieval component already available from training.

6.7 For an unseen drug, one lead is live on the cell-line axis and nothing is closed

No lookup exists for a drug absent from training, so the final gate asks which shared property could stand in for one, testing three channels with partners restricted to training conditions.

The gate reports in four states, and the distance between them is the point. A channel is **LIVE** when its interval clears the pre-set 0.03 relevance margin on the clustering axis the claim rides on, **CLOSED** when the interval lies wholly below that margin, and **INCONCLUSIVE** otherwise. It is **UNTESTABLE** when the matched control could not be built at all: "we could not build the control" must never be written as "closed".

SEE § 4.12 prints the gate's verdicts beside each channel's coverage

THE VERDICTS. Protein target clears the margin under all three constructions clustered on cell line and well; clustered on drug it clears only against the weaker count-matched null and falls short against both estimand-correct nulls. Mechanism excludes zero on the cell-line axis but falls just below the margin and spans zero on the drug axis. Chemistry is narrower and unresolved, excluding zero against the plate-matched null on the cell-line axis but spanning zero in the different-plate construction and under both nulls on the drug axis.

Cell-line clustering asks whether a channel effect is stable across represented cellular contexts; drug clustering asks whether it generalises across the drugs an unseen-drug claim rests on, so target can be live under the first and inconclusive under the second without contradiction. The three nulls answer different questions and the arms differ in support and annotation, so their order is not a biological ranking.

The strongest lead rests on only 18 target-annotated drugs, two held out. Protein-target annotation carries transcriptional signal on that mixed-support arm, but a Leave-Drugs-Out result is not established by it, and mechanism and chemistry are inconclusive rather than closed. The arm's estimated 80%-power detection threshold is about 0.079 in NIR, so effects materially smaller than eight points are not well powered. The next experiment needs enough held-out drugs, preferably held-out mechanism families, for drug-clustered inference.

TWO REPAIRS THAT MUST BE DESIGNED, NOT ASSUMED. One sentinel token cannot be expected to map to a transcriptional residual, so a target or mechanism description is the obvious bridge from an unseen name. It must be read against the backbone's prior knowledge: holding a drug out of Tahoe fine-tuning does not erase medicinal-chemistry text from language pretraining. That arm needs matched description quality, missingness controls, a frozen knowledge probe and enough held-out drugs for drug-clustered intervals. A larger backbone is a replication, not a scaling law, unless the family is held fixed, since Pythia-1b to Gemma-2 changes tokenizer, architecture, corpus and scale together.

6.8 The controls are most of the result

In a literature where uninformative baselines can score well, a negative result is worth what its controls are worth. Each surviving claim rests on a control added after a specific failure. Plate-matched comparison sets followed a zero-information baseline that improved when plate was free to vary; a swap-distance sweep followed a weak similar-drug scramble. A removal gate preceded any ablation null and matched-norm injection any swap interpretation. Intervals were clustered on the units the estimand requires, cell line, drug and treated well, never on cells, and a pre-registered interpretation was allowed to fire its inconclusive branch.

Claims that failed those controls were withdrawn, and the withdrawals recorded, not quietly dropped. That record is why the remaining claims can be read at the strength stated here, and no stronger.

THE CLAIM. Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support. On the 1,192 conditions where both are defined, the training-only per-drug lookup scores 0.913 against the model's 0.549, with the within-well reference at 0.857; restricted to a single other cell line it still scores 0.822. Residual re-encoding does produce a real trained-condition effect against the orth partner, with intervals excluding zero on both clustering axes, and it does not establish transfer: conditions in held-out treated wells give an interval that excludes zero on the cell-line axis and spans zero on the drug axis, which is the axis a transferred residual must generalise over, and on genuinely unseen drugs the gap is -0.0315 $[-0.0836, +0.0206]$ with both axes spanning zero. No drug effect is shown to be zero by any instrument here; the three that bound it fail differently and agree. Cross-context biology leaves a real remainder, $\mathcal{T} = 0.553$ $[0.514, 0.593]$, but $1 - \mathcal{T}$ is an upper bound on what a per-drug constant leaves unshared under this estimator, not a learnable opportunity. Of the three channels open to a drug with no lookup, protein target is live on the cell-line axis and inconclusive on the drug axis on 18 annotated drugs of which two are held out, mechanism and chemistry are inconclusive, and none of the three is closed.

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Appendix

This appendix holds the apparatus the body leans on but does not stop to justify. Each section states its verdict in its heading. None is a new result; several are defects that changed one.

A.1 The bespoke estimators recover a planted answer, in both directions

■ ■ ■ ■ ■ ■ ■ ■ ■ ■

▲ RESID

Three measurements rest on estimators built for the purpose, and a reader has no prior reason to trust them. The instrument is a world with a known answer planted in it, run twice, with the effect and without. All three recover what was planted, and nothing when nothing was.

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

The weight falls on the null direction.

WHAT THE TRANSFER-COEFFICIENT ROWS DO NOT SHOW. They are a data-free unit test whose planted world has cell line as its only unshared axis, so recovery shows the estimator inverting its own generative model and no more. The canonical run, computed with `same_dose_only=false`, `loo_generic=false` and `repro_thr=0.2`, lets cell line, dose, well and estimator noise all enter one number. On synthetic data calibrated to its gene dimension, cell counts and empirical noise, the estimator returns 1.001, 0.747 and 0.494 for planted $\mathcal{T}$ of 1.00, 0.75 and 0.50.

PLANTED WORLD

A synthetic run with the answer built in, no data, no network, one fixed seed.

SCOPE LIMIT | carried into § 4.8

Neither the planted rows nor the calibrated repeat licenses reading $1 - \mathcal{T}$ as a drug $\times$ cell-line interaction share. The quantity measures the fraction not shared across the compared contexts, and in the canonical run those contexts differ in more than cell line.

TWO OF THESE CHECKS CHANGED THE INSTRUMENT. The count-matched null originally drew random partners from a pool *excluding* the channel's own selections, which depletes the null of the partners the channel chose and anti-correlates the two arms. And the useless-channel gap has a standard deviation of roughly 0.09 on one small world, so the original validation, asserting that a single draw fell within 0.10, tested luck; it now averages 24 worlds. That same

Table A.1. All three estimators recover planted answers in both directions. Every row is a synthetic planted-world check, and none of these numbers comes from the evaluation run or is stored in a result artifact. The $\mathcal{T}$ and κ -structure rows reproduce from `variance_decomposition.py --selftest`, the channel-gate rows from `channel_gate.py --selftest`. The κ -structure check must separate two worlds, within-line consistency exceeding across-line where every cell line bends what it is given (+1.17), the excess vanishing where κ is specific to each drug-and-line pairing with no line-level direction (−0.00). The lower row of each pair is the null world, the one that rules out an instrument reporting structure wherever it looks.

Planted truth	Recovered
<i>transfer coefficient $\mathcal{T}$</i>	
$\mathcal{T} = 1.00$ (fully shared)	0.999
$\mathcal{T} = 0.70$	0.703
$\mathcal{T} = 0.40$	0.404
<i>channel gate, over 24 worlds</i>	
a channel that genuinely predicts	+0.397 (sd 0.029)
a channel carrying nothing	−0.016 (sd 0.091)
<i>κ-structure</i>	
a consistent per-line modulation	+1.173
an idiosyncratic interaction	−0.002

variance is why the channel results are gated on a clustered interval and not on a point estimate.

A THIRD DEFECT WAS FOUND AFTER THE CANONICAL RUN. The *retrieval* pool, whose partners’ mean residual forms the prediction, was every retained condition in the cell line except the target’s own drug, so a condition held out of training could enter the prediction for a held-out drug. That is not what § 4.12 says the gate was intended to do. The code now restricts the pool to the split’s training conditions, logs its size, and never falls back silently.

The training subset is 2,523[?] of a retained total logged as 4,131[?] **AUDIT: THE SPLIT TALLY SUMS DIFFERENTLY** conditions, so the old pool was around 64% larger than it was entitled to be. What makes it a defect is that a held-out condition could enter at all, which holds at any tally. Every channel-gate figure in the main text is from the corrected re-run; the planted worlds are unaffected. The repair moved the channels in no common direction. Target fell, mechanism rose, chemistry roughly halved, so the pre-fix figures are not quoted alongside them.

CORRECTION The channel gate’s retrieval pool, re-paired and re-run after the canonical run. § 4.12

Table A.2. No bin in either model has an interval excluding zero in this historical 1200-token run, and the trend correlations are ≈ 0.200 (drug, swap) pairs over 20 cell lines, $n = 40$ per bin. The single-cell column bounds a large distance effect here. The optimal-transport column does not carry into the canonical result, where the same checkpoint clears zero at 1400 tokens under the within-plate stratified test (§ 4.9).

Bin (swap distance)	single-cell model	optimal-transport model
0, nearest (2.75–4.03)	−0.046	−0.001
1 (4.03–4.60)	+0.060	+0.031
2 (4.60–4.93)	−0.006	+0.017
3 (4.93–5.32)	−0.004	+0.043
4, farthest (5.32–6.25)	−0.019	−0.005
trend, corr(distance, Δ_{scr})	+0.051	+0.012

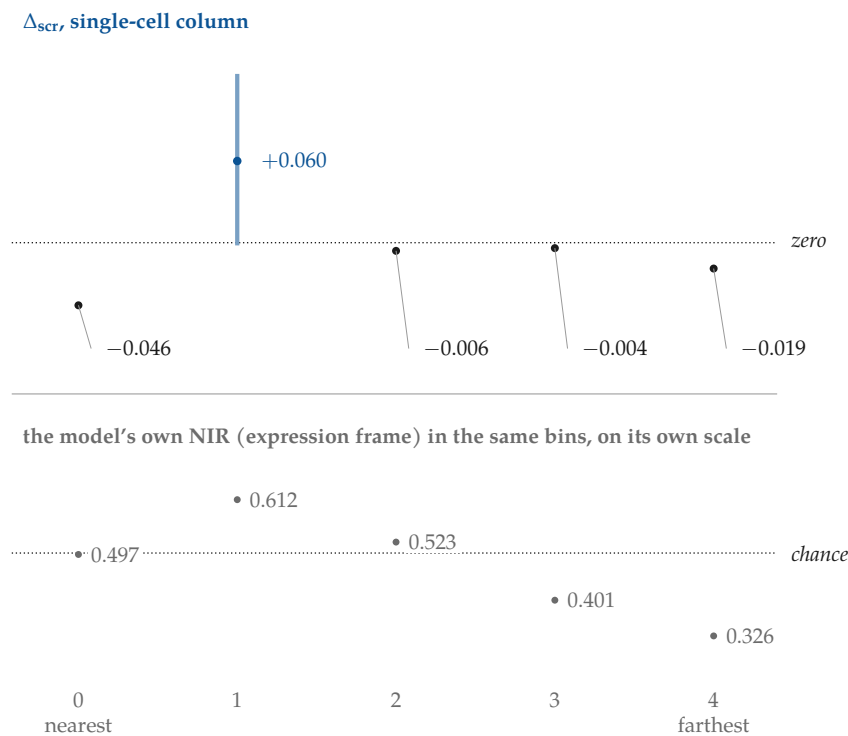


Figure A.1. The one borderline value sits in an inner bin, and the drift that could have faked a trend is in a quantity the gap cancels. Upper strip, the scramble gap in the single-cell column across the five swap-distance bins of Table A.2; the bar marks the only bin for which an interval is quoted, $+0.060 [-0.002, +0.124]$. A genuine dissimilarity effect would be monotone and peak at the right-hand edge, and the one value approaching zero from above is the second bin of five. Lower strip, the model's own NIR in those bins (0.497, 0.612, 0.523, 0.401, 0.326), falling overall but not monotonically, because drugs far from everything populate the far bins. The strips are different quantities on different scales; Δ_{scr} uses the same drug on both arms, so that drift cancels. Neither strip is canonical ($\text{max_new_tokens}=1200$, $n = 40$ per bin).

WHAT THE SWEEP CAN AND CANNOT SUPPORT. $n = 40$ per bin gives half-widths of 0.06–0.10 in the single-cell column and 0.03–0.06 in the optimal-transport column, so the sweep excludes a large effect without establishing a null. The later budget-matched result withdraws the optimal-transport column as evidence of a null, that verdict being generation-budget dependent and non-null at 1400 tokens (§ 4.9).

SEE § 4.2 for the stratified scramble that replaced this design

CORRECTION The optimal-transport column of this sweep, withdrawn as evidence of a null. § 4.9

SCOPE LIMIT | carried into § 4.2

Only the single-cell column remains a bound from this earlier design, and it is a bound and not a null. At this sample size it excludes a large dissimilarity effect and leaves a small one open. The optimal-transport column has been withdrawn as evidence of a null, so nothing in this section rests on it.

A.4 Undertraining is excluded as a trivial explanation, not as a general one

A negative result invites the objection that the model was simply not trained enough, and that objection deserves an answer from the logs. Two instruments give one, what the loss was doing while the learning rate was still large, and what a much longer schedule does to held-out cross-entropy. Under this schedule, longer optimisation does not rescue it.

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

SCOPE LIMIT | carried into § 4.5

The residual arm's two measurements are not on the same weights. The step count recorded above is the rebuilt target build's, which is the build the canonical behavioural results are scored on, whereas the identity test of that arm ran on the earlier target build. A behavioural result for that arm and an activation probe of it therefore come from two different builds, and this qualification travels with any statement that pairs them.

HOW PANEL (A) MUST BE READ. Two properties of the one-epoch logs behind Figure A.2 decide it. The trainer records a *cumulative* epoch mean, not an instantaneous loss, and that flattens by arithmetic whatever the model is doing, so every series is de-averaged to its per-window value before plotting. And the cosine schedule anneals the learning rate to ≈ 0 by the end of the epoch, 2.96×10^{-10} at the last logged step against a post-warmup peak of 10^{-5} , so the final third of each curve is flat partly because the step size has gone to zero, and is not independent evidence of anything.

What is load-bearing is the band between 50% and 70% of the epoch, where the learning rate is still 22–52% of peak, enough to move the weights, and the residual arm's loss has already flattened, travelling only from 1.859 to 1.854.

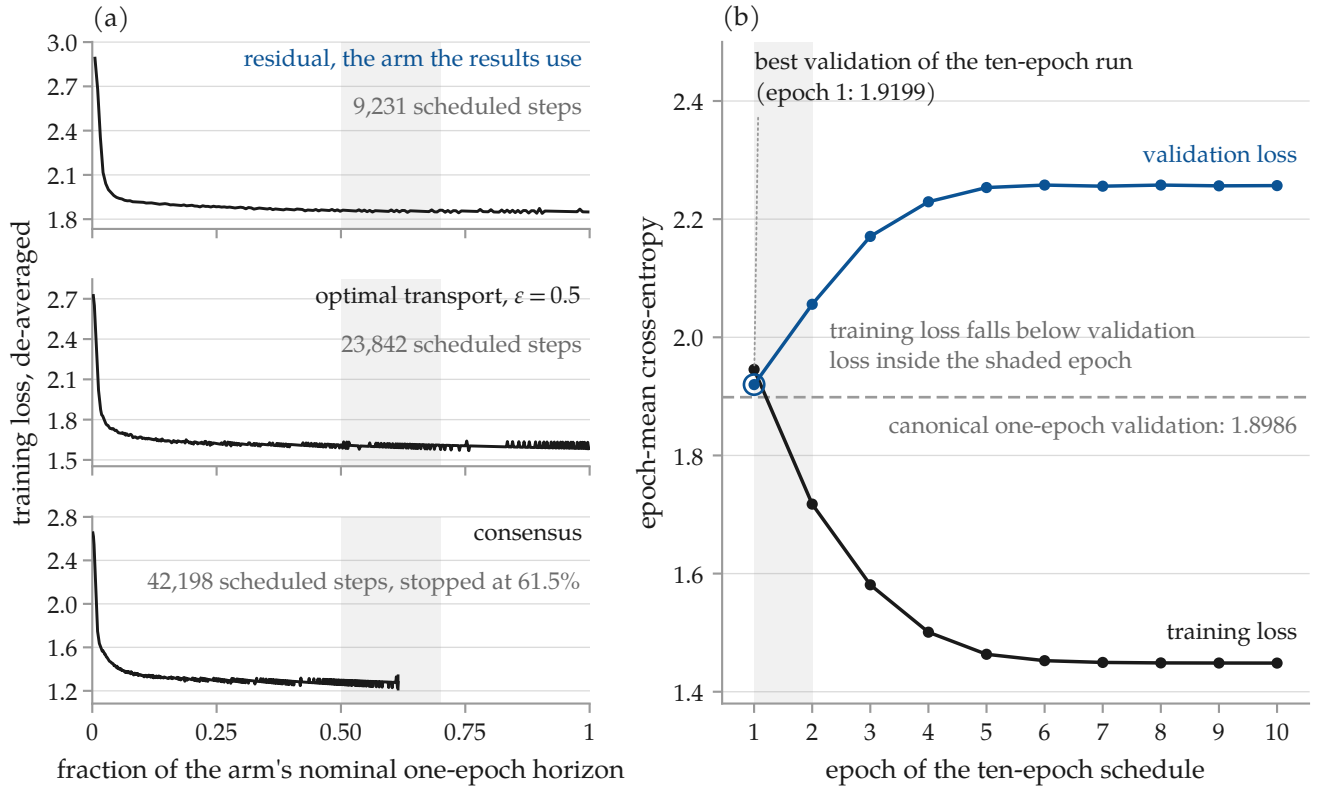


Figure A.2. The canonical short-run arms flatten, while a separate longer residual schedule overfits on held-out cross-entropy. (a) De-averaged training loss against fraction of the nominal one-epoch horizon, three target constructions, each on *its own* y -axis: absolute losses are not comparable between arms because their target strings differ, so no shared scale is offered. The residual and optimal-transport arms complete their epoch; consensus was scheduled for one but terminated by the wall-clock at 61.5%, where its line stops. Three superseded residual builds are omitted, as is the single-cell arm, whose training log was not retained. Every arm runs a cosine schedule annealing to ≈ 0 ; the shaded band marks 50–70% of the horizon, where the residual arm's learning rate is still 22–52% of its post-warmup peak of 10^{-5} and its de-averaged loss travels only from 1.859 to 1.854. (b) Epoch-mean losses for the ten-epoch residual schedule (nominal horizon 92,318; 92,310 optimizer updates). The ringed point is that run's best validation loss, at epoch 1. The dashed line is the canonical one-epoch run's validation loss on the same 3,600-example file, 1.8986, which the ten-epoch run does not reach at any epoch. Training loss is above validation at epoch 1 and below it at epoch 2, so the crossing falls inside the shaded epoch; where inside it is not logged, and no point is drawn for it. Two things this figure does not establish: that all possible learning-rate schedules are ruled out, and that NIR worsens with additional epochs — no long-run checkpoint was scored with NIR.

Its final held-out loss, on a matched validation file of 3,600 examples, is 1.8986 against a training loss of 1.8854.

Table A.3. One base checkpoint, one hyperparameter set, five arms differing only in target. Learning rate 10^{-5} , batch size 1, gradient accumulation 16, maximum length 8192, bf16, weight decay 0.01, warmup ratio 0.03 throughout. The consensus arm stopped early on a flat loss, so it is the ε -ladder’s weakest-evidenced endpoint, and at the residual arm’s generation budget against a neutral partner the optimal-transport endpoint is not null either (§ 4.9); outside Table 4.6 the ladder’s negative rests on the single-cell endpoint alone. These five arms (`single_cell`, `consensus`, `ot_T2`, `residual`, `residual_holdout`) are also the complete family over which the identity test of § 4.5 takes its multiplicity correction, five arms by up to seven depths and 26 measured tests, the family behind the q -values quoted there. One qualification attaches to the residual row. Its step count is the rebuilt target build’s, on which the canonical behavioural results are scored, while the identity test of that arm ran on the earlier target build.

Arm	Target	Training	Note
single cell	the observed treated cell	1 epoch	
consensus	group pseudobulk	$\approx 60\%$ of an epoch	tier-1 NIR unscorable
optimal transport (T2)	barycentric, $\varepsilon=0.5$	1 epoch, 23,842 steps	
residual	signed residual	1 epoch, 9,231 steps	rebuilt build’s step count
residual (holdout)	as above, 738 withheld	1 epoch	

PANEL (B) TESTS ORDINARY LONGER OPTIMISATION DIRECTLY. Both residual logs agree on base model and tokenizer, training and validation files and counts, seed, effective batch, and peak logged learning rate; weight decay and AdamW betas are not logged and are not claimed as verified matches (`training_adequacy_long.json`). The ten-epoch run has a nominal horizon of 92,318 but executes 92,310 optimizer updates, accumulation resetting at every epoch boundary so that each epoch contributes $\lfloor 147,710/16 \rfloor = 9,231$. It is a separate longer schedule, not a continuation; changing `num_epochs` stretches the cosine schedule across all ten epochs.

The result runs the wrong way for the objection. Validation loss is best at epoch 1 (1.9199), worsens to 2.0559 and 2.1708 at epochs 2 and 3, and ends at 2.2568, while training loss falls from 1.9456 to 1.4485. The canonical one-epoch run remains better on the same validation file (1.8986).

SCOPE LIMIT | carried into chapter 5

That result does not show that every possible schedule would fail, and no long-run checkpoint was scored with NIR, so it says nothing about whether NIR itself worsens with additional epochs. Together with the one-epoch traces, it excludes undertraining as a *trivial* explanation, not as a general one.

A.5 Three identity-test designs failed, and each failure is a constraint on the fourth

The substitution test the body reports is the fourth design, and the three before it each produced a result this thesis once quoted as a finding. The instrument here is the algebra of each failure, what the control arm was actually doing.

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

All three failures below are failures of the control arm, not of the readout. Figure A.3 draws the geometry each of them misses, and its three insets are the three designs in the order they are argued here.

FAILED DESIGN ONE, MISMATCHED GEOMETRY. The first test overwrote drug A's code with drug B's inside a ≤ 10 -dimensional purified subspace, and compared the resulting KL against a random vector of identical norm drawn in the *full* 2048-dimensional stream. Roughly 99.5% of the control's energy then lay where the intervention never reached, including the cell-line directions at ≈ 0.6 variance share. "The swap moves the prediction less than noise" is what that geometry predicts, whatever the readout does.

FAILED DESIGN TWO, ABLATE-THEN-ADD. The second drew the control inside the subspace, which looked like a repair. The hook ablates before adding, so what is delivered is (added – removed), and an uncentred class mean shares its global-mean component with the thing removed. The substitution arm was then close to a restoration, the control arm a real displacement. Simulated with no model present, the control delivers $5.3\times$ the perturbation in 100% of draws, predicted gap $2\langle\mu_B, u\rangle = 9.179$ against an observed 9.162, with μ_B the injected class mean and u the cell's component in the slab. The retracted result was geometry, not biology.

FAILED DESIGN THREE, THE NORMALISER. The third injected the displacement and scored $\Delta \log P(g_B)$ against a norm-matched random direction, and was confounded the opposite way. A random direction inside the slab inflates the logsumexp normaliser (measured +2.75 nats) where the observed class-mean displacement does not, so every target is depressed under the control arm whether or not it carries identity. With the scored target stripped of all identity information, 36% of the effect still stood. Norm-matched is not disruption-matched.

That argument also retires the simplest *injection* test, "how far did the prediction move". Distance moved is what norm-matching equalises, so the test has no power in precisely the world where the readout does read the drug. It does not reach the behavioural output-invariance instrument of § 4.2, which perturbs the prompt and matches no norms; that instrument's own

CORRECTION The ablate-then-add identity result, retracted as an artefact of the hook's own algebra.

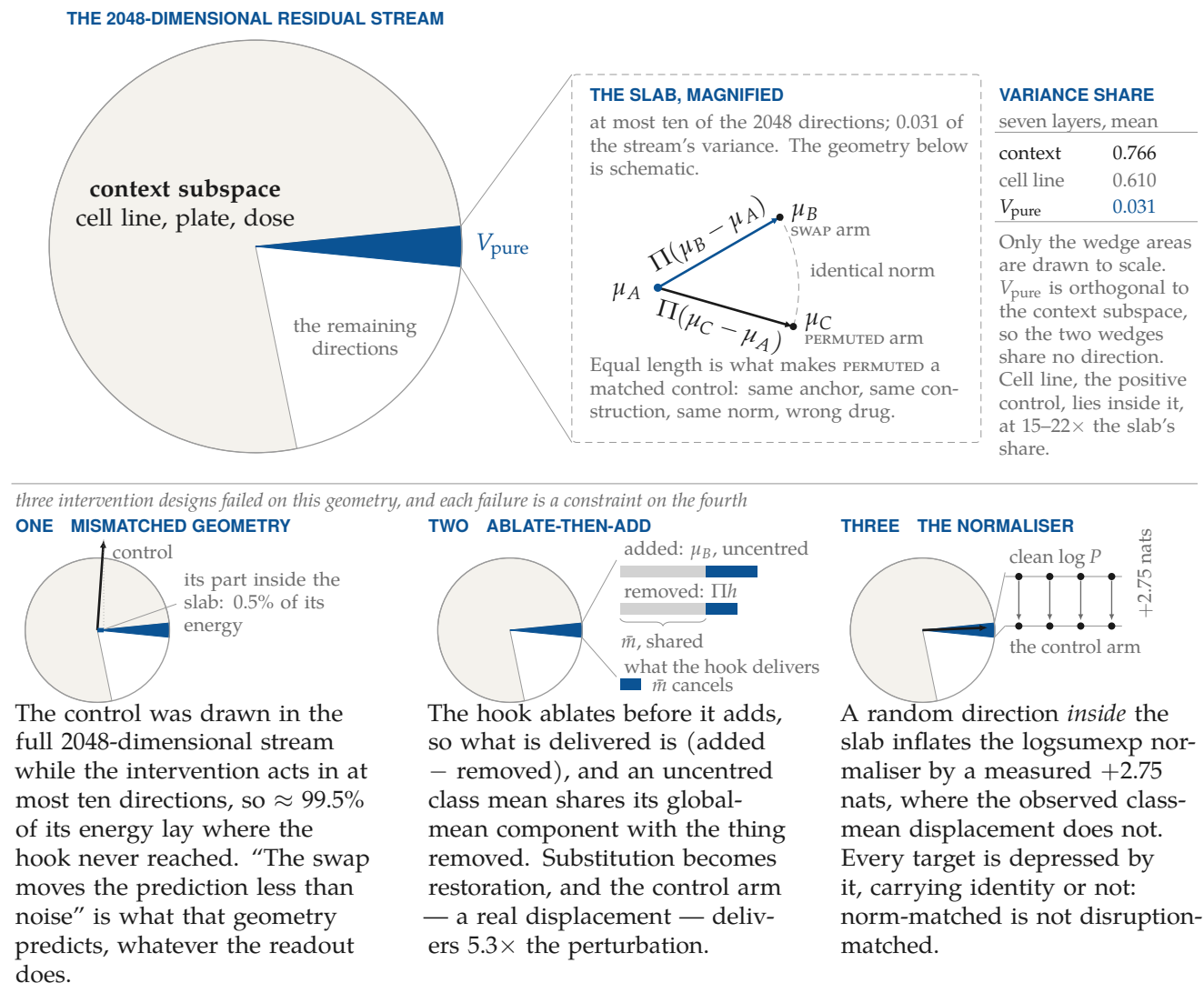


Figure A.3. The purified drug slab is a thin wedge of the residual stream beside the context subspace, and each of the three failed identity tests turns on that geometry. The disc is the 2048-dimensional residual stream at one layer. The two wedge areas are the only scaled marks in the figure, drawn to the measured subspace variance shares in `workspace_probe.json` (twelve drug classes, forty cells each, chance 0.083), averaged over the seven probed layers: 0.766 for the context subspace of cell line, plate and dose, and 0.031 for the purified drug slab V_{pure} . Cell line alone, the positive control, reads 0.610 and lies inside the context subspace, so it is quoted as a number rather than drawn as a third wedge. V_{pure} is what remains after the drug slab — spanned by at most eleven directions, since it is built from twelve centred class means, of which the top ten are kept — is orthogonalised against that context subspace, which is why the two wedges share no direction. The unshaded remainder closes the disc and carries no number. Everything else is schematic and has no scale: the slab is 11.3° wide and cannot hold three class means at true size, so its interior is magnified, where the SWAP and PERMUTED displacements are drawn at equal length because equal length is what makes PERMUTED a matched control. The one exception among the insets is inset one's angle, which is drawn true: a direction drawn isotropically in 2048 dimensions puts 10/2048 of its energy in a ten-dimensional slab, the appendix's $\approx 99.5\%$ lying outside it. No decode accuracy, layer axis or KL divergence appears here; those are Figure 4.8. The three insets are the failed designs of § A.5.

limitation, registering whether the generated text moves and not whether it moves correctly, is stated where it is used.

THREE FURTHER GUARDS ARE ENGINEERING, NOT INFERENCE. A *drug-proxy screen* refuses any context label that is a near-bijection with drug; it caught `dose_float` holding a sample identifier in the residual builders, a label that would have deleted the drug axis had the builders orthogonalised against it. An *injected-norm diagnostic* reports the displacement against the cell's centred slab component (0.69–1.40), so the injection is the size of the thing it displaces and a null cannot be blamed on a whisper. A *selftest with teeth* runs three planted worlds (drug inert, drug read, and drug inert behind a readout sensitive to magnitude and not direction) through the headline code path, separates them (−0.042, +1.520, −0.010), and failed four times in development, each failure a real defect the guard list now covers.

SCOPE LIMIT | carried into § 4.5

The third planted world is not a normaliser control. Each planted response is a single token, so nothing diverges, and the state-dependent part of the normaliser is precisely what the selftest cannot exercise. What separating the three worlds licenses is the code path the selftest runs them through.

A.6 The metric exploit is a property of the metric, not of one baseline

— — — — —

■ EXPR / ▲ RESID

A metric audit whose own instrument is untested proves nothing, and the exploit it turns on was demonstrated with a single construction. The instrument here is a second, independent zero-information baseline put through the same framework. It saturates the standard metric too.

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

The self-test does the deterministic things first (a perfect prediction reaches the metric's maximum, a zero shift and a flat truth return no score), then the thing that matters. It asserts the exploit under a *second, independent* zero-information baseline, where a leave-one-out mean scores $DE-\Delta r = 1.000$ while NIR refuses to place it above chance. The coded assertion is $NIR \leq 0.6$; the recorded value is 0.000.

WHAT THAT 0.000 IS NOT. It is not the metric's chance value, 0.500 (§ 4.1), which is what the generic arm scores in Table 4.11 and Table 4.14. Nor is it the untied degeneracy that section describes, since the self-test uses the same half-tie convention as everything else. It is a property of this construction. The mean of the five other drugs resembles those five more closely than the held-out one, so it loses every comparison.

Two unrelated constructions saturate the standard metric, and the discrimination metric places neither above chance. One would leave the standard metric merely unlucky; two make the exploit a property of the metric.

Reference apparatus

Every term, symbol and convention that carries weight across more than one chapter is fixed here, with the record of the six withdrawn claims.

VERDICT Prompt sensitivity to trained drug names is established; biological transfer is not established, and nothing built here beats the training-only per-drug lookup on common support.

Where an interval exists, a number is quoted as a value followed by its bracketed lower and upper bounds, a two-way cluster-robust 95% interval unless stated otherwise.

Two measurement frames, one chance value

The two frames score different objects against different competitors, and they both put chance in the same place. Comparing a number from one with a number from the other is the most expensive misreading available in this thesis.

In the **■** EXPR *expression frame* the prediction is a cell profile, scored against other drugs on the same plate. In the **▲** RESID *residual frame* it is the drug-specific residual, what survives once the generic programme has been subtracted, scored against other drugs in the same cell line. In both frames the *comparison set J*, the conditions a prediction is ranked against, fixes chance at 0.50, and *J* is declared explicitly at every score.

WHERE THE TRAP IS. Because chance is 0.50 in both frames, two numbers that look alike invite subtraction. They may not be subtracted. The residual frame's within-well precision reference is 0.854 where the expression frame's is 0.576, so the 0.537 in one frame and the 0.498 in the other are not two attempts at one task (§ 4.14).

HOW TO TELL WHICH IS IN FORCE. Three redundant cues say so. Every section declares its frame in its opener as a chip, **■** EXPR or **▲** RESID, shape and word before colour so that the cue survives greyscale; the running head repeats it on every page; and a margin cue marks any change of frame inside a section. A head naming no frame asserts nothing in either, as here and in the apparatus of chapter 3.

SEE Figure 4.1 maps the nine beats in order

FRAME **■** EXPR
▲ RESID in every running head

SEE Figure 3.2 draws the residual construction

SCOPE LIMIT | carried into every chapter

A distance to a reference is comparable within a frame and not between frames. The same restriction holds *within* a frame across supports: populations selected by different rules are not one population. Where this thesis quotes two numbers side by side it names the population each was computed on; where no population is named, they are not being compared.

Three registers of qualification

There are exactly three registers and no fourth.

A scope limit fences in what a number *licenses*, and the reader has to carry it forward. The test: would a later section be entitled to say something different if this sentence did not exist? Its tag names where the limit carries to.

REGISTER The three kinds: a limit carried forward, a correction on record, and ordinary inline care.

A correction marks a withdrawal this thesis makes against itself. Six are withdrawn *claims*, indexed in Table A.4, so that set is bounded and countable; the rest retract a defective piece of apparatus — a voided control, a mis-sampled interval, a replaced instrument, a leaking partner pool. Each mark of either kind stands where the withdrawal is argued.

Routine qualification stays inline and unmarked, in the sentence it qualifies, and most of the qualification in this thesis is of this kind. It is not a lesser statement for being unmarked, only one the reader need not carry anywhere.

TWO PHRASES THAT ARE NOT SYNONYMS. not established and shown to be zero are different statements. A measurement that fails to separate two arms has established nothing about their difference. It has not shown the difference to be zero. Where an interval spans zero, this thesis says which of the two it means.

AND TWO MORE THAT ARE NOT. Excluding zero is not the same as clearing a pre-specified relevance margin, and spanning zero is not evidence of equivalence. Two margins are pre-registered, on different quantities and different instruments.

relevance margin The pre-set 0.03, in NIR units, that a channel's interval has to clear on the gate of § 4.12 before the channel is called live.

materiality margin The pre-registered 10% of the model's own clean preference gap, on the identity test of § 4.5. An effect inside it is too small to be called material.

Where a pre-registered margin exists the verdict vocabulary is fixed in advance, three states plus a fourth for a control that could not be built.

live The interval clears the pre-set 0.03 relevance margin on the clustering axis appropriate to the claim. The axis is written into the verdict, because the word is not usable without it.

closed The interval lies wholly below that margin.

inconclusive Neither. This includes an interval that excludes zero but straddles the margin, and it is not equivalence with the null.

untestable The plate-matched null spans too few cell lines to be built here. “We could not build the control” must never be written as “closed”.

Reading the page

A number a sentence *asserts* goes in the body, always, with its interval, its comparison set and its reference beside it. A number it merely *recalls* may go in the margin column, bare, with a pointer to where it was established. Delete every margin note and the body must still be complete and correctly qualified.

MARGIN COLUMN
Definitions, land-mark constants, frame cues and pointers; never an assertion.

FIVE SHAPES. A section *opener* states the question and the instrument built to answer it, in words, with no numbers; the beat strip that goes with it is the investigation’s nine beats with the current one filled, and no cell filled means the section is not one of the nine, as here. A *claim block*, hairlined above and below, carries the verdict in language that could be shown wrong. § 4.13 closes without one, deliberately: it asks which span of the prompt the model reads, cannot answer, and ends in the open question. A *definition* sits in a tinted panel with no rules. A *scope limit* and a *correction* are the two marked hedge registers above.

THE AUDIT FLAG. A tinted value carrying a raised query mark is a quantity this draft does not settle and a rerun will; three are open. The prose around such a value holds whichever way it moves, and where that could not be done the argument was not made. No such flag should survive into a submitted copy.

Symbols

Fixed for the whole document, never redefined in a section.

NIR	<i>Normalised inverse rank.</i> The fraction of a declared comparison set whose truth a prediction resembles less than it resembles its own, with ties counted at one half. Chance is 0.50 under the exchangeability condition audited in § 4.1.
DRF	<i>Dynamic range fraction.</i> Not a performance score but a meta-metric. It fixes a scale with an uninformative predictor at zero and a perfect predictor at one, then asks how far up that scale a candidate metric places a positive control of known quality (the interpolated duplicate in the expression frame, a raw split-half in the residual frame, chapter 3).
DE- Δr	The field’s standard differential-expression delta-correlation, the correlation between predicted and true change from control, restricted to the genes with the largest true change.
panel- τ	Kendall’s τ between predicted and true rank over the full 946-gene panel. It selects no gene subset.
Δ_{scr}	The scramble gap, NIR of the model minus NIR of the scramble.

$\mathcal{T}$	The disattenuated transfer coefficient, measuring how much of a drug's response is shared across cellular contexts. It is a similarity coefficient, not a variance share.
r	The drug-specific residual, $r(d, c)$: what is left of the shift once the generic programme has been subtracted from it. The scored object in ▲ RESID.
s	The treated-minus-control shift, $s(d, c)$: the treated pseudobulk minus its plate-matched control pseudobulk, and nothing else. An arithmetic object, not a verdict about the drug.
[END_CELL]	The registered atomic token that terminates a cell sentence.
[DOWN]	The registered atomic token marking the up/down boundary in a residual target. Left unregistered it splits into sub-words and the boundary ceases to be a clean symbol.
■ EXPR	Page furniture, not a quantity. The expression frame is in force here.
▲ RESID	Page furniture, not a quantity. The residual frame is in force here.

Terms

Defined once, used in that sense everywhere.

ONE REFERENT, ONE WORD. A synonym reads as a second object. These are the words the drafts converged on, with the ones they gave up.

drug The perturbagen, whatever its chemistry. *Compound*, *molecule* and *agent* are not used as synonyms for it; *small molecule* appears only where the class of thing the atlas contains is what is meant.

cell line The biological context. Two words as a noun, hyphenated only as a compound adjective, as in *cell-line-specific*. The bare *line* stands in where the paragraph has already fixed the referent.

shift, generic, residual Three objects, never interchangeable: the *shift* is treated minus control, the *generic* is the drug-agnostic mean response subtracted from it, and the *residual* is what remains. *Response* keeps its ordinary biological sense and *effect* its ordinary one; neither stands in for the residual.

prediction What the model produces and what gets scored. *Generation* names the act of decoding and only that. *Output* is not used as a noun for it, and survives only inside the fixed instrument name *output invariance*, which names a test and not the thing predicted.

readout Two senses, and the surrounding sentence fixes which. Biologically, what the sequencing reports. Of the model, the decoding stage that turns the residual stream into a sentence, the second term of the *representation and readout* distinction chapter 5 turns on, for which *decoder* is the alternative word. Neither sense names the prediction, which is what that stage emits.

SEE chapter 3
builds the apparatus these terms
name

prompt What the model is given at inference: a cell sentence, the drug name, the dose and the mechanism field.

profile A ranked expression vector. Not a synonym for *sentence*, which is the encoding of one cell.

sample (retired) Earlier drafts used the word for the scored unit and for the physical one, which made two objects look like one. Neither now: the scored unit is a *condition*, the physical one a *treated well*. *Observation* is not used for either.

signature (retired) Bare *signature* names nothing here. *Drug-specific signature* is allowed once, to gloss *residual* on first use; after that the word is *residual*.

UNITS OF ANALYSIS.

condition A (drug, cell line, plate, dose) tuple. It is the unit that gets scored.

treated well The physical treatment assignment. Conditions nest inside it, so a well is not itself a condition and the two are not counted against each other.

batch unit The pair (cell line, plate), never the plate label alone, because plates are numbered within cell line and not globally.

pseudobulk The mean expression profile of a group of cells. A *control pseudobulk* is the same object over plate-matched DMSO cells.

dose A molar concentration resolved from `drugname_drugconc`, carried as a concentration and never as a unitless float.

cell sentence A cell written as its expressed panel genes in descending order of expression, terminated by `[END_CELL]`. The encoding discards expression magnitudes and retains ranks.

FRAMES AND COMPARISON SETS.

expression frame Scores a predicted cell profile against other drugs on the same plate; conditions grouped by (cell line, plate).

residual frame Scores a predicted residual against other drugs in the same cell line, after the generic programme is subtracted; conditions grouped by cell line.

comparison set *J* The conditions a prediction is ranked against. It fixes chance at 0.50 in both frames and is declared wherever a score is quoted.

within-well split-half precision reference The score obtained when a disjoint cell subset of the same treated well is substituted for the prediction, recomputed for every stratum scored. One estimator, not several; what varies between settings is the frame, the comparison set and the cell budget. It is a sampling-precision estimate, optimistic relative to true biological reproducibility: not a biological replicate, and not the maximum predictable fraction of biology.

ceiling A shorthand for the reference above, used wherever a result is stated as a distance from it. The one exception in this document is the DRF positive control of § 4.1, which is not a row of Table 3.2.

noise ceiling (retired) Earlier drafts joined *ceiling* to a word for error, which invited reading the value as a bound on achievable performance. It is a two-cell precision estimate. Where § 4.1 needs the quantity it names the construction, the DRF positive control or a row of Table 3.2.

identifiability A property of a single condition, not of a drug, and distinct from the aggregate references above. The condition’s own within-plate split-half NIR against its plate-mates clears 0.8.

TARGET CONSTRUCTION.

generic The drug-agnostic mean response subtracted to form the residual, operationally the mean shift over the other drugs at the stated scope. It is conventionally read as the stress and cell-cycle programme common to all drugs, a reading used for intuition and nowhere tested.

residual target What remains once the generic is removed. Built *split-before-fit* and *leave-one-drug-out*.

split-before-fit The holdout is fixed from metadata before any statistic is estimated, so no held-out target is defined partly by other held-out conditions.

leave-one-drug-out No condition’s generic contains its own drug, so the two sides of the split are scored against identically defined targets.

scope The set of drugs the generic mean is taken over, either *cell_line* or *plate*. The choice is consequential.

quota The explicit per-split sampling rule, used because a uniform draw reproduces the pool’s proportions and starves the held-out splits.

training arm A model variant differing from the others only in its training target or objective; five are used, cold-started from the same base checkpoint with identical hyperparameters, so that it is the only variable. The bare word *arm* also names any one side of a paired comparison (a split, a run, a channel, a baseline), and is qualified wherever the surrounding sentence does not fix the sense. An unqualified *arm* is not necessarily one of the five.

BASELINES, SPLITS AND REFERENCES.

control_copy The baseline that predicts the untreated cell unchanged. It carries zero drug information by construction, and it is the actual zero-shift predictor.

revert_center A predictor that places every gene at the middle rank. It carries no information about drug, cell or biology, and it is *not* zero-shift.

drug lookup This drug’s mean residual, estimated from training conditions in other cell lines. It needs no model, no prompt and no checkpoint.

moa_lookup The same construction over drugs sharing this drug’s mechanism annotation, on its own narrower support.

channel An annotated drug property — protein target, mechanism, chemistry — an unseen drug might be reached through. For a channel X, a drug’s residual is predicted as the mean residual of the other drugs in the same cell line sharing property X with it: *moa_lookup* generalised from mechanism

to any annotation. It is called live, closed, inconclusive or untestable against a plate-matched null differing in the channel alone (§ 4.12).

scramble The prompt left byte-identical except that the drug name and mechanism are replaced by another drug's, while the control cell and the scoring truth are unchanged. It is the only manipulation that isolates drug use. Its near, orth and opposite strata are fixed by the measured cosine between the partner's residual and the target's, so a stratum's neutrality is an empirical property of these data and not a consequence of its construction.

tiers The three evaluation tiers of the broad characterisation, defined by what the training set contained. *Tier 1*: the (drug, cell line) pairing was seen, so it measures fit, not generalisation. *Tier 2*: the drug never appeared (Leave-Drugs-Out). *Tier 3*: drug and cell line were each seen but never together (Leave-Pairs-Out).

train / unseen_combo / unseen_drug The residual arm's own three-part split. *unseen_combo* is an unseen *well* of a seen drug, so it tests recombination and not full novelty.

held out Held out from Tahoe fine-tuning, with the mechanism annotation still supplied in the prompt. Not absent from the model's prior world knowledge. Every unseen-drug result must be read under that narrower scope.

The record of corrections

Six claims that this thesis printed are withdrawn by this thesis. Table A.4 indexes them, so the set is countable. Each is argued in full in the section the table names. Corrections to the apparatus rather than to a claim carry the same margin mark but are not tabulated here; each stands where the repair is argued.

The same controls that produced these six shape the surviving design, among them a zero-information baseline that improved when plate was free to vary; a swap-distance sweep after a similar-drug scramble proved weak; a removal gate before any ablation null; matched-norm injection before interpreting a swap; intervals clustered on the units that repeat, cell line, drug and treated well, never on cells; and a pre-registered three-state interpretation whose inconclusive branch was allowed to fire. That record is why the remaining claims can be read at the strength stated here, and no stronger.

Table A.4. The six corrections this thesis records against itself. Each row points at the place the withdrawal is argued. Three of the six are argued in a numbered box; the rest in the prose of the section named.

The claim withdrawn	What killed it	Where
The stronger mechanistic story earlier drafts told about the activation swap	Three substitution designs in turn: a control drawn in the full residual stream put almost all its energy where the intervention never acts; ablate-then-add turned the substitution arm into a restoration; and a norm-matched random direction inflated the normaliser. The retracted result was geometry, not biology.	§ 4.5
The retention argument for a cell-line-scoped generic	Both retention figures were computed with a shared control pseudobulk subtracted from both halves, which correlates the halves through their common control error and favours cell-line scope precisely because the cell-line generic is itself less noisy.	§ 4.7
That the optimal-transport arm achieved nothing	The generation budget. Scored at the shorter token cap the arm's gap spanned zero and was printed as inert; re-run at the budget the residual arm had, it does not. The token budget is part of the claim.	§ 4.9
A training-exposure or memorisation advantage on trained conditions	Composition. Restricting the comparison to the drugs present in both arms reverses the sign, and the two arms are scored on populations selected by different rules.	§ 4.10
That the model reads the drug name and not the mechanism field	The comparator. Those arms were scored against an opposite-stratum partner, which inflates every gap read against it, and because that is a property of the swap partner and not of the target build, it survives the re-run.	§ 4.13
That a swing in the held-out-drug control was evidence of seed instability	The comparator, and the draw. Both superseded figures are gaps against the opposite swap, which § 4.10 shows is not a null; and unseeded <code>do_sample</code> advanced from global torch state, so the two arms of a contrast were drawn from different points of one stream. The diagnosis named the trigger and missed the cause.	Chapter 5

Withdrawing a claim is not the same as establishing its negation. Each row records that the evidence offered will not carry the claim, and leaves the underlying question where the section arguing the withdrawal says it leaves it.